ZITIL Master

ZyeemZia™ Information Technology Infrastructure Library Master

Lead IT services that create value, every day.



COURSE BOOK · WORKBOOK · EXAM PREPARATION

8 MODULES · 36 LESSONS · 60 EXAM QUESTIONS

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BEFORE YOU BEGIN

How to use this book

A master-level IT service management track for people who run, design and lead digital services: the service value system, the four dimensions, the core practices from service desk to continual improvement, customer journeys, DevOps and SRE, strategy and governance, and AI-driven operations, finished with complex real-world cases where you must choose, justify and lead.

01
Learn
Work through 36 lessons. Each ends with a knowledge check.

02
Practise
Fill in the templates, answer the scenarios and case studies, review the glossary.

03
Certify
Take the 45-question official exam at zitil.zyeemzia.com, graded on the ZyeemZia server. Pass mark 85%.

04
Get certified
Download your certificate online. Anyone can verify its credential ID at zitil.zyeemzia.com/#verify. 92%+ and the capstone earn Distinction.

Eight-week study plan

WHEN	MODULE	DO THIS	DONE
Weeks 1–2	m1 Value	Value map for your capstone service, principle self-assessment	☐
Weeks 3–4	m2 Model	Four-dimensions assessment, first value-stream map	☐
Weeks 5–7	m3 Restore	Incident and problem design, knowledge and monitoring review	☐
Weeks 8–10	m4 Control	Change model, release plan, CMDB scope and SLA framework	☐
Weeks 11–12	m5 Journey	Customer-journey map, experience measures, team design	☐
Weeks 13–14	m6 Flow	SLOs, error budget, toil audit and flow metrics	☐
Weeks 15–16	m7 Direct	Governance, measurement and digital strategy pack	☐
Weeks 17–18	m8 Lead	AI plan, adoption plan, practice exam	☐
Weeks 19–20	Capstone	Complete the capstone pack and sit the certification exam	☐

Learner name .....

Start date ..... Target exam date .....

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+ Glossary · Capstone guide · Practice exam & answers · Certificate · Notes

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THE FRAMEWORK

The ZITIL Service Value Loop

Eight stages a mature service organisation keeps cycling through: define value, model the system, restore service fast, control change, design the customer journey, speed up the flow, set direction and lead adoption. Masters never leave the loop; each turn makes the services better.



01 Value
What services are for, how an organisation turns demand into value, and the principles that steer every decision

02 Model
Model the whole system: people, information, partners and flows of work

03 Restore
The practices that keep services running and get them back fast when they fail

04 Control
Change, release, deployment, requests, assets, configuration and service levels in one controlled flow

05 Journey
From design and daily delivery to the end-to-end experience customers live

06 Flow
Deliver fast and stay reliable with modern engineering and flow practices

07 Direct
Set direction, govern risk, measure what matters and shape the digital future

08 Lead
Apply AI responsibly, analyse complex situations and lead lasting change

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01

MODULE 1 · VALUE

Service Value System and Guiding Principles

What services are for, how an organisation turns demand into value, and the principles that steer every decision

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Explain services in terms of outcomes, costs and risks, and the roles in a service relationship
- Describe the service value system and diagnose where value is lost
- Apply the seven guiding principles together to real decisions
- Distinguish governance from management and run continual improvement as a habit

LESSONS

1. Services, Value and Co-creation · 30 min
2. The Service Value System · 35 min
3. The Seven Guiding Principles in Practice · 40 min
4. Governance and Continual Improvement in the Value System · 40 min

MODULE 1 · VALUE · LESSON 1 · 30 MIN

Services, Value and Co-creation

Why a service only matters when someone gains an outcome from it

- Define a service in terms of outcomes, costs and risks rather than technology
- Distinguish the roles of customer, user and sponsor in a service relationship
- Explain utility and warranty and why both are needed for value
- Describe how value is co-created by provider and consumer

Most IT departments describe themselves by what they run: servers, networks, applications, a help desk. Consumers do not care about any of these directly. They care about what they can **achieve** because the service exists: a nurse can see a patient's allergy history at the bedside, a garment buyer in Gazipur can confirm shipment dates, a mobile wallet user in Sylhet can pay a utility bill at midnight. Service management is the discipline of organising people, information, partners and work so that those outcomes happen reliably, at an acceptable cost and with acceptable risk.

> DEFINITION · SERVICE, IN PLAIN TERMS
>
> A **service** is a way of helping someone achieve outcomes they want, while the provider takes on specific costs and risks that the consumer would otherwise carry. The consumer does not need to own the servers, hire the engineers or manage the supplier contracts; they get the outcome and pay, directly or indirectly, for the provider to carry the rest.

The three things every service changes for a consumer

O

Outcomes enabled

The results the consumer can now achieve: faster admissions, fewer payment failures, accurate stock counts. Outcomes are what people would describe if you asked why they use the service.

C

Costs

Costs removed (no need to buy and run their own system) and costs added (subscription fees, training time, process changes). Value improves when the balance moves in the consumer's favour.

R

Risks

Risks removed (a provider with redundant data centres) and risks added (dependency on that provider, data leaving the organisation). Mature providers make both visible.

Roles in a service relationship

ROLE	WHAT THEY DO	EXAMPLE: HOSPITAL PATIENT-RECORDS SERVICE
Customer	Defines requirements and agrees service levels; accountable for the outcomes being right	Director of Clinical Services who agrees what the records system must support
User	Uses the service day to day; experiences its quality first-hand	Ward nurses, doctors and admission clerks
Sponsor	Authorises the budget and funds consumption	Chief Financial Officer approving the annual service charge
Provider	Designs, runs and improves the service, taking on costs and risks	The hospital IT department plus its cloud and software partners

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INSIGHT · ONE PERSON, SEVERAL ROLES

In a small clinic the same doctor may be customer, user and sponsor at once. In a large bank the three roles sit in different departments with different priorities: users want speed, customers want features, sponsors want lower cost. Designing a service means listening to all three and making the trade-offs explicit.

Utility versus warranty

Utility: fit for purpose

- What the service does
- Functions and features that support an outcome
- Example: the payroll system calculates overtime and festival bonuses correctly
- Missing utility: the service works but does not help

Warranty: fit for use

- How well the service performs
- Availability, capacity, security, continuity
- Example: payroll runs on the 25th without failing, even at year end
- Missing warranty: the service helps but cannot be trusted

Value is **co-created**. A provider can supply an excellent learning platform to a university, but if lecturers never upload materials and students are not shown how to submit work, no value appears. The consumer contributes data, time, adoption effort and feedback. This is why modern service management treats the relationship, not just the service, as the unit of design: onboarding, training, communication and feedback loops are part of how value is produced.

Products, offerings and services

- A **product** is a configuration of the provider's resources (people, technology, partners, processes) designed to be valuable to a consumer group.
- A **service offering** is how part of a product is presented to a consumer: goods transferred, access to resources granted, and service actions performed.
- One product can support many offerings: a cloud platform product can be offered as a basic tier for start-ups and a regulated tier for banks.
- Consumers see offerings; providers manage products. Confusing the two leads to catalogues full of internal components nobody can order.

SCENARIO · DECIDE · DEFINING THE SERVICE PROPERLY

The IT team of a mobile financial services provider in Dhaka publishes a service catalogue listing 'Core Banking Database Cluster', 'API Gateway' and 'SMS Aggregator Link'. Merchant support staff say they cannot find anything they can ask for. What is the best improvement?

(A) Add detailed technical descriptions, owners and dependency diagrams to each component so that support staff understand what it does

(B) Train the merchant support staff on the technical architecture

(C) Rename entries around consumer outcomes such as 'Merchant settlement', with components mapped underneath

(D) Remove the catalogue and route all requests through email to the IT manager

Circle your choice. Answer and feedback: module answer key.

TEMPLATE **Service value statement**

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Describe one service from the consumer's side before you manage it.

SERVICE	CONSUMER GROUP	OUTCOMES ENABLED	COSTS REMOVED / ADDED	RISKS REMOVED / ADDED	UTILITY AND WARRANTY MUST-HAVES
Merchant settlement	*Retail merchants*	*Next-morning cash in bank account*	*Removed: manual reconciliation / Added: monthly fee*	*Removed: cash handling / Added: platform dependency*	*Correct netting; 99.9% availability 06:00–24:00*

Key terms

Service
A means of enabling outcomes for a consumer while the provider carries specified costs and risks on the consumer's behalf.

Outcome
A result the consumer wants to achieve, such as faster admissions or fewer failed payments.

Co-creation of value
The idea that value appears only when provider and consumer both contribute: the provider supplies resources, the consumer adopts and uses them.

Utility
The functionality of a service: what it does to support an outcome; fitness for purpose.

Warranty
Assurance that a service meets agreed requirements for availability, capacity, security and continuity; fitness for use.

Service offering
A formal description of one or more services as presented to a consumer group, combining goods, access to resources and service actions.

Sponsor
The person or body that authorises budget for service consumption.

Service relationship
The cooperation between provider and consumer, including provision, consumption and relationship management activities.

REFLECT

Pick one service your organisation provides. Write one sentence describing it in your team's language and one in the language of its users. How different are they?

KNOWLEDGE CHECK

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1. A provider says its new HR portal is 'excellent' because it has 40 features. Users still email HR for most requests. Which concept best explains the gap?

   (A) The sponsor has not yet funded enough features to make the portal attractive to users
   (B) Value is co-created, so adoption is part of producing value
   (C) Users are not part of the service relationship
   (D) The portal lacks warranty because it has too many features

2. Which statement describes warranty rather than utility?

   (A) The service is available 99.95% of business hours
   (B) The service lets agents reset customer PINs after verifying identity
   (C) The service shows stock levels for each warehouse
   (D) The service calculates VAT on each invoice

3. In a university, the Registrar defines what the admissions system must do, students use it and the Finance Committee approves its budget. Who is the sponsor?

   (A) The Registrar
   (B) The Finance Committee
   (C) The students
   (D) The IT department as service provider

4. Why should a service catalogue be organised around consumer outcomes rather than technical components?

   (A) Technical components never change, so they need no catalogue
   (B) Consumers recognise and request outcomes, not components
   (C) Components are confidential and must never be documented
   (D) Outcome-based catalogues remove the need for any configuration data about components

5. A cloud migration removes the cost of a hospital's own data centre but adds dependency on one provider's region. How should a service manager present this?

   (A) As a shift in both costs and risks for the customer to weigh
   (B) As a technical matter that the customer need not see
   (C) As a warranty failure that blocks the migration
   (D) As pure cost saving, since the data centre and its staff are no longer needed

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MODULE 1 · VALUE · LESSON 2 · 35 MIN

The Service Value System

How demand becomes value through one connected operating model

- Describe the components of the service value system and how they connect
- Explain why silos destroy value and how the system view counters them
- Relate organisational agility and resilience to the design of the value system

A service organisation is not a list of processes. It is a **system**: a set of parts whose interactions produce something none of them can produce alone. The **service value system** is a model of how all the parts of an organisation work together to turn opportunity and demand into value. Its power is that it forces leaders to see the whole: strategy, governance, everyday work and improvement as one machine.

The service value system at a glance

Value	`Outcomes for consumers and stakeholders` `Measured by benefit, cost and risk`
Service value chain	`Plan` `Improve` `Engage` `Design and transition` `Obtain or build` `Deliver and support`
Practices	`Organisational resources for specific work` `Service desk, incident, change and many more`
Guiding principles and governance	`Seven principles steer decisions` `Governance evaluates, directs and monitors`
Continual improvement	`Runs at every level and in every component`
Opportunity and demand	`Ideas, needs and requests that trigger value creation`

From demand to value

01 Opportunity / demand
A need or an idea arrives: a new regulation, a customer request, an outage

→

02 Principles & governance
Decide whether and how to respond

→

03 Value chain
Activities combine in a value stream to respond

→

04 Practices
Supply the capabilities each activity needs

→

05 Value
Outcomes are realised and measured

Components and what goes wrong without them

COMPONENT	PURPOSE	SYMPTOM WHEN IT IS WEAK
Guiding principles	Shared rules of thumb for decisions in any situation	Every team argues from first principles; decisions are inconsistent
Governance	Direction and oversight by those accountable	Activity without priorities; risk taken without anyone knowing

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COMPONENT	PURPOSE	SYMPTOM WHEN IT IS WEAK
Service value chain	Operating model of six activities	Work handed over the wall between disconnected departments
Practices	Sets of resources for specific types of work	Heroics replace capability; knowledge sits in individuals
Continual improvement	Ongoing alignment with changing needs	The organisation is optimised for last year's problems

WATCH OUT · THE SILO TRAP

Silos form when each department optimises its own measures: the service desk closes tickets fast, development ships features fast, operations keeps change low. Each looks good; the consumer still waits two weeks for a working fix. The value system view asks one question of every team: how does your work move value towards the consumer?

Agility and resilience

Organisational agility

- Ability to move and adapt quickly
- Short feedback loops
- Decisions taken close to the work
- Example: an e-commerce team releasing a new payment option before Eid in two weeks

Organisational resilience

- Ability to absorb shocks and recover
- Redundancy, rehearsed recovery, diverse suppliers
- Example: a bank continuing card payments through a regional power failure

Agility and resilience are often presented as opposites, but a well-designed value system needs both. Automated testing and small changes increase speed **and** reduce failure impact. Clear governance lets teams move fast inside agreed boundaries. The master-level skill is to design the system so that speed and safety reinforce each other instead of competing.

Using the value system as a diagnostic

1. **Start from a real demand** Choose a recent, important piece of demand: a major incident, a new branch opening, a regulatory change.
2. **Trace it through** Follow it from arrival to value realised. Note every handover, queue and decision point.
3. **Check each component** Were principles applied? Did governance set priorities? Which value chain activities and practices were used?
4. **Find the weakest link** Identify where time, quality or information was lost the most.
5. **Improve deliberately** Feed that weakness into the improvement backlog with an owner and a measure.

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SCENARIO · DECIDE · CHOOSING WHERE TO INTERVENE

A power distribution company in Khulna reports that customer complaints about billing errors take 25 days to resolve. The service desk logs them in an hour, billing analysts fix them in two days, but cases wait weeks for approval from a finance committee that meets monthly. What should the service manager do first?

(A) Hire more billing analysts to speed up corrections

(B) Set a stricter target for the service desk to log every complaint within 15 minutes and report compliance weekly to management

(C) Show the end-to-end flow to governance and propose delegated approval limits for small corrections

(D) Automate the complaint form so customers enter more data

Circle your choice. Answer and feedback: module answer key.

Value system health check

☐ We can name our most important services and the outcomes they enable

☐ Governance sets priorities that teams can see and apply

☐ Our guiding principles are known and referenced in decisions

☐ We can trace any important demand end to end

☐ Practices have named owners and measurable capability

☐ Improvement work is visible, prioritised and resourced

☐ We measure agility (lead time) and resilience (recovery time) together

Key terms

Service value system (SVS)
A model of how all components and activities of an organisation work together to turn opportunity and demand into value.

Demand
Need or desire for products and services from internal and external consumers.

Silo
A team or department that optimises its own work and measures at the expense of the end-to-end flow of value.

Organisational resilience
The ability of an organisation to anticipate, absorb and recover from disruption.

Opportunity
A possibility to add value for stakeholders or improve the organisation, not yet requested by a consumer.

Practice
A set of organisational resources designed for performing work or accomplishing an objective.

Organisational agility
The ability of an organisation to move and adapt quickly, flexibly and decisively.

REFLECT

Trace one important piece of demand your organisation handled this year. Where did it wait longest, and which part of the value system owned that wait?

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KNOWLEDGE CHECK

6. What is the main purpose of the service value system model?

   (A) To replace governance with a set of practices
   (B) To define the technical architecture that underpins each service
   (C) To show how all parts of an organisation turn demand into value
   (D) To list every process an IT department must document and audit

7. Each department in a company meets its targets, yet customers still wait weeks for fixes. What is the most likely cause?

   (A) Too few guiding principles
   (B) Silos optimising local measures instead of end-to-end flow
   (C) A lack of service desk staff to log and route customer requests
   (D) Excessive automation of handovers

8. Which statement about agility and resilience is most accurate?

   (A) Good design can raise both, for example with small automated changes
   (B) Resilience matters only to infrastructure teams
   (C) They are opposites, so leaders must consciously choose one over the other
   (D) Agility means removing governance entirely

9. In the value system, what triggers value creation?

   (A) The annual budget cycle
   (B) The service desk
   (C) Continual improvement
   (D) Opportunity and demand

10. A diagnostic finds that a monthly committee is the constraint in a complaint process. Which component of the value system is the weak link?

    (A) Monitoring and event management
    (B) Supplier management
    (C) Governance and decision-making
    (D) The service desk practice

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MODULE 1 · VALUE · LESSON 3 · 40 MIN

The Seven Guiding Principles in Practice

Seven rules of thumb that survive any framework, method or tool

- Explain each guiding principle in your own words with a practical example
- Apply several principles together to a messy, real decision
- Recognise when principles pull in different directions and resolve the tension

Frameworks change; good judgement does not. The guiding principles are seven short pieces of advice that apply to any service decision, at any scale, whether you run a five-person IT team at a clinic or a 2,000-person technology division at a telecom operator. They are not steps. They are lenses: before you commit to a decision, look at it through each one.

The seven guiding principles

P1

Focus on value

Every activity should link, directly or indirectly, to value for someone. Know who the consumer is, what they value and how your work changes it. If you cannot explain the link, question the work.

P2

Start where you are

Before building something new, observe what already exists and what already works. Measure it directly instead of trusting reports or memories. Reuse is cheaper and less risky than reinvention.

P3

Progress iteratively with feedback

Break work into small pieces that each deliver something usable. Gather feedback after each piece and change course when it shows you are wrong.

P4

Collaborate and promote visibility

Involve the right people early and make work, constraints and decisions visible. Hidden work creates mistrust; visible work invites help.

P5

Think and work holistically

No service is delivered by one team or one technology. Consider people, information, partners and processes together, and how changes ripple across the system.

P6

Keep it simple and practical

Use the fewest steps that achieve the objective. Remove anything that adds no value, and design for exceptions rather than burdening every case.

P7

Optimise and automate

First make work as effective and useful as it can be, then automate. Automating a bad process just produces bad results faster.

Principles applied to one situation: replacing a hospital's paging system

PRINCIPLE	WHAT IT CHANGES IN PRACTICE
Focus on value	Define value as 'time for a nurse to reach the on-call doctor', not 'new paging hardware installed'
Start where you are	Measure today's response times and keep the reliable escalation rotas already in use
Progress iteratively	Pilot in one ward, then two, before the whole hospital

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PRINCIPLE	WHAT IT CHANGES IN PRACTICE
Collaborate & visibility	Involve nurses, doctors, switchboard and security; show pilot data on the ward board
Think holistically	Consider Wi-Fi coverage, device charging, shift handover and the supplier's support hours
Keep it simple	One escalation path for emergencies; remove three legacy fallback steps nobody follows
Optimise & automate	Fix the rota data first, then automate escalation when a page is unanswered for two minutes

PRO TIP · USE THE PRINCIPLES AS A CHALLENGE, NOT A CHECKLIST

In a design review, assign one principle to each participant and ask them to argue from it. The person holding 'keep it simple' attacks every extra approval; the person holding 'think holistically' asks who else is affected. Tensions surface early, while they are cheap to resolve.

When principles appear to conflict

Tension

- Keep it simple vs think holistically
- Progress iteratively vs focus on value
- Optimise and automate vs start where you are

How to resolve

- Consider the whole system, then choose the simplest design that respects it
- Make each iteration deliver a small slice of real value, not just activity
- Measure the current state first; automate what has proven to work

WATCH OUT · AUTOMATING THE MESS

A garment exporter automated its purchase-approval workflow without simplifying it. The new tool faithfully reproduced eleven approval steps, three of which existed only because of a fraud case in 2009. Approval time stayed at nine days. Optimise first: remove waste, then automate what remains.

Applying the principles to a decision

1. **State the decision** Write the decision in one sentence and who it affects.
2. **Walk the seven lenses** For each principle, note one risk or improvement it reveals.
3. **Identify tensions** Mark any two principles pulling in different directions.
4. **Choose and justify** Pick the option that best balances them and record why.
5. **Plan the feedback** Decide what evidence will tell you, early, whether you were right.

SCENARIO · DECIDE · A RUSHED TOOL REPLACEMENT

The CIO of a private university wants to replace the IT ticketing tool within one month because staff complain it is slow. Nobody has measured where time is actually lost. Which response best reflects the guiding principles?

(A) Measure where requests wait, fix quick wins, then pilot any new tool in one faculty

(B) Buy the tool with the most features so all future needs are covered

(C) Replace the tool across every faculty at once to avoid the cost and confusion of running two systems in parallel

(D) Refuse the change because the current tool works technically

Circle your choice. Answer and feedback: module answer key.

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TEMPLATE

Principle lens worksheet

Test an important decision against all seven principles before committing.

PRINCIPLE	WHAT THIS DECISION DOES WELL	RISK OR GAP	ACTION
Start where you are	*Reuses existing escalation rota*	*Response times never measured*	*Sample 200 pages before pilot*

Key terms

Guiding principle
A recommendation that guides an organisation in all circumstances, regardless of changes in goals, strategies, type of work or structure.

Start where you are
The principle of observing and reusing what already exists before building something new.

Feedback loop
A mechanism by which the results of an action are observed and used to adjust the next action.

Waste
Any activity or step that consumes resources without adding value for the consumer.

Focus on value
The principle that all activities should link back, directly or indirectly, to value for stakeholders.

Iteration
A small, time-boxed piece of work that produces a usable result and invites feedback.

Holistic thinking
Considering the whole system, including all dimensions and their interactions, rather than one part in isolation.

REFLECT

Which of the seven principles does your organisation apply least? Describe one recent decision that would have turned out better if it had been applied.

KNOWLEDGE CHECK

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11. A team plans to automate a nine-step approval workflow exactly as it is today. Which principle is most at risk?
    - (A) Focus on value
    - (B) Start where you are
    - (C) Optimise and automate
    - (D) Collaborate and promote visibility
12. Before redesigning a service desk, a manager spends a week observing calls and measuring queue times. Which principle is she applying?
    - (A) Start where you are
    - (B) Think and work holistically
    - (C) Keep it simple and practical
    - (D) Progress iteratively with feedback
13. Which choice best reflects 'progress iteratively with feedback' for a nationwide rollout of a new branch system?
    - (A) Let each branch decide its own timing without reporting
    - (B) Pilot in a few branches, learn, then expand in waves
    - (C) Delay launch until every possible feature is complete
    - (D) Train all staff once, then switch every branch on one weekend
14. Two principles appear to conflict: 'keep it simple' and 'think holistically'. What is the best resolution?
    - (A) Always favour holism, since simplicity ignores important dependencies between teams
    - (B) Always favour simplicity, since holism adds complexity
    - (C) See the whole system, then pick the simplest design that respects it
    - (D) Escalate every such conflict to the board
15. A change team publishes its weekly work board, blockers and decisions on the intranet. Which principle does this most directly support?
    - (A) Collaborate and promote visibility
    - (B) Start where you are
    - (C) Think and work holistically
    - (D) Optimise and automate

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 1 · VALUE · LESSON 4 · 40 MIN

Governance and Continual Improvement in the Value System

Who directs the system, and how it keeps getting better

- Explain how governance evaluates, directs and monitors service management
- Apply a structured continual improvement model to a real problem
- Lead improvement as an everyday habit rather than an annual project

Governance answers a simple question: **who decides, on what basis, and how do we know it is working?** In a company the governing body is usually the board; in a public university, the syndicate or council; in a government hospital, a management committee under a ministry. Governance is not management. Managers plan and run work; governance sets direction, approves the boundaries within which managers act, and checks that results match intentions.

The governance cycle

01 Evaluate — Assess the organisation, its strategy, risks and stakeholder needs

→

02 Direct — Set priorities, policies, risk appetite and accountability

→

03 Monitor — Check performance and conformance against what was directed

→

04 Evaluate again — Use what was learned to adjust direction

Governance versus management

ASPECT	GOVERNANCE	MANAGEMENT
Question	Are we doing the right things, the right way, within risk appetite?	How do we do these things well?
Typical bodies	Board, audit committee, IT steering committee	Heads of service, practice owners, team leads
Outputs	Strategy approval, policies, risk appetite, investment decisions	Plans, schedules, staffing, procedures, reports
Time horizon	Quarters and years	Days, weeks and months
Typical failure	Rubber-stamping or micromanaging	Activity without alignment to direction

INSIGHT · GOVERNANCE SHOULD ENABLE SPEED

Good governance makes teams faster, not slower. When a board sets a clear risk appetite and delegates authority within it, a change team can approve low-risk changes in minutes without escalation. When boundaries are unclear, every decision drifts upward and waits for a meeting.

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The continual improvement model

1
What is the vision?
Link the improvement to the organisation's direction and the value it should create.

2
Where are we now?
Assess the current state objectively with data: baselines, maturity, stakeholder views.

3
Where do we want to be?
Define a measurable target state and the gap to close.

4
How do we get there?
Plan the route: options, experiments, iterations and owners.

5
Take action
Execute in small steps, measuring as you go.

6
Did we get there?
Compare results to the target; confirm benefits are real.

7
How do we keep momentum?
Embed the change, celebrate success, and feed lessons into the next cycle.

Making improvement a habit

- Keep one **continual improvement register** (CIR) for the whole organisation, visible to everyone.
- Every team reserves a fixed share of capacity, such as 10–20%, for improvement work.
- Prioritise improvement ideas by value, effort and risk, not by who shouts loudest.
- Measure benefits after implementation; close ideas that produced none and learn why.
- Leaders model the habit by asking 'what did we learn?' after every major event.

TEMPLATE

Continual improvement register

Capture, prioritise and track improvement ideas across the organisation.

ID	IDEA AND PROBLEM IT SOLVES	VALUE / EFFORT / RISK	OWNER	STATUS	MEASURED BENEFIT
CI-014	*Self-service password reset to remove 22% of desk calls*	*High / Medium / Low*	*Service desk lead*	*In pilot*	*Target: 1,800 fewer calls a month*

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ID	IDEA AND PROBLEM IT SOLVES	VALUE / EFFORT / RISK	OWNER	STATUS	MEASURED BENEFIT

CASE STUDY · GOVERNANCE AFTER A PAYMENTS OUTAGE

A mobile financial services provider in Dhaka suffered a seven-hour outage on the last working day before Eid, affecting 4 million wallet users. The board learns that a database upgrade was approved by an operations manager under a general 'routine maintenance' category. There is no defined risk appetite for customer-facing changes during peak periods, and the improvement register holds 140 open items with no owners.

1. What governance failures contributed to the outage?

2. What should the board direct, and what should it delegate?

3. How would you restore the improvement register to a useful tool?

Model answer: module answer key.

SCENARIO · DECIDE · IMPROVEMENT WITHOUT A BASELINE

A hospital IT director wants to report that a new monitoring tool 'improved availability'. Nobody recorded availability before the tool went live. What should she do now?

(A) Estimate the old availability from the memories of senior engineers and report the difference as the improvement

(B) Report only that the tool was delivered on time and on budget

(C) Report current availability, state that no baseline exists, and start measuring now

(D) Delay all reporting until a full year of new data exists

Circle your choice. Answer and feedback: module answer key.

Key terms

Governance
The means by which an organisation is directed and controlled, through evaluating, directing and monitoring.

Risk appetite
The amount and type of risk an organisation is willing to take in pursuit of its objectives.

Continual improvement register (CIR)
A single, prioritised record of improvement ideas with owners, status and measured benefits.

Governing body
The person or group accountable at the highest level for the performance and conformance of the organisation.

Continual improvement
Recurring activity at all levels to keep services and practices aligned with changing needs.

Baseline
A measured starting point used to judge whether an improvement actually changed anything.

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Delegated authority

Decision rights passed by governance to management within defined boundaries.

REFLECT

Who in your organisation can approve a risky change during a peak business period? Is that authority written down, and does the governing body know?

KNOWLEDGE CHECK

16. Which activity belongs to governance rather than management?

(A) Setting the risk appetite for customer-facing changes
(B) Assigning engineers to on-call rotas
(C) Scheduling the weekend maintenance window
(D) Writing a step-by-step procedure for password resets

17. A team wants to start an improvement but cannot say what success looks like. Which step of the improvement model is missing?

(A) Take action
(B) Where do we want to be?
(C) How do we keep momentum?
(D) What is the vision?

18. Why should improvement capacity be reserved in each team's plan?

(A) Because auditors require exactly 20% of every team's time to be left unused
(B) So that teams can avoid operational work
(C) Because improvement must be done only by a central team
(D) Otherwise urgent work consumes all the time available

19. After a major outage, which governance action is most appropriate?

(A) Ask management to stop all changes indefinitely until a full root-cause review is complete
(B) Replace the operations manager and take no further action
(C) Personally approve every future change at board level
(D) Direct a risk policy and delegated authority, then monitor

20. What is the main purpose of measuring benefits after an improvement?

(A) To decide who should be promoted
(B) To justify the cost of the improvement team
(C) To satisfy the supplier who provided the tool and its licences
(D) To confirm the goal was met, or learn why not

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MODULE 1 · VALUE

Answer key

Knowledge checks

1. B Features (utility) do not produce value unless consumers adopt the service. Onboarding, design and communication are part of value co-creation. Warranty, funding and user roles do not explain low adoption here.

2. A Availability is an assurance property: fitness for use. The other three describe functions the service performs: utility.

3. B The sponsor authorises budget. The Registrar acts as customer, students are users, and IT is the provider.

4. B The catalogue is a consumer-facing tool. Components still need documenting, but in configuration data linked to services, not as the items consumers order.

5. A Every service removes some costs and risks and introduces others. Making both visible lets the customer judge value honestly.

6. C The SVS is an operating view of the whole organisation. It is not a process list, a replacement for governance, or a technical architecture.

7. B When every part looks good locally but the outcome is poor, the gaps are in the handovers and local measures. That is the silo pattern.

8. A Small, tested, automated changes increase speed and reduce the blast radius of failure. Governance sets boundaries rather than being removed.

9. D Opportunity and demand are the inputs; the rest of the system responds to them.

10. C The delay sits in a decision point owned by governance. Desk, monitoring and suppliers are not involved in that wait.

11. C The principle says to optimise before automating. Automating an unexamined process locks in its waste.

12. A Direct observation and measurement of the current state is the heart of starting where you are.

13. B Small, staged deliveries with learning between them reduce risk and improve the result.

14. C The principles work together: holistic thinking reveals what matters, simplicity removes everything else.

15. A Making work and decisions visible builds trust and invites collaboration.

16. A Setting risk appetite is a direction decision for the governing body. The others are management activities within that direction.

17. B Defining the measurable target state is the 'where do we want to be' step. Without it, success cannot be judged.

18. D Without protected capacity, the urgent always beats the important. Improvement is everyone's work, not a reason to avoid operations.

19. D Governance sets direction and boundaries and then monitors. Approving each change, blaming one person or freezing change permanently all fail to fix the system.

20. D The 'did we get there?' step verifies value and creates learning. Justifying budgets or careers is not its purpose.

Scenarios & case studies

Scenario · Defining the service properly
C is best. Correct. Consumers recognise outcomes. Components still matter, but they belong in the provider's internal view and in configuration data, linked to the services they support.

Scenario · Choosing where to intervene
C is best. Correct. The constraint is a governance decision point, not the desk or the analysts. Delegating low-value corrections within agreed limits addresses the real delay.

Scenario · A rushed tool replacement
A is best. Correct. This starts where you are, focuses on the real value problem, and progresses iteratively with feedback before a full commitment.

Case study · Governance after a payments outage
The board had not set a risk appetite for peak periods, so a high-risk change was approved under a routine category: a failure to **direct**. Nobody reported change risk to the board: a failure to **monitor**. The board should direct a risk appetite (for example, no non-emergency changes to payment systems during declared peak windows), define change categories and delegated authority, and require regular reporting on change success and incidents. It should not approve individual changes. The register should be triaged: close stale items, merge duplicates, assign owners to the top 15 by value, and review progress monthly with measured benefits.

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Scenario · Improvement without a baseline

C is best. Correct. Honesty about evidence protects credibility. Establishing measurement now allows real improvement claims later.

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02

MODULE 2 · MODEL

Four Dimensions and the Service Value Chain

Model the whole system: people, information, partners and flows of work

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Apply the four dimensions and external factors to any service design
- Build roles, culture and information practices that support service outcomes
- Choose sourcing models and integrate multiple suppliers around one service
- Map value streams across the six value chain activities and remove waste

LESSONS

1. The Four Dimensions of Service Management · 35 min
2. People, Culture, Information and Technology · 40 min
3. Partners, Suppliers and Service Integration · 40 min
4. The Service Value Chain and Value Streams · 45 min

MODULE 2 · MODEL · LESSON 1 · 35 MIN

The Four Dimensions of Service Management

A holistic lens that stops you designing only the technology

- Name the four dimensions and explain what each covers
- Use the dimensions to find blind spots in a service design
- Analyse external factors that shape every dimension

When an IT project fails, the post-mortem rarely blames the technology alone. More often the system worked, but the staff were not trained, the supplier contract did not cover weekend support, or nobody redesigned the approval workflow. The **four dimensions** are a simple discipline for avoiding this: every service, practice and change must be considered from four angles at once.

The four dimensions

D1

Organisations and people

Structures, roles, responsibilities, culture, skills and capacity. Who does the work, who decides, and do they have the competence and motivation to do it well?

D2

Information and technology

The information a service needs and creates, and the technology that stores, processes and protects it: applications, platforms, networks, data, security and emerging tools.

D3

Partners and suppliers

The other organisations involved in designing, building, delivering and supporting the service, and the contracts and relationships that govern them.

D4

Value streams and processes

How the parts of the organisation work together: the sequence of activities, handovers and controls that turn demand into value.

The four dimensions applied to a hospital e-prescription service

DIMENSION	QUESTIONS TO ASK	EXAMPLE FINDING
Organisations and people	Who owns the service? Are doctors and pharmacists trained? Is there out-of-hours support?	No named service owner; night pharmacists never trained
Information and technology	Which data is needed? Is it accurate, secure, available on wards?	Drug formulary updated monthly by spreadsheet; Wi-Fi weak in two wards
Partners and suppliers	Which vendors are involved? What do contracts promise?	Software vendor support 09:00–17:00 only; prescriptions run 24/7
Value streams and processes	How does a prescription flow from doctor to patient? Where are the handovers?	Paper fallback process undocumented; delays when system is down

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INSIGHT · DIMENSIONS ARE NOT DEPARTMENTS

It is tempting to map each dimension to a team: HR for people, IT for technology, procurement for suppliers. That defeats the purpose. Each dimension is a **lens on the same service**, and every team must consider all four. A network engineer designing Wi-Fi for wards should think about who supports it at night and how nurses report faults.

External factors that shape every dimension (PESTLE)

P

Political

Government policy, public-sector priorities, stability. Example: a national digital-government programme mandating online services.

E

Economic

Exchange rates, inflation, funding. Example: dollar shortages delaying imported hardware for Bangladeshi firms.

S

Social

Demographics, expectations, digital literacy. Example: rapid growth of smartphone banking among rural users.

T

Technological

New platforms, AI, cloud, obsolescence. Example: legacy systems losing vendor support.

L

Legal

Data protection, telecom regulation, labour law. Example: personal-data rules requiring local storage.

E

Environmental

Energy, climate, sustainability. Example: monsoon flooding and power cuts affecting data centres.

A design with and without the four dimensions

Technology-only design

- New cloud platform selected
- Migration plan for servers
- Go-live date set
- Staff 'will learn on the job'
- Supplier contract copied from old one

Four-dimension design

- Platform selected against service outcomes
- Roles, skills and on-call model defined
- Data quality and security controls planned
- Contract aligned with 24/7 service hours
- Processes for request, incident and change redesigned

SCENARIO · DECIDE · A BLIND SPOT IN A ROLLOUT

A garment manufacturer in Narayanganj is rolling out a new production-tracking app to 6,000 line workers. The technology is tested and the vendor is ready. Line supervisors have not been consulted, and the factory network team works only day shift while production runs two shifts. What is the most important action before go-live?

(A) Run a further round of technical load testing to be sure the app can handle 6,000 users

(B) Ask the vendor to add more features before launch

(C) Delay go-live indefinitely until every risk is removed

(D) Close the people and support gaps: involve supervisors and arrange night-shift support

Circle your choice. Answer and feedback: module answer key.

TEMPLATE **Four-dimensions review**

Review a service or change from all four angles before approving it.

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DIMENSION	CURRENT STATE	GAP OR RISK	ACTION AND OWNER
Partners and suppliers	*Vendor support weekdays 09:00–17:00*	*Service runs 24/7*	*Negotiate 24/7 severity-1 cover – Procurement lead*

Key terms

Four dimensions
Organisations and people; information and technology; partners and suppliers; value streams and processes: the perspectives that must all be considered for any service.

Information and technology
The dimension covering the information a service uses and creates and the technology that supports it.

Value streams and processes
The dimension covering how work is organised and coordinated to create value.

Service owner
The person accountable for a service's performance and value across its life.

Organisations and people
The dimension covering structures, roles, culture, skills and capacity.

Partners and suppliers
The dimension covering other organisations involved in the service and how relationships with them are managed.

PESTLE
A checklist of external factors: political, economic, social, technological, legal and environmental.

REFLECT

Think of a recent change that disappointed its users. Which of the four dimensions was neglected, and what would have exposed the gap earlier?

KNOWLEDGE CHECK

1. A new system is technically excellent but staff were never trained. Which dimension was neglected?
   - (A) Information and technology
   - (B) Value streams and processes for system testing
   - (C) Partners and suppliers
   - (D) Organisations and people
2. Why should the four dimensions not be assigned one each to separate departments?
   - (A) Because there are more than four departments
   - (B) Because only IT is allowed to manage dimensions
   - (C) They are lenses on the same service that every team must apply
   - (D) Because auditors require a single dimension owner for each service

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3. A regulation requires personal data of citizens to be stored inside the country. Which external factor is this?

(A) Legal
(B) Economic
(C) Technological change in hosting and storage options
(D) Social

4. Which finding belongs to the partners-and-suppliers dimension?

(A) Ward Wi-Fi coverage is weak
(B) The paper fallback process for outages is undocumented
(C) The vendor's support contract covers weekdays only
(D) Night staff are untrained

5. What is the main benefit of applying the four dimensions before go-live?

(A) Allowing the project to skip supplier negotiations entirely
(B) Proving to auditors that the technology was tested
(C) Finding gaps outside technology while they are cheap to fix
(D) Reducing the number of stakeholders who must be consulted

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MODULE 2 · MODEL · LESSON 2 · 40 MIN

People, Culture, Information and Technology

The human and digital foundations every service stands on

- Design roles, competencies and culture that support service outcomes
- Assess information quality and technology choices against service needs
- Balance standardisation and innovation in the technology estate

The organisations-and-people dimension is where most service management transformations succeed or fail. Processes can be drawn in a week and tools bought in a month, but changing how people collaborate, decide and learn takes years. Equally, the information-and-technology dimension is no longer just infrastructure: data is now a product in its own right, and technology choices lock in costs and capabilities for a decade.

Elements of the people dimension

S

Structure

How teams are organised: by function, by product, by customer or by value stream. Structure shapes communication paths and handovers.

R

Roles and accountability

Who is responsible, who is accountable, who must be consulted. Clear accountability ends the 'not my job' reflex.

C

Competence

Technical, business, management and interpersonal skills. Modern service roles need breadth across areas plus depth in one.

Cu

Culture

Shared values and behaviours: how people react to failure, information and change.

Ca

Capacity

Enough people with enough time, including time to learn and improve.

Culture types and how they handle information

Power-oriented

- Information is hoarded
- Messengers are blamed
- Failure leads to scapegoats
- New ideas are crushed

Rule-oriented

- Information follows channels
- Messengers are tolerated
- Failure leads to justice and rules
- New ideas create problems

Performance-oriented

- Information is actively sought
- Messengers are trained
- Failure leads to enquiry
- New ideas are welcomed

PRO TIP · THE T-SHAPED PROFESSIONAL

Service teams need people with **depth** in one area (database administration, networking, service desk) and **breadth** across others (business understanding, customer communication, automation basics). T-shaped people reduce handovers because they can resolve more of a problem themselves and understand the next team's needs.

Information qualities to assess

QUALITY	QUESTION	EXAMPLE OF FAILURE
Accuracy	Is it correct?	Patient weight recorded in pounds instead of kilograms

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QUALITY	QUESTION	EXAMPLE OF FAILURE
Completeness	Is anything missing?	Asset records missing 30% of laptops
Timeliness	Is it available when needed?	Stock data refreshed weekly for a daily decision
Accessibility	Can the right people reach it?	Knowledge articles locked in one engineer's folder
Security and privacy	Is it protected appropriately?	Customer national ID numbers visible to all support agents
Relevance	Does it support the decision?	Reports with 80 metrics nobody reads

Technology decisions that shape services for years

- Build, buy or subscribe: custom software, packaged products or cloud services.
- Degree of standardisation: one platform for all, or best-of-breed tools per team.
- Hosting: on-premises, private cloud, public cloud or hybrid, considering data residency and latency.
- Integration approach: point-to-point links versus APIs and event streams.
- Lifecycle and obsolescence: supported versions, upgrade cadence, end-of-life risk.
- Security and resilience built in, not bolted on.

Standardise or innovate?

Standardise when...

- The capability is a commodity (email, laptops, identity)
- Scale and cost efficiency matter most
- Regulation demands consistent controls

Allow variety when...

- The capability differentiates the business
- Teams are experimenting with new products
- Learning speed matters more than unit cost

SCENARIO · DECIDE · CULTURE AFTER A FAILED CHANGE

At a private bank in Dhaka, a junior engineer's change caused a 40-minute ATM outage. The head of operations wants to issue a written warning and require two senior approvals for all future changes by junior staff. As the service manager, what do you recommend?

(A) Run a blameless review to find system causes, then fix the checks that let the change through

(B) Issue the warning as proposed so that all staff understand the consequences of careless changes

(C) Ban junior engineers from making changes

(D) Take no action, since outages are unavoidable

Circle your choice. Answer and feedback: module answer key.

TEMPLATE

Competency matrix

Map team skills against service needs and plan development.

TEAM MEMBER	DEEP SKILL	BROAD SKILLS	GAP FOR SERVICE NEEDS	DEVELOPMENT ACTION
Network engineer A	*Routing and switching*	*Scripting basics, customer comms*	*Cloud networking*	*Cloud networking course and shadow cloud team for 4 weeks*

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TEAM MEMBER	DEEP SKILL	BROAD SKILLS	GAP FOR SERVICE NEEDS	DEVELOPMENT ACTION

Key terms

Organisational culture
The shared values, beliefs and behaviours that shape how people work, decide and react to failure.

Blameless review
A review of a failure that seeks system causes and improvements rather than individuals to punish.

Information quality
The fitness of information for its purpose, judged by accuracy, completeness, timeliness, accessibility, security and relevance.

Standardisation
Using common platforms, tools and methods across an organisation to reduce cost, risk and complexity.

Performance-oriented culture
A culture in which information flows freely, failures lead to enquiry and new ideas are welcomed.

T-shaped professional
A person with deep expertise in one area and broad knowledge across several others.

Technical debt
The future cost created by choosing a quick or limited technical solution now instead of a better one.

REFLECT

How does your team react when someone reports a mistake? What one leadership behaviour would move it closer to a performance-oriented culture?

KNOWLEDGE CHECK

6. In which culture are messengers of bad news trained rather than blamed?
   - (A) Power-oriented
   - (B) Rule-oriented
   - (C) Performance-oriented
   - (D) Hierarchy-oriented with strict escalation

7. Why are T-shaped professionals valuable in service teams?
   - (A) They avoid customer contact
   - (B) They replace the need for documentation and knowledge bases
   - (C) They specialise so deeply that they never need help
   - (D) They reduce handovers by covering more of a problem

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8. Support agents can see every customer's national ID number. Which information quality is weakest?

   (A) Security and privacy
   (B) Timeliness
   (C) Completeness
   (D) Relevance of the data shown to each agent's task

9. When is standardisation usually the best choice?

   (A) For experimental products that need rapid learning
   (B) For capabilities that differentiate the business from competitors
   (C) Whenever a team requests a new tool
   (D) For commodity capabilities such as email and identity

10. After a junior engineer's change causes an outage, which response best supports long-term reliability?

    (A) Banning junior staff from making production changes
    (B) A blameless review that fixes weak controls
    (C) Doubling approvals for all changes made by every team
    (D) A written warning to deter future mistakes

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MODULE 2 · MODEL · LESSON 3 · 40 MIN

Partners, Suppliers and Service Integration

Managing a service that many organisations deliver together

- Choose sourcing models that fit service needs and risk
- Design supplier relationships around outcomes, not just price
- Explain service integration and when a multi-supplier model needs it

Almost no modern service is delivered by one organisation. A mobile banking app may depend on a cloud provider, a telecom SMS gateway, a card scheme, a fraud-scoring vendor, an outsourced contact centre and a local systems integrator. The consumer sees one service; the provider must make a dozen organisations behave like one team. The partners-and-suppliers dimension is about choosing those organisations wisely and managing the relationships so value flows across company boundaries.

Sourcing options

MODEL	WHAT IT MEANS	TYPICAL FIT	MAIN RISK
Insourcing	Own staff and assets deliver the service	Core, differentiating or highly sensitive services	High fixed cost, slower to scale
Outsourcing	A supplier delivers the service under contract	Commodity services with clear requirements	Loss of knowledge and control
Single sourcing	One supplier for a whole area	Simple governance, volume discounts	Dependency and lock-in
Multi-sourcing	Several suppliers for different parts	Best capability per area, competition	Integration complexity and blame-shifting
Partnership	Shared goals and often shared risk and reward	Innovation, long-term strategic services	Unclear accountability if poorly defined

Factors in a sourcing decision

St
Strategic focus
Is the capability core to how we compete? Core capabilities are rarely handed over completely.

Co
Corporate culture
Does the supplier's way of working fit ours? Culture clashes destroy many outsourcing deals.

Re
Resource scarcity
Can we hire and keep the skills ourselves? Scarce skills often push towards partners.

Ct
Cost
Total cost of ownership over several years, including transition and exit costs, not just the monthly price.

Ri
Risk
Concentration, financial stability of the supplier, regulatory exposure and data location.

Ag
Agility
How quickly can the arrangement adapt to changing needs?

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WATCH OUT · WATERMELON CONTRACTS

A supplier report can show every service level green on the outside while users experience red on the inside. This happens when targets measure the supplier's activity (tickets answered within 30 minutes) rather than outcomes (problems resolved, users able to work). Design supplier measures around the consumer's experience and the end-to-end service.

Transactional versus strategic supplier relationships

Transactional

- Focus on price and contract compliance
- Short-term, easily replaced
- Formal communication through contract managers
- Suited to commodities: printing, cabling, licences

Strategic

- Focus on joint outcomes and innovation
- Long-term, high switching cost
- Joint governance, shared roadmaps
- Suited to core platforms, managed services, cloud

When several suppliers deliver parts of one service, someone must integrate them: coordinate incidents that cross supplier boundaries, run a single change calendar, consolidate reporting and settle disputes about who owns a problem. This **service integration** role can be kept in-house, given to a lead supplier or given to a specialist integrator. Keeping it in-house is usually best for organisations whose services are strategic, because integration is where accountability for the end-to-end result lives.

A multi-supplier operating model

Customer organisation	`Strategy and governance` `Service ownership and budget`
Service integration	`End-to-end incident and problem coordination` `Single change calendar` `Consolidated performance reporting`
Suppliers	`Cloud hosting` `Network and telecom` `Application support` `Service desk and field support`

SCENARIO · DECIDE · BLAME BETWEEN SUPPLIERS

A university's student portal fails every Monday morning. The hosting supplier says the application is at fault; the application supplier blames the network; the network supplier says its links are fine. Each supplier is meeting its own service levels. What should the university do?

(A) Run a joint problem investigation and make cross-supplier cooperation a contract duty

(B) Replace all three suppliers with new ones at the next renewal, since none is taking responsibility

(C) Ask each supplier to send a written root-cause report independently

(D) Accept the situation because each supplier meets its SLA

Circle your choice. Answer and feedback: module answer key.

TEMPLATE **Supplier evaluation scorecard**

Compare potential suppliers on more than price.

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CRITERION	WEIGHT %	SUPPLIER A SCORE (1–5)	SUPPLIER B SCORE (1–5)	EVIDENCE
24/7 support capability	*20*	*4*	*2*	*Reference call with a Chattogram port customer*
Exit and data return terms	*10*	*3*	*5*	*Draft contract clause 14*

Key terms

Sourcing model
The way an organisation obtains a capability: insourcing, outsourcing, single sourcing, multi-sourcing or partnership.

Service integration
Coordinating multiple suppliers so they deliver one coherent end-to-end service with clear accountability.

Total cost of ownership (TCO)
The full cost of a service over its life, including acquisition, operation, transition and exit.

Exit plan
A pre-agreed approach for moving a service away from a supplier, including data return and knowledge transfer.

Multi-sourcing
Using several suppliers for different parts of a service or portfolio.

Watermelon reporting
Reports that look green because component targets are met while the consumer's experience is red.

Strategic supplier
A supplier whose services are critical and hard to replace, managed through joint governance and shared goals.

Vendor lock-in
Dependence on a supplier that makes switching costly or impractical.

REFLECT

List your three most critical suppliers. For each, could you exit within six months if you had to? What would stop you?

KNOWLEDGE CHECK

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11. A hospital wants to keep accountability for an end-to-end clinical service delivered by five suppliers. Who should usually perform service integration?

(A) Each supplier for its own component only
(B) The cheapest supplier
(C) The hospital itself
(D) Whichever supplier has the most staff on site

12. What is a 'watermelon' service report?

(A) A report that combines red and green targets equally
(B) A report written jointly by two or more suppliers
(C) Green on targets while users experience failure
(D) A report produced only at the end of a contract

13. Which factor most strongly suggests keeping a capability in-house?

(A) The capability is a commodity such as printing
(B) It is core to how the organisation competes
(C) The skills are widely available in the market
(D) A supplier offers a lower monthly price

14. Why should total cost of ownership include exit costs?

(A) Because exit costs replace transition costs
(B) Auditors require exit costs to be estimated at zero
(C) Leaving a supplier can be expensive and must be planned
(D) Exit costs are always paid by the supplier

15. Which relationship type suits a long-term managed cloud platform supporting core banking?

(A) Strategic
(B) Spot purchase
(C) Informal relationship based on goodwill and no contract
(D) Transactional

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MODULE 2 · MODEL · LESSON 4 · 45 MIN

The Service Value Chain and Value Streams

Six activities that combine, in many ways, to create value

- Describe the six value chain activities and their main inputs and outputs
- Model a value stream as a sequence of value chain activities and practices
- Map a value stream and identify waste and improvement opportunities

The **service value chain** is the operating heart of the value system. It describes six activities that every service organisation performs. The chain is not a linear process; it is a set of building blocks. Different types of demand flow through different combinations of these activities, called **value streams**. A password reset might touch only engage and deliver-and-support; a new product launch touches all six.

The six value chain activities

PL
Plan
Create shared understanding of the vision, current status and direction for all four dimensions and all products and services.

IM
Improve
Ensure continual improvement of products, services and practices across every activity.

EN
Engage
Understand stakeholder needs, maintain transparency and good relationships with all stakeholders.

DT
Design and transition
Ensure products and services meet stakeholder expectations for quality, cost and time to market.

OB
Obtain/build
Make sure service components are available when and where they are needed and meet agreed specifications.

DS
Deliver and support
Ensure services are delivered and supported according to agreed specifications and stakeholder expectations.

Typical inputs and outputs

ACTIVITY	EXAMPLE INPUTS	EXAMPLE OUTPUTS
Plan	Policies, demand forecasts, performance data	Strategic and tactical plans, portfolio decisions
Improve	Performance data, feedback, improvement ideas	Improvement initiatives, status reports
Engage	Requests, feedback, incidents from users; market data	Requirements, contracts, customer agreements, user requests
Design and transition	Requirements, architecture, service components	New or changed services, release plans, performance data
Obtain/build	Specifications, contracts, design	Built or acquired components, knowledge
Deliver and support	Released services, user requests, incidents	Services delivered, resolved incidents, satisfaction data

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DEFINITION · VALUE STREAM

A **value stream** is the series of steps an organisation takes to create and deliver a product or service to a consumer, in response to a specific type of demand. Each step uses one value chain activity and draws on one or more practices.

Value stream: restoring a failed payment service

01 Engage
Users report failures through the app and contact centre

→

02 Deliver and support
Service desk logs, major incident declared, workaround applied

→

03 Design and transition
Emergency fix tested and released

→

04 Deliver and support
Service confirmed restored, users informed

→

05 Improve
Post-incident review feeds problem management

Mapping a value stream

1. **Choose the demand** Pick one specific trigger, such as 'new employee needs laptop and access'.
2. **Define start and end** Start when demand arrives; end when the consumer realises value, not when IT closes a ticket.
3. **Walk the steps** Record each step, the team, the value chain activity and the practices used.
4. **Measure time** For each step record touch time (actual work) and wait time (queues).
5. **Find waste** Highlight waits, rework, handovers, approvals with no value and duplicated data entry.
6. **Design the future state** Remove or automate waste, merge steps and set new flow targets.

Value stream measures

MEASURE	FORMULA	WHAT IT TELLS YOU
Lead time	`Lead time = time value realised − time demand arrived`	The total elapsed time the consumer experiences.
Flow efficiency	`Flow efficiency = total touch time ÷ lead time × 100%`	Share of lead time spent actually working; values under 15% are common and show large waiting waste.
Percent complete and accurate	`%C&A = items received usable without rework ÷ items received × 100%`	Quality of handovers into a step; low values reveal rework loops.

CASE STUDY · ONBOARDING AT A PHARMACEUTICAL COMPANY

A pharmaceutical manufacturer in Gazipur takes 19 working days to make a new sales representative productive. Mapping shows 11 steps across HR, IT, security and the sales office. Actual work totals 6 hours. Laptops wait 5 days for a security image, access requests wait for three separate approvals, and HR sends start dates by email that IT re-enters into its tool, often wrongly.

1. What is the flow efficiency of this value stream?

2. Which value chain activities are involved?

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3. Propose three changes that would cut lead time most.

Model answer: module answer key.

SCENARIO · DECIDE · CHOOSING THE STREAM TO MAP FIRST

A telecom operator wants to start value stream mapping. Candidates: annual budgeting, the enterprise customer new-connection stream (6 weeks, top complaint), printer toner replacement, and the internal travel approval process. Which should be mapped first?

(A) The enterprise new-connection stream

(B) Annual budgeting, because it happens only once a year and therefore affects every department and every service

(C) Printer toner replacement

(D) Internal travel approval

Circle your choice. Answer and feedback: module answer key.

TEMPLATE **Value stream map worksheet**

Record each step of a value stream with time and waste.

STEP	TEAM	VALUE CHAIN ACTIVITY	PRACTICES USED	TOUCH TIME	WAIT TIME	WASTE OBSERVED
Laptop imaging	*End-user computing*	*Obtain/build*	*IT asset management, deployment*	*1 h*	*5 days*	*Waiting for security image approval*

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Key terms

Service value chain
An operating model of six interconnected activities that respond to demand and create value.

Value stream
A series of steps, combining value chain activities and practices, that turns a specific type of demand into value.

Touch time
The time during which work is actually being done on an item.

Value stream mapping
A technique for visualising the steps, times and waste in a value stream to design a better future state.

Value chain activity
One of plan, improve, engage, design and transition, obtain/build, and deliver and support.

Lead time
Total elapsed time from the arrival of demand to the realisation of value.

Flow efficiency
Touch time as a percentage of lead time.

Engage
The value chain activity of understanding stakeholder needs and maintaining relationships with them.

REFLECT

Choose one value stream you are part of. Where does it really start and end from the consumer's point of view, and where does your team's part begin and end?

KNOWLEDGE CHECK

16. Which value chain activity ensures components are available when and where needed and meet specifications?

(A) Deliver and support
(B) Obtain/build
(C) Design and transition
(D) Plan, through the annual sourcing and capacity forecast

17. A value stream's lead time is 20 days and touch time is 2 days. What is the flow efficiency?

(A) 2%
(B) 20%
(C) 90%
(D) 10%

18. Where should a value stream end?

(A) When the supplier invoices for the work
(B) When the consumer realises value
(C) When the last IT team closes its ticket
(D) When the change is approved by the change authority

19. What is the relationship between the value chain and value streams?

(A) Value streams replace the value chain entirely
(B) The value chain is one fixed sequence that every stream follows
(C) Each activity belongs to exactly one stream
(D) Streams combine chain activities in response to specific demand

20. Which value stream is the best first candidate for mapping?

(A) A slow, high-volume stream that customers complain about
(B) A rare annual process with few stakeholders
(C) A process owned entirely by one supplier
(D) The simplest process in the organisation, whatever its value

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MODULE 2 · MODEL

Answer key

Knowledge checks

1. D Training, skills and capacity belong to the organisations-and-people dimension.

2. C Each dimension applies to every service and team. Splitting them by department recreates silos.

3. A Data-residency rules are legal factors, even though they have technological consequences.

4. C Contract terms with a vendor are part of partners and suppliers. The others are technology, people and process findings.

5. C The dimensions expose people, supplier and process gaps early, when fixing them costs least.

6. C Performance-oriented cultures seek information actively and treat bad news as valuable.

7. D Breadth plus depth means fewer handovers and better collaboration with adjacent teams.

8. A Excessive access to sensitive data is a security and privacy weakness.

9. D Commodities benefit from scale and consistency; differentiating or experimental areas may need variety.

10. B Blameless reviews reveal system weaknesses and keep people reporting problems honestly.

11. C For strategic services, keeping integration in-house keeps end-to-end accountability with the customer.

12. C Component targets look green while the end-to-end experience is red.

13. B Core, differentiating capabilities are rarely handed over entirely; the others point towards outsourcing.

14. C Data migration, dual running and knowledge transfer can be large; ignoring them understates real cost.

15. A Critical, hard-to-replace services need strategic relationships with joint governance.

16. B Obtain/build makes components available to specification. Design and transition ensures services meet expectations for quality, cost and time.

17. D Flow efficiency = touch time ÷ lead time = 2 ÷ 20 = 10%.

18. B Ending at value realised keeps the focus on the consumer, not internal milestones.

19. D The chain provides building blocks; each stream combines them in its own order, sometimes using one activity several times.

20. A High value and visible pain give the best return and momentum.

Scenarios & case studies

Scenario · A blind spot in a rollout
D is best. Correct. The organisations-and-people dimension and the support model are the weakest. Technology readiness alone will not deliver value on the night shift.

Scenario · Culture after a failed change
A is best. Correct. Blame discourages reporting and learning. Most failures come from weak system controls, such as missing tests or poor peer review, which a blameless review will reveal.

Scenario · Blame between suppliers
A is best. Correct. The university needs end-to-end accountability through integration. Joint investigation finds the cause; contractual collaboration clauses prevent the next stand-off.

Case study · Onboarding at a pharmaceutical company
Lead time is about 19 working days, roughly 152 working hours, so flow efficiency is about 6 ÷ 152 ≈ 4%. The stream involves **engage** (HR request), **obtain/build** (laptop and image), **design and transition** (role-based access design), and **deliver and support** (handover and first-day support), with **improve** for the redesign. Highest-impact changes: (1) pre-imaged laptop stock so the device step becomes same-day; (2) role-based access packages approved once per role, removing three individual approvals; (3) an integration from the HR system to the IT tool so start dates flow automatically, removing re-entry errors. Together these could bring lead time below five days.

Scenario · Choosing the stream to map first
A is best. Correct. It is high value, slow and a top complaint: improvement there will be visible to customers and leadership and builds momentum.

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03

MODULE 3 · RESTORE

Restoring Service: Desk, Incident, Problem, Monitoring and Knowledge

The practices that keep services running and get them back fast when they fail

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Design a service desk and channel strategy that resolves at first contact
- Run incidents and major incidents with clear roles, priorities and communication
- Use problem management and root-cause techniques to prevent repeat incidents
- Build actionable monitoring and observability across all service layers
- Turn everyday work into reusable knowledge

LESSONS

1. Service Desk: The Front Door of Every Service · 35 min
2. Incident Management and Major Incidents · 45 min
3. Problem Management: Removing Causes · 45 min
4. Monitoring, Event Management and Observability · 40 min
5. Knowledge Management for Service Teams · 35 min

MODULE 3 · RESTORE · LESSON 1 · 35 MIN

Service Desk: The Front Door of Every Service

Designing a desk that resolves, listens and builds trust

- Explain the purpose of the service desk and its channels
- Design a desk model for an organisation's size, hours and languages
- Measure desk performance with balanced, outcome-focused metrics

The **service desk** is the single point of contact between a service provider and its users. For many users it *is* IT: their opinion of every system is shaped by the last conversation they had with the desk. A modern desk is not a call-logging team. It captures demand, resolves as much as possible at first contact, coordinates the rest, and feeds what it learns about users back into the organisation.

What a modern service desk does

C

Capture demand

Receive incidents, service requests, questions and feedback through every channel and record them consistently.

R

Resolve and fulfil

Fix what can be fixed now, using knowledge, scripts and automation, and fulfil standard requests.

Co

Coordinate

Route what cannot be solved to the right team, track it and keep the user informed.

E

Empathise

Understand the user's situation and business impact, not just the technical symptom.

L

Learn

Spot trends, recurring issues and knowledge gaps and feed them to problem, knowledge and improvement work.

Channels and when they work best

CHANNEL	STRENGTHS	WATCH OUT FOR
Phone	Urgent, complex or emotional issues; users with low digital literacy	Expensive per contact; peaks at shift start
Chat (human or virtual agent)	Quick questions, parallel handling, transcripts	Bots that trap users without a route to a person
Self-service portal	Standard requests, knowledge search, status tracking	Poor search and out-of-date articles drive users back to phone
Email	Non-urgent requests with attachments	Unstructured; slow; hard to classify
Walk-up / tech bar	Device issues, VIP or executive support, hospitals and campuses	Needs staffing at the right location and time
Messaging apps	Users already living in mobile messaging, field staff	Data protection and record-keeping must be designed in

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Desk models

Centralised

- One desk for the whole organisation
- Efficient and consistent
- May lack local knowledge or language coverage

Virtual / follow-the-sun

- Agents in several locations act as one desk
- 24/7 coverage across time zones
- Needs strong shared tools and knowledge

Local / embedded

- Desk staff located with users (wards, factories, campuses)
- Strong relationships and local context
- Higher cost; risk of inconsistent practice

EXAMPLE · A HYBRID DESK IN A HOSPITAL GROUP

A hospital group with sites in Dhaka and Chattogram runs a central desk for phone, chat and portal 24/7, plus 'floor walkers' on each clinical site during day shifts. Floor walkers handle device issues on wards and capture knowledge; the central desk handles everything else and owns the record. Clinicians get human help where they work, and the group keeps one source of truth.

Core service desk measures

MEASURE	FORMULA	WHAT IT TELLS YOU
First-contact resolution (FCR)	`FCR = contacts resolved during first contact ÷ contacts that could be resolved by the desk × 100%`	How often users get their issue solved without a callback or handover.
Average speed of answer (ASA)	`ASA = total waiting time of answered contacts ÷ number of answered contacts`	How long users wait; useful for staffing, not as a quality measure on its own.
Abandonment rate	`Abandonment = contacts abandoned before answer ÷ contacts offered × 100%`	Users giving up; high rates signal under-staffing or long queues.
Cost per contact	`Cost per contact = total desk cost ÷ total contacts handled`	Efficiency; must be balanced with quality and shift-left to self-service.

WATCH OUT · METRICS THAT BACKFIRE

A desk measured mainly on average handling time will rush users off the phone and transfer problems it could have solved. A desk measured mainly on ticket closure will close tickets before users confirm the fix. Balance speed with resolution quality, user satisfaction and the effort users had to make.

Shift-left: moving resolution closer to the user

- Level 0: users solve the issue themselves through knowledge and self-service automation.
- Level 1: desk agents solve it with good knowledge and appropriate access rights.
- Level 2/3: specialists are used only for genuinely complex work.
- Each shift left lowers cost and shortens resolution time, but only if knowledge is accurate and users trust it.

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SCENARIO · DECIDE · CHOOSING THE DESK IMPROVEMENT

A mobile financial services provider's internal service desk handles 18,000 contacts a month for agents and staff. 35% are password resets and PIN unlocks, FCR is 48% and abandonment is 22% at 09:00. What should the desk manager do first?

(A) Hire more agents for the 09:00 peak and keep processes unchanged, since the peak is the most visible problem

(B) Introduce secure self-service reset with identity verification and promote it to users

(C) Set a target to reduce average handling time by 30%

(D) Close the phone channel and require all contacts by email

Circle your choice. Answer and feedback: module answer key.

Key terms

Service desk
The single point of contact between the service provider and users for incidents, requests and communication.

Shift-left
Moving resolution of work to lower-cost, faster levels, ideally to users themselves through self-service.

Omnichannel
Providing consistent support across phone, chat, portal, email and other channels with a shared record.

Follow-the-sun
A desk model in which teams in different time zones hand over work to provide 24-hour support.

First-contact resolution (FCR)
The percentage of contacts resolved during the first interaction without a callback or handover.

Abandonment rate
The share of contacts where the user gives up before being answered.

Virtual agent
An automated conversational assistant that answers questions or fulfils requests through chat or voice.

REFLECT

If your users rated your service desk on effort rather than speed, how would it score? What causes them the most effort today?

KNOWLEDGE CHECK

1. Which measure best shows whether users get their issue solved without being passed on?
   (A) First-contact resolution
   (B) Number of contacts handled per agent per shift
   (C) Average speed of answer
   (D) Cost per contact

2. A desk is measured mostly on average handling time. What behaviour is most likely?
   (A) Abandonment falls because each call takes longer to complete
   (B) Agents spend more time on knowledge articles
   (C) Agents end contacts quickly rather than resolving them
   (D) User satisfaction rises steadily

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3. What does 'shift-left' mean?

(A) Starting the support shift earlier in the morning

(B) Moving resolution closer to users, such as self-service

(C) Handing all complex incidents to a supplier

(D) Moving work to the left-hand team in an organisation chart

4. When is a local, embedded desk most justified?

(A) When support is needed only by email

(B) When users need in-person help where they work, as on wards

(C) When all users work remotely in different countries

(D) When the organisation wants the lowest possible cost per contact

5. Abandonment is 22% at 09:00 but 3% the rest of the day. What does this most likely indicate?

(A) A demand peak that capacity does not cover

(B) The desk has too many agents in the afternoon

(C) Poor quality knowledge articles

(D) Users prefer email to phone for urgent problems at that hour

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MODULE 3 · RESTORE · LESSON 2 · 45 MIN

Incident Management and Major Incidents

Restoring normal service fast, calmly and visibly

- Classify, prioritise and route incidents consistently
- Run a major incident with clear roles, communication and decision-making
- Measure and improve restoration performance

An **incident** is an unplanned interruption to a service or a reduction in its quality. The purpose of incident management is to **minimise the negative impact** of incidents by restoring normal operation as quickly as possible. The emphasis is on restoration, not diagnosis: if a workaround gets wards printing again in five minutes, use it now and find the cause later through problem management.

Incident lifecycle

01 Detect — Monitoring alert, user report or automated test

→

02 Log & classify — Record, categorise, link to service and configuration items

→

03 Prioritise — Impact × urgency gives priority

→

04 Diagnose & route — Match known errors, swarm or escalate

→

05 Restore — Workaround or fix applied, service verified

→

06 Close & learn — User confirms, data captured for problem and knowledge

A simple priority matrix

IMPACT \ URGENCY	HIGH URGENCY	MEDIUM URGENCY	LOW URGENCY
High impact (many users or critical service)	P1 – Critical	P2 – High	P3 – Medium
Medium impact (a department or important function)	P2 – High	P3 – Medium	P4 – Low
Low impact (one user, workaround available)	P3 – Medium	P4 – Low	P4 – Low

INSIGHT · IMPACT IS ABOUT BUSINESS, NOT TECHNOLOGY

A failed server in a test environment is technically dramatic and business-trivial. A small configuration error that stops the hospital pharmacy printing labels is technically minor and clinically urgent. Define impact using service and business consequences: users affected, revenue, safety, regulatory exposure and reputation.

Tiered escalation versus swarming

Tiered escalation

- Level 1 → Level 2 → Level 3
- Each tier tries before passing on
- Efficient for routine incidents
- Slow for novel or complex ones; context lost at each handover

Swarming

- The right people work the incident together early
- No ticket ping-pong
- Faster for complex incidents
- Needs culture and tooling for quick collaboration; can over-use experts

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Major incident roles

IC

Incident commander

Owns the response and makes decisions. Does not fix things personally; keeps the whole picture.

TL

Technical lead(s)

Lead diagnosis and restoration in their domain and report progress to the commander.

CL

Communications lead

Sends regular, plain-language updates to users, executives, regulators and the contact centre.

SC

Scribe

Keeps a timestamped log of facts, decisions and actions for the response and the review.

BL

Business liaison

Represents affected business units, confirms impact and tests workarounds from the user side.

Running a major incident bridge

1. **Declare early** If criteria might be met, declare. Standing down a major incident costs little; declaring late costs a lot.
2. **Assign roles** Name the commander, communications lead and scribe in the first five minutes.
3. **Establish facts** What is broken, since when, who is affected, what changed recently?
4. **Set a rhythm** Agree update intervals, for example every 30 minutes, and keep them even when there is no news.
5. **Prefer restoration** Roll back the recent change, fail over or use a workaround before chasing root cause.
6. **Verify with users** Confirm restoration with real users and monitoring, not just the technical team.
7. **Hand over to problem** Close the incident and open a problem record with the timeline and evidence.

Restoration measures

MEASURE	FORMULA	WHAT IT TELLS YOU
Mean time to restore service (MTRS)	`MTRS = total downtime across incidents ÷ number of incidents`	Average time from service loss to restoration; the key incident measure.
Mean time to detect (MTTD)	`MTTD = Σ(detection time − start time) ÷ number of incidents`	How quickly incidents are noticed; long values show monitoring gaps.
Reassignment rate	`Reassignments ÷ incidents`	Ticket ping-pong; high values show poor routing or missing skills.

SCENARIO · DECIDE · A PAYMENTS OUTAGE AT 20:00

At 20:00 on the day before Eid, transaction success for a mobile wallet drops from 99% to 60%. A database patch was deployed at 19:30. The database team wants two hours to find the root cause before deciding anything. You are the incident commander. What do you do?

- (A) Give the database team two hours to find the root cause so the right fix is applied
- (B) Ask the communications lead to say that nothing is wrong to avoid panic
- (C) Decide to roll back the patch now while diagnosis continues in parallel
- (D) Escalate to the CEO and wait for a decision

Circle your choice. Answer and feedback: module answer key.

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TEMPLATE Major incident communication template

Send short, regular updates to stakeholders during a major incident.

TIME	WHAT IS AFFECTED	WHO IS AFFECTED	CURRENT ACTION	WORKAROUND FOR USERS	NEXT UPDATE AT
20:15	*Wallet payments failing for ~40% of transactions*	*All retail users and merchants*	*Rolling back 19:30 database patch*	*Retry after 5 minutes; cash-out unaffected*	*20:45*

CASE STUDY · MAJOR INCIDENT AT A CONTAINER TERMINAL

At 06:40 on a Monday, the terminal operating system at a busy container terminal in Chattogram stops allocating truck slots. Trucks queue for three kilometres, customs inspections stall and shipping lines start calling. Four engineering teams join a call; each tests its own theory. By 08:30 nobody has told port management what is happening, and a network engineer has restarted a core switch without telling anyone, briefly making things worse.

1. What went wrong in the way the incident was run?

2. How should the first 30 minutes have looked?

3. What should happen after service is restored?

Model answer: module answer key.

Key terms

Incident
An unplanned interruption to a service or reduction in its quality.

Major incident
An incident with significant business impact that needs a separate, coordinated response.

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Impact
The effect of an incident on the business: users affected, revenue, safety, compliance or reputation.

Priority
The relative importance of an incident, usually derived from impact and urgency.

Incident commander
The person who leads a major incident response and makes decisions without personally fixing components.

Urgency
How quickly an incident must be resolved to avoid greater impact.

Swarming
An approach in which the people needed to resolve an incident collaborate on it early, instead of escalating in tiers.

Mean time to restore service (MTRS)
The average elapsed time from loss of service to its restoration.

REFLECT

Think of your last major incident. Who was in charge, and did everyone know it within ten minutes? What would you change?

KNOWLEDGE CHECK

6. What is the primary goal of incident management?
   - (A) Restoring normal service as quickly as possible
   - (B) Recording all incidents for audit purposes
   - (C) Assigning accountability to the team that caused the incident
   - (D) Finding the root cause of every incident

7. A test server fails and no production service is affected. What priority is appropriate?
   - (A) Low, because business impact is minimal
   - (B) High, because the infrastructure team is affected
   - (C) Major incident, because the failure is total
   - (D) Critical, because a server has failed completely

8. Which role in a major incident should avoid hands-on technical work?
   - (A) Database engineer
   - (B) Network engineer working on the failing link
   - (C) Incident commander
   - (D) Technical lead

9. When is swarming most valuable?
   - (A) When only one specialist team is involved and the fix is already known
   - (B) For routine password resets
   - (C) For incidents with a documented workaround
   - (D) For complex or novel incidents needing several skills at once

10. Three incidents last 30, 90 and 60 minutes. What is the MTRS?
    - (A) 180 minutes
    - (B) 90 minutes
    - (C) 60 minutes
    - (D) 45 minutes

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MODULE 3 · RESTORE · LESSON 3 · 45 MIN

Problem Management: Removing Causes

From firefighting to prevention through analysis and known errors

- Distinguish incidents, problems and known errors
- Apply problem identification, control and error control
- Select and use root-cause analysis techniques appropriately

If incident management is the fire brigade, problem management is fire prevention. A **problem** is a cause, or potential cause, of one or more incidents. Organisations that only fight fires see the same incidents week after week; their best engineers spend their time restoring the same failures. Problem management reduces the likelihood and impact of incidents by finding causes, recording workarounds and driving permanent fixes.

Incident, problem, known error

TERM	WHAT IT IS	EXAMPLE
Incident	An interruption or degradation users experience now	Ward label printers stop printing at 07:10
Problem	The underlying cause, often unknown at first	Why do label printers fail every morning?
Known error	A problem that has been analysed but not yet resolved	Print spooler memory leak; workaround: restart service at 06:30

The three phases of problem management

01
Problem identification
Spot trends, recurring incidents, major incidents and risks

→

02
Problem control
Analyse, prioritise, document workarounds and known errors

→

03
Error control
Manage known errors, assess fixes and raise changes

Reactive versus proactive problem management

Reactive

- Triggered by incidents that already happened
- Major incident reviews
- Trend analysis of repeat incidents
- Fixes what hurts now

Proactive

- Triggered before incidents occur
- Supplier vulnerability notices
- Capacity trends, near misses, test failures
- Prevents what would hurt later

Root-cause analysis techniques

5W
Five whys
Ask 'why?' repeatedly until you reach a cause you can act on. Simple and fast, but can follow one path and miss others.

FB
Fishbone (Ishikawa)
Group possible causes into categories such as people, process, technology, supplier and environment. Good for workshops.

KT
Kepner-Tregoe style analysis
Describe what the problem is and is not: where, when, extent. Differences point to the cause. Strong for complex technical faults.

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TL

Timeline analysis

Build a precise timeline of events, changes and alerts. Essential after major incidents.

FT

Fault tree

Work backwards from the failure through logical AND/OR conditions. Useful for safety-critical systems.

PRO TIP · CONTRIBUTING FACTORS, NOT A SINGLE ROOT

Complex systems rarely fail for one reason. A payment outage might need a code bug, a missing alert, an untested failover and a change made during peak hours, all at once. Look for **contributing factors** and fix several of them; each fix reduces the chance of a repeat.

Five whys in practice: hospital label printers

1. **Why did printers fail at 07:10?** The print service stopped responding.
2. **Why did it stop?** It ran out of memory.
3. **Why did it run out of memory?** A driver leaks memory with every label printed.
4. **Why was the leaking driver installed?** An automatic update applied a new driver version last month.
5. **Why did the update reach production untested?** Printer drivers were excluded from the change process. Action: include drivers in change control and test updates; roll back driver now.

Prioritising problems

- Frequency and impact of related incidents (past and expected).
- Cost of the workaround: staff time, user disruption, risk.
- Risk if the problem causes a future major incident.
- Effort and risk of the fix itself.
- Problems with good workarounds and expensive fixes may be accepted, documented and reviewed later.

SCENARIO · DECIDE · CHOOSING WHAT TO ANALYSE

A university's IT team has 40 open problems and capacity to analyse three this month. Options include: Wi-Fi drops in the exam hall (6 incidents, each during exams), slow login to the library catalogue (300 minor incidents, workaround available), a cosmetic error message on the staff portal (2 incidents), and a vendor bulletin warning of a storage firmware fault that can corrupt data.

(A) The slow library login first, because it has generated the most incidents of any problem in the register this year

(B) The cosmetic error, because it is quick to fix

(C) Exam-hall Wi-Fi and the storage firmware risk first, then the slow library login

(D) Whichever problems the most senior academics complain about

Circle your choice. Answer and feedback: module answer key.

TEMPLATE **Problem record**

Document a problem from identification to resolution.

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PROBLEM ID	DESCRIPTION AND LINKED INCIDENTS	IMPACT AND PRIORITY	ROOT CAUSE / CONTRIBUTING FACTORS	WORKAROUND	PERMANENT FIX AND CHANGE REF	STATUS
PRB-0231	*Ward label printers fail daily 07:00–07:30 (14 incidents)*	*High: delays medication rounds*	*Leaking driver; drivers outside change control*	*Restart spooler at 06:30*	*Roll back driver; add drivers to change process (CHG-5520)*	*Known error*

Key terms

Problem
A cause, or potential cause, of one or more incidents.

Known error
A problem that has been analysed and has a documented cause or workaround but is not yet resolved.

Workaround
A solution that reduces or eliminates the impact of an incident or problem for which a full resolution is not yet available.

Problem control
Analysing problems, documenting workarounds and known errors, and prioritising them.

Error control
Managing known errors, including periodic reassessment and initiating permanent fixes through change.

Proactive problem management
Identifying and resolving problems before they cause incidents.

Contributing factor
A condition that, together with others, allowed an incident to happen or made it worse.

Five whys
A root-cause technique that repeatedly asks why a failure happened until an actionable cause is found.

REFLECT

Which incident does your organisation resolve every week without ever asking why it keeps happening? What would it take to open a problem record for it?

KNOWLEDGE CHECK

11. What is a known error?

(A) An error that users know about before IT does

(B) A change that failed and was rolled back by the change authority

(C) An incident that has been resolved

(D) An analysed problem with a documented cause or workaround

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12. A vendor warns about a firmware fault that could corrupt data. No incidents have occurred yet. What kind of problem management is this?

    (A) Proactive
    (B) Reactive
    (C) Major incident management
    (D) Error control after a failed change

13. Which technique describes what the problem is and is not, to isolate the cause?

    (A) Fishbone diagram
    (B) Kepner-Tregoe style analysis
    (C) Five whys
    (D) Swarming with the major incident team

14. Why might a problem with a good workaround remain unresolved deliberately?

    (A) Problem management never fixes problems with workarounds
    (B) Workarounds are always permanent solutions
    (C) The fix may cost more or carry more risk than the workaround
    (D) Because auditors forbid changes to resolve known errors

15. Which phase raises a change request to implement a permanent fix?

    (A) Problem identification
    (B) Incident logging
    (C) Error control
    (D) Problem control, when the workaround is first documented

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MODULE 3 · RESTORE · LESSON 4 · 40 MIN

Monitoring, Event Management and Observability

Seeing trouble before users do

- Classify events and design responses to each class
- Design monitoring that reflects user experience as well as components
- Reduce alert noise and use telemetry to speed diagnosis

Monitoring and event management observes services and components, records changes of state that matter (**events**), and decides the right response. Done well, it detects incidents before users notice, triggers automated fixes, and provides evidence for capacity planning and problem analysis. Done badly, it floods engineers with thousands of alerts a day until they ignore all of them, including the one that mattered.

Event classes

CLASS	MEANING	EXAMPLE	TYPICAL RESPONSE
Informational	Normal operation, logged for reference	Nightly backup completed	Log only; use for trends and audit
Warning	Approaching a threshold; no impact yet	Disk 85% full on database server	Ticket or automated clean-up; investigate before it becomes an incident
Exception	Service or component failing or degraded	Payment API error rate above 5%	Create incident; alert on-call; run automated recovery if safe

Layers of monitoring

User experience	`Synthetic transactions (scripted logins and payments)` `Real-user monitoring of page and app performance`
Service / application	`Request rate, error rate, latency` `Business KPIs such as orders per minute`
Platform	`Containers, databases, message queues` `Cloud service health`
Infrastructure	`Servers, storage, network links` `Power and cooling in data centres`

Monitoring versus observability

Monitoring

- Asks known questions
- Predefined dashboards and thresholds
- Tells you that something is wrong
- Good for known failure modes

Observability

- Lets you ask new questions of the system
- Metrics, logs and traces with rich context
- Helps you find why something is wrong
- Essential for distributed, cloud-native services

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The golden signals for any service

La

Latency

How long requests take, separating successful and failed requests.

Tr

Traffic

Demand on the service: requests per second, transactions per minute, active sessions.

Er

Errors

Rate of failed requests, explicit or implicit (for example wrong content).

Sa

Saturation

How full the service is: CPU, memory, queue depth, connection pools.

Quick checks on a Linux application host

Linux host

```
# Is the payments service running?
systemctl status payments-api --no-pager
# Recent errors in the last 10 minutes
journalctl -u payments-api --since "10 min ago" -p err --no-pager
# Synthetic check: HTTP status and total time of the health endpoint
curl -s -o /dev/null -w "%{http_code} %{time_total}s\n" https://localhost:8443/health
# Disk saturation on data volumes
df -h /var/lib/payments
```

Manual checks like these are useful during diagnosis, but the same signals should be collected continuously by your monitoring platform and alerted on automatically.

WATCH OUT · ALERT FATIGUE

One operations centre in Dhaka received 14,000 alerts a day; engineers acknowledged them in bulk without reading. When a real storage failure began, its alert was buried. Every alert should be **actionable**: if nobody needs to do anything, it should not page anyone. Correlate related events, suppress duplicates and route by service, not by component.

Reducing noise in six weeks

1. **Inventory** Export a month of alerts; count by source, service and action taken.
2. **Delete the useless** Remove alerts nobody has acted on in 30 days, after confirming with owners.
3. **Tune thresholds** Base thresholds on service impact and observed baselines, not vendor defaults.
4. **Correlate** Group events from the same cause (one switch failure, not 200 server-unreachable alerts).
5. **Automate responses** For known safe fixes, such as clearing temp files, act automatically and log it.
6. **Review weekly** Track alerts per on-call shift and the share that were actionable.

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SCENARIO · DECIDE · A DASHBOARD THAT STAYED GREEN

A hospital's patient-portal dashboard showed all servers, databases and network links green, yet patients could not book appointments for three hours because an external SMS verification provider was failing. What monitoring change would have detected this fastest?

(A) A synthetic booking transaction that runs end to end every few minutes

(B) More detailed CPU and memory thresholds on every web server and database in the portal estate

(C) Asking the SMS provider for a monthly availability report

(D) Increasing the dashboard refresh rate

Circle your choice. Answer and feedback: module answer key.

Key terms

Event
Any change of state that has significance for the management of a service or component.

Exception event
An event indicating that a service or component is failing or not operating normally.

Synthetic monitoring
Running scripted transactions against a service to test its behaviour as a user would experience it.

Observability
The ability to understand a system's internal state from its external outputs, typically metrics, logs and traces.

Golden signals
Latency, traffic, errors and saturation: four measures that describe the health of most services.

Alert fatigue
Desensitisation caused by too many non-actionable alerts, leading to missed real issues.

Event correlation
Grouping related events to identify a single underlying cause.

REFLECT

What percentage of alerts in your environment lead to an action? How would you find out within a week?

KNOWLEDGE CHECK

16. Disk usage on a database server reaches 85%. No users are affected yet. Which event class is this?

(A) Warning
(B) Exception
(C) Major incident requiring an immediate bridge call
(D) Informational

17. Which monitoring approach best detects failures in third-party dependencies of a user journey?

(A) Monthly supplier availability reports reviewed by procurement
(B) Synthetic end-to-end transactions
(C) Network link utilisation
(D) Server CPU thresholds

18. Which of these is NOT one of the golden signals?

(A) Saturation
(B) Licence cost
(C) Latency
(D) Errors

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19. What is the strongest rule for deciding whether an event should page an engineer?

(A) Someone needs to act on it
(B) The vendor's default threshold was crossed
(C) It has occurred more than three times in the past hour on any component
(D) It comes from a production server

20. How does observability differ from traditional monitoring?

(A) It is a dashboard that shows only green and red
(B) It replaces the need for any alerting
(C) It focuses only on infrastructure components
(D) It lets you explore unknown failures

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MODULE 3 · RESTORE · LESSON 5 · 35 MIN

Knowledge Management for Service Teams

Making the right knowledge available at the moment it is needed

- Explain why knowledge management drives resolution speed and quality
- Apply a capture-as-you-solve approach to keep knowledge current
- Measure knowledge use and value

Every incident resolved, change implemented and request fulfilled creates knowledge. In most organisations that knowledge stays in the heads of a few experienced engineers or disappears in closed tickets. **Knowledge management** maintains and improves the effective, efficient and convenient use of information and knowledge across the organisation. For service teams, it is the engine behind shift-left, faster restoration and consistent quality.

From data to wisdom

Wisdom	`Judgement about what to do and why` `Example: deciding to roll back during peak hours`
Knowledge	`Information combined with experience and context` `Example: 'this error after a patch usually means the cache must be rebuilt'`
Information	`Data with meaning` `Example: error rate rose to 12% after 19:30`
Data	`Raw facts` `Example: HTTP 500 count per minute`

Explicit versus tacit knowledge

Explicit

- Written down and easy to share
- Articles, runbooks, diagrams
- Can be searched and reused at scale

Tacit

- In people's heads: experience and intuition
- Shared through mentoring, shadowing, pairing
- Lost when people leave unless deliberately transferred

Capture-as-you-solve knowledge loop

01 Search — Search the knowledge base at the start of every interaction →

02 Reuse — If an article helps, link it to the ticket →

03 Improve — If it is wrong or incomplete, flag or fix it now →

04 Create — If nothing exists, draft an article in the user's words →

05 Publish & evolve — Review, publish and keep improving with use

PRO TIP · WRITE IN THE USER'S WORDS

Users search for 'cannot print labels on ward', not 'spooler service failure'. Titles and keywords should use the language of the person searching. Include the symptom, environment, cause if known and the steps to resolve, and keep each article to one issue.

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Signs your knowledge base is failing

- Agents prefer to ask a senior colleague than to search.
- Many articles have not been viewed or updated in a year.
- Self-service search returns long lists with no clear answer.
- The same question appears in tickets every week.
- Experts are a bottleneck and cannot take leave without escalations rising.

Knowledge measures

MEASURE	FORMULA	WHAT IT TELLS YOU
Link rate	`Link rate = tickets with a linked article ÷ tickets resolved × 100%`	How much the knowledge base is used in resolution.
Self-service success	`Self-service success = sessions ending without a ticket ÷ self-service sessions × 100%`	How often users solve their own issue (use with care: some users simply give up).
Article reuse	`Reuse = times articles were linked ÷ number of active articles`	Whether articles are actually helpful or just stored.

SCENARIO · DECIDE · LOSING AN EXPERT

The only engineer who understands a 15-year-old billing system at a power distribution company is retiring in four months. What is the most effective knowledge action?

(A) Ask him to write a complete manual of the system before he leaves, working alone in the final month

(B) Record a video tour of the system on his final day

(C) Pair two engineers with him on real work and capture runbooks as they go

(D) Rely on the vendor's original documentation from 2011

Circle your choice. Answer and feedback: module answer key.

TEMPLATE Knowledge article template

Create consistent, searchable articles.

TITLE (USER'S WORDS)	SYMPTOMS	ENVIRONMENT / SERVICE	CAUSE (IF KNOWN)	RESOLUTION STEPS	OWNER AND REVIEW DATE
Ward label printer prints blank labels	*Labels feed but are blank after 07:00*	*Clinical printing – wards 3–9*	*Driver memory leak (PRB-0231)*	*Restart print service from desk console; if repeated, raise P3*	*Desk knowledge lead – review 30 June*

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TITLE (USER'S WORDS)	SYMPTOMS	ENVIRONMENT / SERVICE	CAUSE (IF KNOWN)	RESOLUTION STEPS	OWNER AND REVIEW DATE

Key terms

Knowledge management
The practice of maintaining and improving the effective, efficient and convenient use of information and knowledge.

Tacit knowledge
Knowledge held in people's experience and intuition, hard to write down and shared mainly through practice.

Runbook
A step-by-step guide for performing an operational task or responding to a known situation.

Knowledge base
A searchable repository of articles, runbooks and other explicit knowledge.

Explicit knowledge
Knowledge that has been written down or recorded and can be easily shared.

Capture as you solve
An approach in which knowledge is searched, reused, improved and created as part of everyday work rather than as a separate project.

Link rate
The share of resolved tickets that reference a knowledge article.

REFLECT

If your most knowledgeable colleague left tomorrow, which service would suffer first? What is one practical step you could take this month?

KNOWLEDGE CHECK

21. Which approach best keeps a knowledge base current?

(A) A yearly project to rewrite all articles
(B) Allowing only one author to publish articles
(C) Locking articles once they are approved by a review board so they never drift
(D) Capturing and improving knowledge as part of solving each issue

22. Mentoring and job shadowing are especially effective for transferring which kind of knowledge?

(A) Tacit
(B) Explicit
(C) Regulatory information in published policy documents
(D) Data

23. Users search for 'cannot print labels', but the article title is 'Spooler service memory exhaustion'. What should change?

(A) Add more technical detail and logs to the article body
(B) Remove the article from self-service
(C) Train users on spooler terminology
(D) Write titles and keywords in the user's language

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24. Self-service success is rising, but desk calls are not falling. What might explain this?

(A) Users may give up on self-service and call

(B) The desk has too many virtual agents answering chat

(C) The knowledge base is too small

(D) Agents are linking too many articles

☐ Lesson complete · Date _____ · Knowledge check score ____ / 4

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MODULE 3 · RESTORE

Answer key

Knowledge checks

1. A FCR directly measures resolution at the first interaction.

2. C Speed-only targets push agents to end contacts quickly, often at the expense of resolution.

3. B Shift-left moves resolution closer to the user while keeping quality.

4. B Embedded staff add local context and in-person help; they cost more, so they suit environments where that matters.

5. A A time-specific spike in abandonment points to a demand peak not matched by capacity or alternatives.

6. A Restoration comes first; root cause belongs to problem management.

7. A Priority follows business impact and urgency, not technical drama.

8. C The commander keeps the whole picture and decides; getting absorbed in fixing loses that view.

9. D Swarming avoids slow tiered handovers for complex work; routine work fits tiered handling.

10. C MTRS = (30 + 90 + 60) ÷ 3 = 60 minutes.

11. D A known error has been analysed but not yet fully resolved.

12. A Acting on risks before incidents occur is proactive problem management.

13. B Is/is-not comparison is the core of Kepner-Tregoe style analysis.

14. C Prioritisation weighs fix cost and risk against workaround cost; some known errors are accepted and reviewed.

15. C Error control manages known errors and initiates permanent fixes through change enablement.

16. A It approaches a threshold without impact: a warning.

17. B Synthetic journeys exercise the full path, including external services.

18. B The golden signals are latency, traffic, errors and saturation.

19. A Only actionable alerts should page people; everything else is logged or handled automatically.

20. D Observability supports exploring unknown failure modes using rich telemetry.

21. D Knowledge stays fresh when it is improved in the flow of work, not in occasional projects.

22. A Tacit knowledge lives in experience and is shared through practice.

23. D Findability depends on matching the words people actually search with.

24. A Sessions ending without a ticket can include users who gave up; the measure must be read with care.

Scenarios & case studies

Scenario · Choosing the desk improvement
B is best. Correct. Removing a third of demand through safe self-service frees capacity, lowers abandonment and lets agents spend time on issues they can resolve at first contact.

Scenario · A payments outage at 20:00
C is best. Correct. Restoration comes first during peak demand. The recent change is the most likely cause, rollback is the fastest restoration path, and diagnosis can continue from logs.

Case study · Major incident at a container terminal
The response lacked a commander, a communication rhythm and decision control: teams worked in parallel without coordination, and an uncontrolled action (the switch restart) added risk. In the first 30 minutes, the on-call manager should have declared a major incident, named an incident commander, a communications lead and a scribe, established facts (what changed overnight, which functions fail, which users are affected), agreed a 30-minute update cycle for port management and shipping lines, and offered a workaround such as manual slot allocation for priority trucks. Technical actions should be proposed to and approved by the commander. After restoration, the incident should be closed only once users confirm, and a problem record opened with the full timeline. A blameless review should examine contributing factors, including why the switch restart happened without coordination.

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Scenario · Choosing what to analyse
C is best. Correct. These combine high impact (exams, data loss) with real likelihood. The library login has a workaround and can follow. The cosmetic error can wait.

Scenario · A dashboard that stayed green
A is best. Correct. Synthetic user journeys test the service as patients experience it, including third-party dependencies that component monitoring misses.

Scenario · Losing an expert
C is best. Correct. Tacit knowledge transfers best through shared practice. Capturing runbooks during real work makes the explicit knowledge accurate and tested.

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04

MODULE 4 · CONTROL

Controlling Change and Service Levels

Change, release, deployment, requests, assets, configuration and service levels in one controlled flow

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Run change enablement that is both faster and safer, using risk-based authority
- Plan releases and deployments with progressive, reversible approaches
- Design service requests and catalogue entries for effortless fulfilment
- Scope asset and configuration data by the decisions it supports
- Agree and review service levels that reflect real user experience

LESSONS

1. Change Enablement: Fast and Safe Together · 45 min
2. Release and Deployment Management · 40 min
3. Service Request Management and the Service Catalogue · 35 min
4. IT Asset and Service Configuration Management · 45 min
5. Service Level Management and Experience · 45 min

MODULE 4 · CONTROL · LESSON 1 · 45 MIN

Change Enablement: Fast and Safe Together

Managing risk without strangling the flow of change

- Classify changes as standard, normal or emergency and choose the right authority
- Assess change risk using evidence rather than opinion
- Redesign a slow change process to increase both speed and safety

Every improvement, fix and new feature reaches users through a **change**: the addition, modification or removal of anything that could affect services. Change enablement exists to maximise the number of successful changes by assessing risk properly, authorising changes at the right level and managing the change schedule. The old picture of change control as a weekly committee that says 'no' is outdated. Modern change enablement is mostly about designing safe paths that let the right changes flow quickly.

Three types of change

S

Standard

Low-risk, well-understood and pre-authorised. Follows a defined procedure, often fully automated. Example: adding a user to a standard access group, scaling a web tier within limits.

N

Normal

Needs risk assessment and authorisation according to its risk. Low-risk normal changes can be approved by a peer or automated checks; high-risk ones by a senior authority.

E

Emergency

Must be implemented as soon as possible, usually to resolve an incident or close a serious vulnerability. Uses an expedited authority and documentation is completed after implementation.

Matching change authority to risk

RISK LEVEL	EXAMPLE	CHANGE AUTHORITY	TYPICAL LEAD TIME
Very low (standard)	Approved server patch via tested pipeline	Pre-authorised model; no individual approval	Minutes
Low	Configuration change to one internal application	Peer review plus automated tests	Hours
Medium	Database schema change for a customer-facing service	Service owner and technical lead	1–2 days
High	Core banking platform upgrade	Change advisory group including business representatives	Planned weeks ahead
Emergency	Patch for actively exploited vulnerability	Emergency authority (small, named, reachable 24/7)	As fast as safely possible

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Evidence for change risk assessment

- Blast radius: which services and users could be affected if it fails?
- History: how often have similar changes failed, for this team and this system?
- Testing: what automated and manual tests have passed, and in which environments?
- Reversibility: can the change be rolled back quickly, or is it one-way (for example, a data migration)?
- Timing: does it collide with peak business periods or other changes?
- Dependencies: what other services, suppliers or changes rely on it?

INSIGHT · APPROVAL DOES NOT CREATE SAFETY

Research on high-performing technology organisations has repeatedly found that heavyweight external approval boards do not reduce change failure, but they do slow delivery. Safety comes from small changes, automated testing, peer review, progressive rollout and fast rollback. Use approval boards for genuinely high-risk, cross-cutting changes and for oversight of the change system itself.

Change performance measures

MEASURE	FORMULA	WHAT IT TELLS YOU
Change success rate	`Change success rate = changes implemented without causing an incident or rollback ÷ total changes × 100%`	Reliability of the change system.
Change failure rate	`Change failure rate = changes causing degraded service or needing remediation ÷ total changes × 100%`	The complement of success rate; a core delivery-performance metric.
Change lead time	`Lead time = deployment time − time change request raised (or code committed)`	How quickly changes flow; long times encourage risky batching.
Standard change share	`Standard share = standard changes ÷ total changes × 100%`	How much change has been made routine and safe.

Redesigning a slow change process

1. **Measure** Collect a quarter of change data: volume, type, lead time, failures and their causes.
2. **Find the safe majority** Identify change types with high volume and near-zero failure. Make them standard changes.
3. **Delegate** Move low and medium-risk approval to peers and service owners with automated checks.
4. **Automate evidence** Pull test results, scan results and deployment history into the change record automatically.
5. **Refocus the board** The change advisory group reviews only high-risk changes and trends, not every request.
6. **Review monthly** Track lead time and failure rate together; adjust models when either worsens.

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SCENARIO · DECIDE · A CHANGE BOARD UNDER PRESSURE

A telecom operator's weekly change advisory board reviews 260 changes per meeting. Average lead time is 11 days, change failure rate is 18%, and engineers increasingly bundle many changes into large monthly releases. What should the change manager propose?

- (A) Create standard changes for proven low-risk types and delegate other approvals by risk
- (B) Add a second weekly meeting and require more detailed documentation for every change to reduce failures
- (C) Freeze all changes for a month to stabilise the environment
- (D) Let engineers deploy anything without records to save time

Circle your choice. Answer and feedback: module answer key.

CASE STUDY · CHANGE FAILURES AT A MOBILE OPERATOR

A mobile operator's quarterly data shows 1,900 changes, a change failure rate of 14% and an average lead time of nine days. Analysis of the 266 failed changes shows that 70% were large releases bundling 20 or more changes, 18% were made outside agreed windows during traffic peaks, and only 3% were standard changes. Engineers say the process is so slow that they save up changes for the monthly window.

1. What does the data say about the causes of change failure?

2. Which changes would you make to the change model first?

3. Which measures would show that your changes are working?

Model answer: module answer key.

TEMPLATE **Standard change model**

Define a change type that can be pre-authorised.

CHANGE TYPE	SCOPE AND LIMITS	PROCEDURE / AUTOMATION	TESTS REQUIRED	ROLLBACK METHOD	SUCCESS HISTORY
Web tier scale-out	*Add up to 4 nodes in existing cluster*	*Pipeline job SCALE-WEB*	*Health check passes on new nodes*	*Remove nodes via same job*	*212 runs, 0 failures in 12 months*

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CHANGE TYPE	SCOPE AND LIMITS	PROCEDURE / AUTOMATION	TESTS REQUIRED	ROLLBACK METHOD	SUCCESS HISTORY

Key terms

Change
The addition, modification or removal of anything that could have a direct or indirect effect on services.

Standard change
A low-risk, pre-authorised change that is well understood and follows a defined procedure.

Emergency change
A change that must be implemented as soon as possible, using an expedited authorisation path.

Change schedule
A calendar of planned and past changes used to coordinate work and avoid conflicts.

Change enablement
The practice of maximising successful changes by assessing risk, authorising changes and managing the change schedule.

Normal change
A change that must be scheduled, assessed and authorised according to its risk.

Change authority
The person or group that authorises a change, chosen according to the change's risk.

Change failure rate
The percentage of changes that lead to degraded service or require remediation.

REFLECT

Look at last month's changes. What share could safely have been standard changes? What stops them from being standard today?

KNOWLEDGE CHECK

1. A well-tested, pre-authorised procedure for adding users to a standard access group is which type of change?
   - (A) Emergency
   - (B) Major change requiring board authorisation
   - (C) Standard
   - (D) Normal
2. Which statement about change approval is best supported by evidence?
   - (A) More approvers always reduce change failure
   - (B) Heavy external approval slows delivery but rarely cuts failure
   - (C) Change boards should approve every change individually, however small
   - (D) Automated testing increases change risk
3. Who should authorise an emergency patch for an actively exploited vulnerability at 02:00?
   - (A) The full change advisory board at its next weekly meeting
   - (B) Nobody; emergency changes need no authorisation
   - (C) A small, named emergency authority reachable 24/7
   - (D) The vendor who released the patch
4. Out of 400 changes last quarter, 24 caused incidents or needed remediation. What is the change failure rate?
   - (A) 24%
   - (B) 94%
   - (C) 16%
   - (D) 6%

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5. Why do long change lead times often increase risk?

(A) Long lead times give too much testing time

(B) Approvers become too familiar with each change and approve it too readily

(C) Teams batch many changes into large, risky releases

(D) Short changes are harder to roll back

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 4 · CONTROL · LESSON 2 · 40 MIN

Release and Deployment Management

Making new functionality available safely, and moving it into production smoothly

- Distinguish release management from deployment management
- Choose deployment approaches that fit risk and architecture
- Design a pipeline that makes releases routine

Two practices are often confused. **Deployment management** moves new or changed hardware, software, documentation or any other component into live environments. **Release management** makes new and changed services and features available for use. With modern techniques the two can be separated: code can be deployed to production weeks before a feature is released to users with a switch, and a release can be targeted at a small group first.

Deployment versus release

Deployment

- Technical movement of components into environments
- Owned by engineering and platform teams
- Measured by deployment frequency, success and speed
- Can happen many times a day

Release

- Making features or services available to users
- Involves business, training, communication, support readiness
- Measured by adoption and user outcomes
- Timed for business readiness

Deployment approaches

APPROACH	HOW IT WORKS	BEST FOR	RISK PROFILE
Big bang	Everything switches at once	Tightly coupled systems with no alternative	High: failure affects everyone at once
Phased	Groups of users or sites move in waves	Branches, hospitals, factories	Medium: issues found early in small waves
Blue-green	Two identical environments; traffic switches from old to new	Web and API services	Low: instant switch back
Canary	A small share of traffic goes to the new version, then grows	High-traffic digital services	Low: limited blast radius, data-driven expansion
Feature flags / dark launch	Code deployed but hidden behind a switch	Separating deploy from release; A/B tests	Low, if flags are governed and cleaned up

A modern delivery pipeline

01 **Commit** — Code and configuration in version control →

02 **Build** — Compile, package, create immutable artefact →

03 **Test** — Unit, integration, security and performance tests →

04 **Stage** — Deploy to production-like environment →

05 **Deploy** — Progressive rollout with health checks →

06 **Release** — Feature enabled for target users

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EXAMPLE · A PHASED RELEASE AT A BANK

A Bangladeshi commercial bank rolled out a new teller system to 180 branches. It started with two branches in Dhaka for four weeks, then 20 branches across divisions, then the rest in four waves. Each wave had floor-support staff, a daily defect review and go/no-go criteria. Issues with Bangla receipt printing were found in the first wave and fixed before they reached 178 more branches.

Release readiness: beyond the code

- Users trained and job aids ready, in the right languages.
- Service desk briefed with known issues and knowledge articles.
- Support model and escalation paths agreed, including suppliers.
- Monitoring and dashboards updated for new components.
- Communication sent: what changes, when, and where to get help.
- Rollback or fallback plan rehearsed and go/no-go criteria agreed.

Canary rollout with health gate (pipeline script excerpt)

Linux host

```
# deploy v2.4.1 to 5% of traffic
deployctl rollout start payments-api --version 2.4.1 --weight 5
# watch error rate and latency for 15 minutes against the SLO
deployctl rollout gate payments-api --metric error_rate --max 0.5% --window 15m
deployctl rollout gate payments-api --metric p95_latency --max 300ms --window 15m
# promote if gates pass, otherwise roll back automatically
deployctl rollout promote payments-api --steps 25,50,100 --pause 10m
```

`deployctl` *is a generic placeholder for whichever deployment tool your platform uses; the pattern of weights, gates and automatic rollback is what matters.*

SCENARIO · DECIDE · CHOOSING A DEPLOYMENT APPROACH

A hospital is replacing its laboratory results interface, which feeds results to wards 24/7. The new interface can run alongside the old one, and traffic can be switched between them in seconds. Which approach is best?

(A) Big bang on a Sunday night, because traffic is lowest then and fewer clinicians would be affected

(B) Blue-green with a rehearsed switch-back

(C) Phased by ward over three months

(D) Deploy without a plan because the vendor has tested it

Circle your choice. Answer and feedback: module answer key.

TEMPLATE **Go/no-go checklist for a release**

Agree objective criteria before release day.

CRITERION	MEASURE / EVIDENCE	THRESHOLD	OWNER	STATUS
Critical defects open	*Defect tracker*	*Zero severity-1, ≤3 severity-2 with workarounds*	*Test lead*	*Met*

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CRITERION	MEASURE / EVIDENCE	THRESHOLD	OWNER	STATUS

Key terms

Release management
The practice of making new and changed services and features available for use.

Canary release
Routing a small share of users or traffic to a new version and expanding only if health measures stay within limits.

Feature flag
A configuration switch that turns functionality on or off without redeploying code.

Deployment pipeline
An automated sequence of build, test and deployment steps from code commit to production.

Deployment management
The practice of moving new or changed components into live environments.

Blue-green deployment
Running two identical environments and switching traffic from the old to the new, with the ability to switch back instantly.

Go/no-go criteria
Agreed, objective conditions that must be met before a release proceeds.

REFLECT

In your organisation, is deployment the same moment as release? What would change if you could separate them?

KNOWLEDGE CHECK

6. Which practice is concerned with making features available to users, including training and communication?

(A) IT asset management for the new release components
(B) Monitoring and event management
(C) Release management
(D) Deployment management

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7. Which approach routes a small percentage of traffic to a new version first?

(A) Canary
(B) Blue-green
(C) Phased rollout by geographic region across all branches
(D) Big bang

8. What is the main benefit of feature flags?

(A) Removing the need for testing
(B) Allowing users to edit production code directly from the browser
(C) Separating deployment from release
(D) Making rollback impossible

9. Which item is part of release readiness rather than deployment?

(A) Artefact built and signed
(B) Database migration script executed in staging
(C) Service desk briefed with known issues
(D) Container image scanned

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MODULE 4 · CONTROL · LESSON 3 · 35 MIN

Service Request Management and the Service Catalogue

Making standard requests effortless for users and efficient for providers

- Distinguish service requests from incidents and changes
- Design request models and catalogue entries users can understand
- Automate fulfilment while keeping appropriate controls

A **service request** is a user's request for something that is a normal part of service delivery: access to a system, a new laptop, information, a password reset, a meeting room set-up. Nothing is broken. Service request management supports the agreed quality of a service by handling these requests in an effective and user-friendly way. Because requests are predictable, they are the best candidates for standardisation and automation.

Request, incident or change?

SITUATION	CLASSIFICATION	WHY
New employee needs email and finance system access	Service request	Normal, pre-defined part of service delivery
Email has stopped working for one department	Incident	Unplanned interruption
Finance wants a new approval step added to the purchasing system	Change (via demand/engagement)	Modifies the service; needs assessment
User asks how to export a report to a spreadsheet	Service request (information)	Request for information or advice

Elements of a good request model

Sc
Scope
Exactly what is offered, for whom, and what is not included.

Ap
Approvals
Who must approve, if anyone. Many requests need none if access is role-based or low-cost.

Fu
Fulfilment steps
Tasks, teams, automation and expected durations.

SL
Service level
Target fulfilment time and how progress is communicated.

Co
Cost
Charge or budget impact, if any, visible to the requester.

PRO TIP · WRITE THE CATALOGUE FOR THE PERSON ORDERING

A catalogue entry titled 'AD group membership modification' means nothing to a nurse. 'Get access to the lab results system' does. Each entry should say what you get, who can ask, how long it takes and what it costs, in plain language, with a single button to order.

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Approval design

Over-controlled

- Line manager, IT manager and finance approve a mouse
- Approvals wait for absent managers
- Users email IT directly to avoid the portal
- Approval adds days but catches nothing

Right-sized

- Low-cost items pre-approved within a budget
- Role-based access granted automatically
- Approvals only where risk or cost justifies them
- Delegates and timeouts prevent stalled requests

Automating a request end to end

1. **Standardise** Agree one model for the request, removing local variations.
2. **Simplify** Remove approvals and data fields that add no value.
3. **Integrate** Connect the portal to identity, HR, asset and finance systems.
4. **Automate** Fulfil automatically where possible: create accounts, assign licences, send instructions.
5. **Monitor** Track volume, fulfilment time, failures and satisfaction for each model.

Request measures

MEASURE	FORMULA	WHAT IT TELLS YOU
Fulfilment time	`Fulfilment time = time delivered − time requested`	The user's experience of waiting; report median and 90th percentile.
Automation rate	`Automation rate = requests fulfilled with no manual step ÷ total requests × 100%`	Progress in removing toil from routine work.
First-time-right rate	`First-time-right = requests fulfilled without rework or return ÷ total × 100%`	Quality of the request model and data captured.

SCENARIO · DECIDE · APPROVALS THAT BLOCK ONBOARDING

At a pharmaceutical company, a request for standard sales-app access needs approval from the line manager, the sales director and IT security. Median fulfilment is 9 days; 97% of requests are approved unchanged. What is the best redesign?

(A) Keep all approvals but send automatic reminder emails to each approver every 24 hours until they respond

(B) Grant access automatically by role from the HR record, with periodic access reviews

(C) Remove all controls and let anyone request any access

(D) Move the request to email so it is faster to send

Circle your choice. Answer and feedback: module answer key.

TEMPLATE

Catalogue entry design

Describe a request so users can order it without help.

ENTRY NAME (USER WORDS)	WHO CAN REQUEST	WHAT YOU GET	APPROVAL	TARGET TIME	COST
Get access to the lab results system	*Doctors, nurses, lab staff*	*Read access to results for your ward*	*None – role-based*	*Same working day*	*No charge*

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ENTRY NAME (USER WORDS)	WHO CAN REQUEST	WHAT YOU GET	APPROVAL	TARGET TIME	COST

Key terms

Service request
A request from a user for something that is a normal, pre-defined part of service delivery.

Request model
A predefined approach for fulfilling a type of request, including scope, approvals, steps and service levels.

Role-based access
Granting access automatically based on a person's job role rather than individual approvals.

Automation rate
The share of requests fulfilled with no manual intervention.

Service request management
The practice of supporting agreed service quality by handling all pre-defined, user-initiated requests effectively and in a user-friendly way.

Service catalogue
Structured information about the services and offerings available to consumers.

Fulfilment
The delivery of what was requested.

REFLECT

Which request in your organisation needs the most approvals? Do those approvals ever change the outcome?

KNOWLEDGE CHECK

10. A user asks how to export a report to a spreadsheet. How should this be classified?

(A) A problem record for the reporting application
(B) An incident
(C) A service request for information
(D) A normal change

11. 97% of requests for a type of access are approved unchanged. What does this suggest?

(A) More approvers are needed to catch problems
(B) The approval adds delay with little control value
(C) The approvers are not reading the requests carefully
(D) The access should be removed from the catalogue entirely

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12. Which catalogue entry title is most user-friendly?

(A) AD group membership modification
(B) LIS-RO-ACL change
(C) Get access to the lab results system
(D) Directory object permission update for a security group

13. What is the best order for improving a request?

(A) Automate first, then simplify
(B) Standardise, simplify, integrate, then automate
(C) Automate everything immediately without any changes
(D) Integrate first, then standardise

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MODULE 4 · CONTROL · LESSON 4 · 45 MIN

IT Asset and Service Configuration Management

Knowing what you have, what it costs and how it connects to services

- Distinguish IT asset management from service configuration management
- Scope a configuration management database by the decisions it supports
- Use discovery, lifecycle and licence data to reduce cost and risk

Two practices answer two different questions. **IT asset management** asks: what do we own or use, what does it cost, and is it compliant? It manages assets through their lifecycle, from planning and purchase to disposal. **Service configuration management** asks: how are components configured and how do they relate to the services they support? It provides the map that incident, change and problem teams need to understand impact.

Asset versus configuration item

IT asset

- Anything of financial value that contributes to an IT product or service
- Managed for cost, contracts, licences and lifecycle
- Example: 1,200 laptops, 300 software licences, a support contract

Configuration item (CI)

- Any component that needs to be managed to deliver a service
- Managed for relationships and configuration
- Example: payments API, its database, the load balancer and the service they form

Asset lifecycle

01	→	02	→	03	→	04	→	05
Plan Demand, standards, budget		**Acquire** Purchase, lease or subscribe		**Deploy** Register, configure, assign		**Operate** Maintain, patch, track usage		**Retire** Recover, wipe data, dispose or recycle

WATCH OUT · THE CMDB THAT TRIED TO HOLD EVERYTHING

A university tried to record every cable, mouse and software setting in its configuration management database. After 18 months, data was 40% inaccurate and nobody trusted it. Scope configuration data by the decisions it must support: if no incident, change or audit decision needs a piece of data, do not collect it.

Scoping configuration data by decisions

1. **List the decisions** Impact analysis for changes, routing incidents, audit evidence, capacity planning.
2. **Identify required CIs** Which services, applications, platforms and infrastructure do those decisions need?
3. **Define relationships** Runs on, depends on, connects to, supported by: only the types decisions use.
4. **Choose sources** Prefer automated discovery and cloud APIs over manual entry.
5. **Assign owners** Each CI class has an owner accountable for accuracy.
6. **Verify and audit** Reconcile discovery with records; measure accuracy monthly.

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Service model from a configuration view

Business service	`Mobile wallet payments`
Application services	`Payments API` `Customer authentication` `Notification service`
Platforms	`Container cluster` `Managed database` `Message queue`
Infrastructure and suppliers	`Cloud region A and B` `SMS gateway supplier` `Network links`

Asset and configuration measures

MEASURE	FORMULA	WHAT IT TELLS YOU
CI accuracy	`Accuracy = CIs whose recorded attributes match discovered reality ÷ CIs sampled × 100%`	Trustworthiness of configuration data.
Licence compliance position	`Compliance = licences owned − licences deployed (by product)`	Negative values show exposure to audit fines; large positives show waste.
Asset utilisation	`Utilisation = assets in active use ÷ assets owned × 100%`	Idle hardware and unused subscriptions waste money.

Practical wins from asset data

- Reclaim unused software subscriptions every quarter.
- Plan hardware refresh before end-of-support dates, not after failures.
- Wipe and certify disposal of devices holding personal data.
- Find shadow IT and unmanaged cloud spending through finance and network data.
- Give incident teams the owner and support contract of any failing component in seconds.

SCENARIO · DECIDE · A LICENCE AUDIT LETTER

A garment group receives an audit notice from a software publisher. The asset register lists 800 licences; network discovery finds the software on 1,140 devices, including 210 that have not been used in 90 days. What should the IT asset manager do first?

(A) Buy 340 additional licences immediately so that the group is fully compliant before the audit begins
(B) Reconcile discovery with entitlements, remove unused installs and quantify the remaining gap
(C) Uninstall the software from all devices until the audit ends
(D) Tell the publisher the register shows 800 and ignore discovery data

Circle your choice. Answer and feedback: module answer key.

TEMPLATE **CI class definition**

Define what configuration data to keep and why.

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CI CLASS	DECISIONS IT SUPPORTS	KEY ATTRIBUTES	RELATIONSHIPS	DATA SOURCE	OWNER
Application service	*Change impact, incident routing*	*Name, owner, criticality, support group*	*Runs on platform; supports business service*	*Service catalogue + discovery*	*Application portfolio manager*

Key terms

IT asset
Any financially valuable component that can contribute to the delivery of an IT product or service.

Configuration item (CI)
Any component that needs to be managed in order to deliver an IT service.

Configuration management database (CMDB)
A database used to store configuration records and their relationships throughout their lifecycle.

Licence compliance
The match between software entitlements owned and software actually deployed or used.

IT asset management
The practice of planning and managing the full lifecycle of IT assets to maximise value, control costs and manage risks.

Service configuration management
The practice of ensuring accurate, reliable information about the configuration of services and the CIs that support them is available when needed.

Discovery
Automated scanning of networks and platforms to identify devices, software and relationships.

Shadow IT
Technology acquired or used without the knowledge or control of the IT function.

REFLECT

Which decision in your organisation would most benefit from trustworthy configuration data today? What is the smallest scope that would support it?

KNOWLEDGE CHECK

14. Which question does service configuration management mainly answer?

(A) Which supplier invoice is overdue
(B) Who approved each purchase order in the finance system
(C) How components relate to the services they support
(D) How much each laptop cost when purchased

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15. Why do many CMDB initiatives fail?

(A) They are scoped by decisions
(B) They use automated discovery
(C) They collect more data than any decision needs
(D) They assign an owner to every CI class in the database

16. Licences owned: 500. Licences deployed: 620. What is the compliance position?

(A) 120 licences surplus
(B) Fully compliant
(C) 620 licences short
(D) 120 licences short

17. What should happen to a laptop holding patient data at the end of its life?

(A) Disposal with general office waste and recycling
(B) Certified data wiping before disposal or reuse
(C) Immediate donation to staff
(D) Storage indefinitely in a cupboard

18. Which data source is preferred for keeping configuration data accurate?

(A) Engineers' memories
(B) Annual manual surveys
(C) Spreadsheets maintained by each team separately
(D) Automated discovery and platform APIs

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MODULE 4 · CONTROL · LESSON 5 · 45 MIN

Service Level Management and Experience

Agreeing, measuring and improving what users really experience

- Write service level agreements that reflect outcomes users care about
- Link SLAs to internal agreements and supplier contracts
- Combine operational metrics with experience measures in service reviews

Service level management sets clear, business-based targets for service performance so that delivery can be properly assessed, monitored and managed. Its most visible artefact is the **service level agreement (SLA)**, but the real work is the conversation: understanding what customers need, agreeing realistic targets, measuring honestly and improving together. An SLA nobody discusses is just paperwork.

The agreement chain

Service level agreement (SLA)	`Between provider and customer`	`Targets for the end-to-end service`
Operational level agreement (OLA)	`Between internal teams`	`Targets that support the SLA, such as database restore time`
Underpinning contract (UC)	`Between provider and supplier`	`Legally binding targets that support the SLA, such as 4-hour hardware replacement`

Good and poor SLA targets

POOR TARGET	WHY IT IS WEAK	BETTER TARGET
Server availability 99.9%	Users do not use servers; components can be up while the service fails	Patients can book appointments online 99.5% of the time, 06:00–23:00
Respond to incidents in 15 minutes	Response is not resolution	90% of P2 incidents restored within 4 hours
Average response time under 2 seconds	Averages hide bad experiences	95% of payment confirmations within 3 seconds
All tickets closed in 5 days	Mixes requests and incidents; easily gamed	Standard access requests fulfilled within 1 working day (90%)

Service level calculations

MEASURE	FORMULA	WHAT IT TELLS YOU
Availability	`Availability = (agreed service time − downtime) ÷ agreed service time × 100%`	Share of agreed time the service was usable; agree what counts as 'down' in advance.
Allowed downtime	`Allowed downtime = agreed service time × (1 − availability target)`	Example: 30 days × 24 h × 0.1% ≈ 43 minutes a month for 99.9%.
SLA compliance	`Compliance = transactions or periods meeting target ÷ total × 100%`	How often the target was met.

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SLA versus experience level agreement (XLA)

SLA focus

- Operational outputs: availability, response, resolution
- Measured from provider systems
- Can be met while users are frustrated

XLA focus

- User outcomes and sentiment: productivity lost, effort, satisfaction
- Measured from users and digital experience data
- Complements SLAs to expose watermelon effects

INSIGHT · MEASURE WHAT USERS FEEL, THEN EXPLAIN IT WITH WHAT YOU CONTROL

A strong service review starts with the user's experience (sentiment, effort, business outcomes) and then uses operational data to explain it. If satisfaction drops while SLAs are green, the SLAs measure the wrong things.

Running a service review

1. **Prepare** Combine SLA results, experience data, major incidents, changes and improvement status.
2. **Start with outcomes** How did the service support the business this period? What did users say?
3. **Explain performance** Use operational data to explain good and bad periods.
4. **Agree actions** Improvement items with owners, linked to the improvement register.
5. **Revisit targets** Are targets still right for the business? Adjust when needs change.

SCENARIO · DECIDE · GREEN SLA, ANGRY CUSTOMER

A university's IT department meets every SLA target for the student portal, yet the Pro-Vice-Chancellor complains that students cannot submit assignments during peak deadlines. Monitoring shows the portal is 'available' but pages take 40 seconds to load at peak. What should the service level manager do?

(A) Show the Pro-Vice-Chancellor the green SLA report to prove the service is performing as agreed

(B) Agree a peak-time response target with the customer and add student-experience measures

(C) Increase server capacity permanently without changing the agreement

(D) Remove the portal from SLA reporting

Circle your choice. Answer and feedback: module answer key.

TEMPLATE **SLA target worksheet**

Design targets that reflect customer outcomes.

SERVICE	CUSTOMER OUTCOME	MEASURE	TARGET AND HOURS	DATA SOURCE	LINKED OLAS / UCS
Online appointment booking	*Patients book without calling*	*Successful synthetic bookings*	*99.5%, 06:00–23:00; 95% in <5 s*	*Synthetic monitoring*	*OLA: database team; UC: SMS gateway 99.9%*

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SERVICE	CUSTOMER OUTCOME	MEASURE	TARGET AND HOURS	DATA SOURCE	LINKED OLAS / UCS

Key terms

Service level management
The practice of setting business-based targets for service performance and assessing, monitoring and managing delivery against them.

Service level agreement (SLA)
A documented agreement between a provider and a customer that identifies services and expected service levels.

Operational level agreement (OLA)
An agreement between internal teams that supports the delivery of an SLA.

Underpinning contract (UC)
A contract with an external supplier that supports the delivery of agreed service levels.

Experience level agreement (XLA)
An agreement focused on user experience and outcomes, complementing operational SLA targets.

Availability
The ability of a service to perform its agreed function when required, usually expressed as a percentage of agreed service time.

Service review
A regular meeting between provider and customer to review performance, experience and improvements.

REFLECT

Pick one SLA you are responsible for. Would its users recognise the targets as describing their experience? What would you change?

KNOWLEDGE CHECK

19. Which agreement is between two internal teams?

(A) Operational level agreement
(B) Service level agreement
(C) Experience level agreement signed with the customer
(D) Underpinning contract

20. For a 30-day month with a 24/7 service, approximately how much downtime does a 99.9% availability target allow?

(A) About 4 minutes
(B) About 7 hours
(C) About 4.3 hours
(D) About 43 minutes

21. Why are percentile targets often better than averages?

(A) Averages cannot be measured automatically
(B) Averages hide the bad experiences of a significant minority
(C) Percentiles are easier to calculate
(D) Percentiles always make the provider's performance look better in reports

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22. SLAs are all green but satisfaction is falling. What is the most likely explanation?

(A) Users are being unreasonable

(B) The SLAs measure things users do not experience as value

(C) The satisfaction survey is irrelevant

(D) The service desk is closing tickets too slowly to meet targets

23. What should a service review start with?

(A) Business outcomes and user experience

(B) A list of all tickets closed

(C) A technical description of every component of the service

(D) The supplier's invoice

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MODULE 4 · CONTROL

Answer key

Knowledge checks

1. C Low-risk, repeatable and pre-authorised describes a standard change.

2. B Safety comes from small, tested, reversible changes; heavy approval mainly adds delay.

3. C Emergency changes still need authorisation, but from a small, reachable authority.

4. D 24 ÷ 400 = 6%.

5. C Slow processes encourage batching; large batches are harder to test and roll back.

6. C Release management covers availability to users and the readiness around it.

7. A Canary releases expose a small share of traffic and expand based on health data.

8. C Flags let code be deployed but released later, to selected users, or switched off.

9. C Desk readiness concerns users and support; the others are technical deployment steps.

10. C Requests for information or advice are service requests.

11. B Near-universal approval signals that a role-based or pre-approved model may be better.

12. C Users recognise outcomes in their own words.

13. B Optimise before automating: standardise and simplify, then integrate and automate.

14. C Configuration management focuses on components and relationships; costs and contracts belong to asset management.

15. C Over-scoping makes data impossible to keep accurate; scoping by decisions keeps it useful.

16. D 500 − 620 = −120: the organisation is under-licensed by 120.

17. B The retire stage must protect data through certified wiping or destruction.

18. D Automation reduces manual effort and keeps data current.

19. A OLAs are internal agreements supporting the SLA.

20. D 30 × 24 × 60 × 0.001 ≈ 43 minutes.

21. B Averages can look fine while many users suffer; percentiles show the spread.

22. B This is the watermelon effect: operational targets miss what users actually experience.

23. A Starting with outcomes keeps the review relevant; operational data then explains the results.

Scenarios & case studies

Scenario · A change board under pressure

A is best. Correct. This removes most of the board's volume, shortens lead time, and lets the board focus on genuinely risky changes. Smaller, faster changes reduce the batching that drives failure.

Case study · Change failures at a mobile operator

Failure is concentrated in large, bundled releases, a direct result of slow lead times that encourage batching, and in poorly timed changes. Standard changes are almost never the problem. First, convert proven low-risk change types into standard changes and delegate low and medium-risk approvals to peers and service owners with automated evidence, so engineers no longer need to batch. Second, publish a peak-period policy, enforced by the change schedule, for customer-facing systems. Third, ask the advisory group to focus on high-risk changes and on the trend data. Measure change failure rate, lead time, average changes per release and the share of standard changes monthly; success means smaller releases, shorter lead time and a falling failure rate together.

Scenario · Choosing a deployment approach

B is best. Correct. Running both environments allows an instant switch and an instant return if results stop flowing, protecting patient safety.

Scenario · Approvals that block onboarding

B is best. Correct. Near-universal approval shows the approvals add delay but little control. Role-based provisioning plus reviews gives speed and assurance.

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Scenario · A licence audit letter

B is best. Correct. Reconciliation establishes the true position, removing unused installs reduces exposure legitimately, and the remaining gap informs negotiation or purchase.

Scenario · Green SLA, angry customer

B is best. Correct. Availability alone missed the real experience. Agreeing targets that reflect peak use and measuring experience closes the gap.

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05

MODULE 5 · JOURNEY

Creating, Delivering and Supporting Services Along the Customer Journey

From design and daily delivery to the end-to-end experience customers live

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Design services that can be run, supported and changed from day one
- Manage work, queues and priorities so value keeps flowing
- Map customer journeys and improve moments of truth
- Measure experience and value realisation and build trusted relationships

LESSONS

1. Creating Services That Can Be Run · 45 min
2. Delivering and Supporting at Scale · 40 min
3. Customer Journeys and Moments of Truth · 45 min
4. Stakeholder Value, Relationships and Experience · 40 min

MODULE 5 · JOURNEY · LESSON 1 · 45 MIN

Creating Services That Can Be Run

Designing, building and transitioning services with operations in mind from day one

- Design services for operability, supportability and change, not just functionality
- Organise teams and skills around the flow of value
- Plan a service transition that protects users and support teams

Many services are built by one group and thrown over the wall to another to run. The builders optimise for features and deadlines; the operators inherit missing monitoring, undocumented dependencies and fragile manual steps. **Creating, delivering and supporting** services as one continuous responsibility closes this gap. The questions 'how will we run it, support it and change it?' are asked at the first design workshop, not the week before go-live.

Design qualities beyond features

Op
Operability
Can it be deployed, monitored, scaled and recovered with little manual effort? Are health checks and logs built in?

Su
Supportability
Can the desk diagnose common issues? Are error messages meaningful, and is knowledge ready at launch?

Ch
Changeability
Can it be updated safely and often? Is it modular, tested automatically and free of hard-coded settings?

Se
Security and privacy
Are access, data protection and threat controls designed in from the start?

Re
Resilience
What happens when a dependency fails? Are there timeouts, retries, fallbacks and graceful degradation?

Co
Cost awareness
Are running costs understood and visible, especially for cloud consumption?

Service design package: what operations needs at handover

ITEM	WHY IT MATTERS
Service description and owner	Everyone knows what the service is for and who decides
Architecture and dependency map	Incident and change teams can assess impact
Service levels and SLOs	Teams know what 'good' looks like and when to act
Monitoring, alerts and dashboards	Failures are detected before users report them
Runbooks and knowledge articles	Common issues can be resolved at the desk or by on-call staff
Support model and escalation paths	Including suppliers and their hours
Capacity and cost model	Growth and budget are planned, not discovered

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Team organisation options

Functional teams

- Teams by technology: network, database, service desk
- Deep expertise, efficient specialist use
- Many handovers along a value stream

Product / service teams

- Long-lived teams owning a service end to end: build and run
- Fast flow, strong ownership
- Need platform support and shared standards

Platform and enabling teams

- Platform teams provide self-service infrastructure
- Enabling teams coach others in new skills
- Reduce cognitive load on product teams

INSIGHT · YOU BUILD IT, YOU SUPPORT IT, WITH HELP

Teams that support what they build fix the causes of operational pain quickly, because they feel it. But full end-to-end ownership can overload small teams. A balanced model gives product teams on-call ownership for their services, with platform teams providing paved roads (standard pipelines, monitoring, security) and a service desk handling first contact.

Planning a service transition

1. **Define readiness criteria** Operational, support, security and business criteria agreed with all parties.
2. **Build support capability** Train desk and on-call staff; publish knowledge; run a support rehearsal.
3. **Test operational procedures** Backup, restore, failover and scaling tested, not just functions.
4. **Plan early-life support** Builders and operators together for an agreed period, with elevated monitoring.
5. **Agree exit criteria** Incident rates, open defects and knowledge completeness that end early-life support.

SCENARIO · DECIDE · A HANDOVER WITH GAPS

A development team at a telecom operator plans to hand a new self-care app to operations next week. There are no runbooks, alerts cover only server CPU, and the service desk has not seen the app. The launch date has been announced publicly. What is the best course?

(A) Delay the launch until every operational document is complete, however long it takes and whatever the public reaction

(B) Launch with an early-life support period led jointly by developers and operations

(C) Hand over as planned and let operations learn as incidents occur

(D) Ask the desk to redirect all app issues to the developers' email

Circle your choice. Answer and feedback: module answer key.

Operational readiness checklist

☐ Service owner named and service in the catalogue
☐ SLOs agreed and dashboards live
☐ Actionable alerts route to the right on-call team
☐ Runbooks tested by someone other than the author
☐ Backup and restore tested end to end
☐ Desk trained and knowledge articles published
☐ Supplier support hours match service hours
☐ Early-life support period and exit criteria agreed

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Key terms

Operability
The ease with which a service can be deployed, monitored, scaled and recovered.

Early-life support
A period after go-live in which extra resources and monitoring support a new or changed service until it stabilises.

Platform team
A team that provides internal self-service capabilities, such as pipelines and monitoring, to other teams.

Graceful degradation
Designing a service to keep providing reduced but useful functionality when a component fails.

Service design package
The set of information handed to those who will run and support a service, describing how it works and how to operate it.

Product team
A long-lived team that owns a product or service end to end, including building and running it.

Cognitive load
The amount of mental effort a team must spend to understand and operate what it owns.

REFLECT

Think of the last service your organisation launched. Who was on call in its first month, and did they help design it?

KNOWLEDGE CHECK

1. Which design quality concerns how easily a service can be deployed, monitored and recovered?
   - (A) Operability
   - (B) Changeability
   - (C) Cost awareness for cloud consumption
   - (D) Utility
2. What is the main purpose of early-life support?
   - (A) Replace the service desk during the first year
   - (B) Delay go-live until all defects are fixed
   - (C) Stabilise a new service through joint ownership
   - (D) Transfer all support costs to the development team permanently
3. Which team type provides self-service pipelines and monitoring to other teams?
   - (A) Service desk
   - (B) Functional database team
   - (C) An enabling team that coaches other teams in new skills
   - (D) Platform team
4. Why does a 'build it and throw it over the wall' model cause operational problems?
   - (A) Operations teams are always less skilled
   - (B) It produces too many runbooks and alerts for the desk to use
   - (C) Developers write too much documentation
   - (D) Builders don't feel the pain of running what they create

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MODULE 5 · JOURNEY · LESSON 2 · 40 MIN

Delivering and Supporting at Scale

Managing work, queues and priorities so value keeps flowing

- Prioritise competing work across demand, incidents and improvement
- Manage queues and work in progress to protect flow
- Balance planned and unplanned work in service teams

Service teams juggle four kinds of work at once: new features and projects, changes, user requests and incidents, and improvement or technical debt. Without deliberate management, unplanned work crowds out everything else and teams spend their days firefighting. Delivering and supporting at scale means making all work visible, limiting how much is in progress, and prioritising openly.

Types of work in a service team

TYPE	EXAMPLE	RISK IF UNMANAGED
Business projects and features	New merchant onboarding flow	Crowds out support; deadlines slip
Internal projects	Database platform upgrade	Postponed indefinitely until a crisis
Changes and requests	Firewall rules, access, configuration	Endless queue of small tasks
Unplanned work	Incidents and urgent escalations	Consumes all capacity; nothing else finishes

Flow laws every service manager should know

MEASURE	FORMULA	WHAT IT TELLS YOU
Little's law	`Average lead time = average work in progress ÷ average throughput`	More items in progress means longer waits, unless throughput rises.
Utilisation and waiting	`Waiting grows sharply as utilisation approaches 100%: wait ∝ utilisation ÷ (1 − utilisation)`	Teams loaded at 95% have queues many times longer than teams at 70%.

WATCH OUT · THE 100% UTILISATION TRAP

Managers often try to keep every engineer 100% busy. Queue theory shows that this makes lead times explode and leaves no capacity for incidents or improvement. High-performing teams deliberately plan at 70–80% of capacity, leaving slack that absorbs unplanned work.

Prioritisation approaches

CoD

Cost of delay

Ask what each week of delay costs: lost revenue, risk, penalties. Do first what is most expensive to delay.

WSJF

Weighted shortest job first

Divide cost of delay by job size; small, valuable items go first.

Cl

Classes of service

Expedite, fixed date, standard and intangible items get different policies and limits.

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Ca

Capacity allocation

Reserve fixed shares of capacity for each work type, for example 60% features, 20% improvement, 20% unplanned.

Making work visible and limiting WIP

1. **Map the work types** Agree the categories of work and where each enters the team.
2. **Build one board** All work, planned and unplanned, on one visible board with clear columns.
3. **Set WIP limits** Limit items in each column; stop starting, start finishing.
4. **Make policies explicit** Write down how items are prioritised, expedited and declared done.
5. **Review flow weekly** Look at lead time, throughput, blocked items and ageing work.

Reducing unplanned work

- Track the share of unplanned work over time; aim to bring it down steadily.
- Route recurring incidents to problem management rather than repeatedly restoring.
- Pay down technical debt that causes frequent failures.
- Automate toil that interrupts engineers repeatedly.
- Protect focus time; batch non-urgent requests into scheduled windows.

SCENARIO · DECIDE · OVERLOADED INFRASTRUCTURE TEAM

An infrastructure team in a Dhaka university has 140 open items, average lead time of 31 days, and engineers working on 6–8 items each. The CIO asks them to 'work harder' to clear the backlog before semester starts. What is the best advice?

(A) Start all 140 items now so that everything moves forward at least a little before the semester begins

(B) Ask engineers to work weekends until the backlog is cleared

(C) Hire contractors for every open item

(D) Limit work in progress, finish the most valuable items first and make priorities visible

Circle your choice. Answer and feedback: module answer key.

TEMPLATE **Capacity allocation plan**

Agree how team capacity is shared between types of work.

WORK TYPE	TARGET SHARE %	ACTUAL SHARE LAST MONTH %	POLICY	ACTION IF EXCEEDED
Unplanned (incidents, escalations)	*20*	*41*	*P1/P2 incidents always first*	*Problem reviews for top 3 incident sources*

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WORK TYPE	TARGET SHARE %	ACTUAL SHARE LAST MONTH %	POLICY	ACTION IF EXCEEDED

Key terms

Work in progress (WIP)
Work that has started but is not yet finished.

Throughput
The number of work items completed per unit of time.

Weighted shortest job first (WSJF)
A prioritisation method that divides cost of delay by job size.

Capacity allocation
Reserving fixed shares of a team's capacity for different types of work.

Little's law
A queueing relationship: average lead time equals average work in progress divided by average throughput.

Cost of delay
The economic impact of delaying a piece of work by a unit of time.

Unplanned work
Work that arrives without being scheduled, such as incidents and urgent escalations.

REFLECT

What share of your team's time last month was unplanned work? What single change would reduce it most?

KNOWLEDGE CHECK

5. A team has 60 items in progress and finishes 20 a week. What is the average lead time according to Little's law?
   - (A) 1.5 weeks
   - (B) 6 weeks
   - (C) 20 weeks
   - (D) 3 weeks

6. What happens to waiting time as a team's utilisation approaches 100%?
   - (A) It falls slowly because everyone is busy and working efficiently
   - (B) It falls to zero
   - (C) It increases sharply
   - (D) It stays constant

7. Which prioritisation method divides cost of delay by job size?
   - (A) Capacity allocation by work type and team
   - (B) Highest-paid person's opinion
   - (C) Weighted shortest job first
   - (D) First in, first out

8. What is the best first step to reduce a team's long lead times?
   - (A) Start more items in parallel
   - (B) Limit work in progress
   - (C) Add more columns to the board so work is tracked in more detail
   - (D) Increase utilisation to 100%

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MODULE 5 · JOURNEY · LESSON 3 · 45 MIN

Customer Journeys and Moments of Truth

Seeing the service the way customers live it, from first contact to realised value

- Map a customer journey across stages and touchpoints
- Identify moments of truth and pain points that shape perception
- Design improvements that change the journey, not just a team's output

Customers do not experience your org chart. They experience a **journey**: a series of interactions with your organisation from the first time they hear of a service until long after they start using it. Each interaction is a **touchpoint**; some touchpoints matter far more than others. A master service manager designs and improves the whole journey, because value is won or lost in the gaps between departments.

Stages of a customer journey

01 Explore — Understand markets, needs and potential customers; the customer discovers you

→

02 Engage — Build relationships and trust; understand needs in depth

→

03 Offer — Shape demand and design offerings that fit

→

04 Agree — Agree value, terms, service levels and expectations

→

05 Onboard — Bring the customer and their users into the service

→

06 Co-create — Use the service together, day to day

→

07 Realise — Confirm and grow the value achieved

DEFINITION · MOMENT OF TRUTH

A **moment of truth** is a touchpoint at which the customer forms or changes their opinion of the service strongly. For a mobile wallet user it might be the first failed cash-out at an agent; for a hospital patient, the moment results are delayed; for a merchant, the first late settlement. Moments of truth deserve disproportionate design attention.

Journey map: merchant onboarding for a payments provider

STAGE	TOUCHPOINT	CUSTOMER FEELING	PAIN POINT	OPPORTUNITY
Explore	Sales visit, website	Curious	Fees unclear	Simple fee calculator
Agree	Contract signing	Cautious	12-page contract in English only	Bangla summary, digital signature
Onboard	Account and QR set-up	Frustrated	Documents requested three times	One document upload, status tracking
Co-create	Daily payments	Satisfied	Settlement report hard to read	Clear daily statement in app
Realise	Quarterly review	Unsure of value	No view of growth	Sales insights dashboard

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Mapping a journey

1. **Pick a persona** Choose a specific customer type, such as 'small shop owner in Rajshahi with basic smartphone'.
2. **Gather evidence** Interview customers, observe, analyse tickets, complaints and digital analytics.
3. **Lay out stages and touchpoints** Include every channel: app, call centre, agent, branch, email.
4. **Add feelings and effort** Where is the customer anxious, delighted, confused or forced to repeat themselves?
5. **Mark moments of truth** Identify where opinions are formed most strongly.
6. **Connect to back-stage** Show the internal teams, systems and value streams behind each touchpoint.

Front stage and back stage

Front stage

- What the customer sees and does
- Apps, staff interactions, messages, statements
- Where experience is felt

Back stage

- What happens behind the scenes
- Approvals, systems, suppliers, internal handovers
- Where experience is usually caused

EXAMPLE · FIXING THE JOURNEY, NOT THE TEAM

A private hospital in Dhaka measured each department separately: registration in 5 minutes, consultation on time, laboratory within target. Patients still spent 3.5 hours per visit. Journey mapping showed waits between departments, and repeated data entry at each desk. A single patient ID captured at registration and a queue display linking departments cut the visit to 2 hours without any department changing its own targets.

SCENARIO · DECIDE · CHOOSING THE JOURNEY IMPROVEMENT

A mobile financial services provider finds that 30% of new customers never complete a second transaction. Journey research shows the first cash-out at an agent often fails because agents lack float, and customers blame the app. What should the service manager prioritise?

(A) Redesign the app's welcome screens and add tutorial videos so customers understand the service better

(B) Offer cashback on second transactions

(C) Treat first cash-out as a moment of truth and fix agent float with the agent network team

(D) Increase marketing to acquire more customers

Circle your choice. Answer and feedback: module answer key.

TEMPLATE **Journey map worksheet**

Map one persona's journey and find improvement opportunities.

STAGE	TOUCHPOINT / CHANNEL	CUSTOMER ACTION	FEELING (−2 TO +2)	BACK-STAGE TEAM / SYSTEM	PAIN POINT	IMPROVEMENT IDEA
Onboard	*App registration*	*Uploads national ID*	*−1*	*KYC team / document system*	*Photo rejected without reason*	*Instant quality check with clear guidance*

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STAGE	TOUCHPOINT / CHANNEL	CUSTOMER ACTION	FEELING (−2 TO +2)	BACK-STAGE TEAM / SYSTEM	PAIN POINT	IMPROVEMENT IDEA

Key terms

Customer journey
The complete end-to-end experience a customer has with a service provider across all stages and touchpoints.

Moment of truth
A touchpoint at which the customer forms or changes their opinion of the service strongly.

Journey map
A visual representation of a customer's stages, touchpoints, feelings and pain points.

Back stage
The internal activities, systems and partners that support front-stage interactions.

Touchpoint
Any interaction between a customer and the provider's people, systems or communications.

Persona
A realistic description of a type of customer or user, used to guide design.

Front stage
The parts of a service that customers see and interact with directly.

Onboarding
Bringing a customer and their users into a service so they can start realising value.

REFLECT

Which moment of truth in your service do you think customers judge you on most? What evidence would confirm or challenge your guess?

KNOWLEDGE CHECK

9. A patient waits an hour between departments even though each department meets its own target. What technique best reveals this?

(A) Journey mapping across departments

(B) A separate satisfaction survey run by each department

(C) Departmental KPI dashboards

(D) Server performance monitoring

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10. What is a moment of truth?

(A) The moment the contract is signed
(B) The final stage of every customer journey
(C) Any interaction recorded in the CRM system by front-line staff
(D) A touchpoint where the customer's opinion is strongly formed

11. In which journey stage does a customer and their users start using the service for the first time?

(A) Onboard
(B) Agree
(C) Realise
(D) Explore

12. A front-stage pain point usually has its cause where?

(A) In the design of the user interface and its screens alone
(B) Back stage: internal processes, systems or partners
(C) In the marketing team
(D) Always in the customer's own behaviour

13. Why should journey maps use a specific persona?

(A) So that one average journey can represent every customer type
(B) Different customers have different journeys and pain points
(C) Regulators require personas
(D) Personas replace the need for real customer research

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MODULE 5 · JOURNEY · LESSON 4 · 40 MIN

Stakeholder Value, Relationships and Experience

Building trusted relationships and proving the value services create

- Manage relationships with customers and stakeholders strategically
- Measure customer, user and employee experience with suitable metrics
- Track value realisation and communicate it credibly

Value is co-created, so relationships matter. **Relationship management** establishes and nurtures links between the organisation and its stakeholders at strategic and tactical levels. A strong relationship lets you hear about needs early, negotiate trade-offs honestly and recover trust after failures. Weak relationships turn every incident into a dispute and every request into a negotiation.

Relationship maturity

1
Order taker
IT receives requests and delivers what is asked, with little understanding of why.

2
Service provider
IT delivers agreed services with SLAs and regular reporting.

3
Trusted adviser
IT understands business strategy and proposes solutions proactively.

4
Strategic partner
IT co-creates business strategy and shares accountability for outcomes.

Three lenses on experience

Customer experience (CX)

- How customers perceive the whole relationship
- Contracts, value, trust, communication
- Measured by NPS, retention, account reviews

User experience (UX)

- How easy and pleasant the service is to use
- Interfaces, speed, clarity, accessibility
- Measured by task success, effort, usability tests

Employee experience (EX)

- How staff experience their tools and workplace
- Devices, systems, support, collaboration
- Measured by digital experience scores and sentiment

Common experience measures

MEASURE	FORMULA	WHAT IT TELLS YOU
Net Promoter Score (NPS)	`NPS = % promoters (scores 9–10) − % detractors (scores 0–6)`	Willingness to recommend; ranges from −100 to +100.
Customer satisfaction (CSAT)	`CSAT = satisfied responses (e.g. 4–5 of 5) ÷ total responses × 100%`	Satisfaction with a specific interaction.
Customer effort score (CES)	`CES = average rating of 'how easy was it to…' across responses`	Effort is a strong predictor of loyalty in service interactions.

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WATCH OUT · SURVEY TRAPS

Surveys sent only after closed tickets miss users who gave up. Response rates of 5% represent the most delighted and the most angry. Combine surveys with behavioural data (repeat contacts, abandonment, adoption) and qualitative conversations before drawing conclusions.

Tracking value realisation

1. **Define expected outcomes** Agree measurable benefits with the customer before investing.
2. **Assign benefit owners** Business owners, not IT, are accountable for realising benefits.
3. **Baseline** Measure the current state of each benefit.
4. **Track and report** Report benefits alongside service performance at reviews.
5. **Adjust** If benefits lag, investigate adoption, design or assumptions and act.

Communicating value credibly

- Use the customer's language and measures, not IT jargon.
- Show trends over time, not single snapshots.
- Be honest about failures and what you learned.
- Connect service performance to business outcomes with evidence.
- Keep reports short; offer detail for those who want it.

CASE STUDY · REBUILDING TRUST WITH A HOSPITAL BOARD

The IT department of a 900-bed hospital in Chattogram is seen by the board as slow and expensive. Clinicians complain about logins, printers and the patient-record system. IT reports show 98% of SLAs met. The new head of IT has six months to change the board's view.

1. Why is there a gap between IT reports and the board's perception?

2. What relationship and measurement changes would you make first?

3. How would you show progress at the six-month board meeting?

Model answer: module answer key.

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SCENARIO · DECIDE · LOW NPS, HIGH CSAT

A cloud services provider in Dhaka sees ticket CSAT of 92% but an NPS of −15 from enterprise customers. What is the most likely explanation and best action?

(A) NPS is unreliable and should be ignored in favour of CSAT

(B) Explore relationship-level issues in account reviews, since NPS reflects the whole relationship

(C) Increase the ticket survey frequency to raise response numbers and confirm that customers are in fact satisfied

(D) Offer discounts to all enterprise customers at their next renewal so that their scores improve quickly

Circle your choice. Answer and feedback: module answer key.

Key terms

Relationship management
The practice of establishing and nurturing links between an organisation and its stakeholders at strategic and tactical levels.

Net Promoter Score (NPS)
A measure of loyalty: the percentage of promoters minus the percentage of detractors.

Customer effort score (CES)
A measure of how easy customers found it to get something done.

Employee experience (EX)
How employees perceive their tools, workplace and support.

Benefit owner
A business person accountable for realising a defined benefit.

Value realisation
Confirming that expected outcomes and benefits have actually been achieved.

Trusted adviser
A relationship level in which the provider understands the customer's strategy and proposes solutions proactively.

REFLECT

At what relationship maturity level does your main customer see your team? What would move you one level up?

KNOWLEDGE CHECK

14. Survey: 50% promoters, 30% passives, 20% detractors. What is the NPS?

(A) +20 (B) +50
(C) +30 (D) +80

15. Who should usually be accountable for realising business benefits from an IT investment?

(A) The supplier who delivered the system and its support (B) The service desk manager
(C) A business benefit owner (D) The IT project manager

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16. Which measure best predicts loyalty in service interactions according to many studies?

(A) Customer effort
(B) Average speed of answer
(C) Ticket volume
(D) The number of SLA targets in the agreement

17. What characterises a strategic-partner relationship?

(A) Delivering exactly what is requested
(B) Negotiating the lowest price for each individual request
(C) Co-creating strategy and sharing outcomes
(D) Reporting SLAs monthly

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MODULE 5 · JOURNEY

Answer key

Knowledge checks

1. A Operability covers deployment, monitoring, scaling and recovery.

2. C Early-life support is a temporary period of joint ownership and elevated monitoring.

3. D Platform teams build paved roads; enabling teams coach.

4. D Separating build from run removes the feedback that drives operability.

5. D Lead time = WIP ÷ throughput = 60 ÷ 20 = 3 weeks.

6. C Queueing effects make waits grow rapidly at high utilisation.

7. C WSJF favours small, valuable items.

8. B Reducing WIP directly shortens lead time.

9. A Only an end-to-end journey view shows gaps between departments.

10. D Moments of truth are the touchpoints with outsized influence on perception.

11. A Onboarding brings customers and users into the service.

12. B Most experience problems are caused by back-stage handovers and systems.

13. B Specific personas keep the map grounded; averaging customers hides real pain.

14. C NPS = 50 − 20 = +30.

15. C Benefits are realised in the business; a business owner must be accountable.

16. A Low effort strongly predicts continued loyalty.

17. C Strategic partners shape strategy together and share accountability.

Scenarios & case studies

Scenario · A handover with gaps
B is best. Correct. The public date can be kept while risk is controlled: builders stay engaged, operational gaps are closed quickly and the desk is supported.

Scenario · Overloaded infrastructure team
D is best. Correct. Little's law shows that cutting work in progress shortens lead time. Finishing valuable items first delivers more value before semester.

Scenario · Choosing the journey improvement
C is best. Correct. The pain is a back-stage cause (agent float) showing up at a front-stage moment of truth. Fixing it addresses the real reason customers leave.

Case study · Rebuilding trust with a hospital board
IT measures its own outputs, while clinicians experience slow logins and failures at moments of truth on wards, so the reports are 'watermelon' green. First, appoint relationship managers for clinical directorates, run journey walks with nurses and doctors, and add experience measures (login time, clinician effort score, time lost per shift). Agree three outcome-based priorities with the medical director, such as single sign-on on wards. At six months, report in clinical terms: minutes saved per shift, incidents affecting patient care, clinician sentiment trend and the next priorities, with honest notes on what did not improve.

Scenario · Low NPS, high CSAT
B is best. Correct. Customers can be pleased with individual tickets yet unhappy with reliability, pricing or strategic fit. Account reviews reveal relationship-level issues.

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06

MODULE 6 · FLOW

High-Velocity IT: DevOps, SRE and Value Streams

Deliver fast and stay reliable with modern engineering and flow practices

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Explain the outcomes and metrics of high-velocity IT
- Apply DevOps culture, pipelines and automated controls
- Run reliability with SLIs, SLOs, error budgets and toil reduction
- Manage value streams with flow metrics, constraints and platforms

LESSONS

1. High-Velocity IT: Speed with Safety · 40 min
2. DevOps in Practice · 45 min
3. Site Reliability Engineering · 45 min
4. Value Streams and Flow at Scale · 40 min

MODULE 6 · FLOW · LESSON 1 · 40 MIN

High-Velocity IT: Speed with Safety

Why digital organisations must deliver fast, reliably and responsibly at the same time

- Explain why high-velocity delivery matters in digital organisations
- Describe the outcomes high-velocity IT aims for and the trade-offs between them
- Use the four key delivery metrics to assess delivery performance

In a digital business, technology is not a support function; it is the product. A mobile bank's app, a ride-hailing platform's dispatch engine and an online university's learning platform are the business. When competitors release improvements weekly, a monthly release cycle and a three-week change process become a strategic weakness. **High-velocity IT** is the application of service management to organisations where speed of delivery, reliability and responsible use of technology all matter at once.

Five outcomes of high-velocity IT

V

Valuable investments

Choosing the right things to build: innovative, well-prioritised and tied to real outcomes. Speed without direction only produces waste faster.

F

Fast development

Shortening the time from idea to working software in users' hands, through small batches and automation.

R

Resilient operations

Services that stay available and recover quickly, even as change accelerates.

C

Co-created value

Working closely with users and customers so that what is delivered is actually used and valued.

A

Assured conformance

Meeting governance, risk, security and regulatory requirements, ideally automatically, without slowing delivery.

The four key delivery metrics

METRIC	WHAT IT MEASURES	ELITE TEAMS TYPICALLY	STRUGGLING TEAMS TYPICALLY
Deployment frequency	How often changes reach production	On demand, many times a day	Monthly or less
Lead time for changes	Commit to running in production	Less than a day	One to six months
Change failure rate	Share of deployments causing failure	Low single digits to ~15%	Much higher, often over 40%
Time to restore service	Recovery time after a failure	Less than an hour	A week or more

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INSIGHT · SPEED AND STABILITY ARE NOT A TRADE-OFF

Year after year, large studies of software delivery find that the best teams are both faster **and** more stable. Small, frequent changes are easier to test, understand and roll back. Teams that deploy rarely tend to deploy large, risky batches and fail more often.

Traditional versus high-velocity operating patterns

Traditional

- Large annual projects
- Separate build and run teams
- Manual testing and approvals
- Change control by committee
- Success measured by on-time delivery

High-velocity

- Small, continuous increments
- Product teams that build and run
- Automated tests, security and compliance checks
- Risk-based, often automated change
- Success measured by outcomes and flow

Behaviours that support high velocity

- Embrace experimentation and treat failure as information.
- Make decisions close to the work with clear guardrails.
- Share knowledge openly and avoid hero culture.
- Automate repetitive work so people focus on judgement.
- Design for reversibility: favour decisions that can be undone cheaply.
- Keep the customer and the outcome visible in daily work.

WATCH OUT · HIGH VELOCITY IS NOT 'MOVE FAST AND BREAK THINGS'

In payments, healthcare and power systems, breaking things harms people. High-velocity IT in regulated industries means building safety, security and compliance into automated pipelines, so that speed never depends on skipping controls.

SCENARIO · DECIDE · ASSESSING DELIVERY PERFORMANCE

A digital bank deploys once a month, lead time is 45 days, change failure rate is 32% and restoring service takes about two days. Leadership wants to deploy weekly. What should the delivery lead focus on first?

(A) Deploy the same monthly batch weekly by splitting the release calendar into four

(B) Add a second approval board to reduce the change failure rate before increasing speed

(C) Focus only on restoring faster and leave deployment unchanged

(D) Reduce batch size and automate testing and deployment so changes become smaller and safer

Circle your choice. Answer and feedback: module answer key.

TEMPLATE **Delivery performance baseline**

Record the four key metrics for one service and set improvement targets.

SERVICE	DEPLOYMENT FREQUENCY	LEAD TIME FOR CHANGES	CHANGE FAILURE RATE	TIME TO RESTORE	TARGET IN 6 MONTHS
Mobile banking app	*Monthly*	*45 days*	*32%*	*~2 days*	*Weekly; <7 days; <15%; <4 hours*

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SERVICE	DEPLOYMENT FREQUENCY	LEAD TIME FOR CHANGES	CHANGE FAILURE RATE	TIME TO RESTORE	TARGET IN 6 MONTHS

Key terms

High-velocity IT
The application of service management in digital organisations where fast delivery, resilience and responsible technology use matter together.

Lead time for changes
Time from code being committed to it running successfully in production.

Batch size
The amount of change delivered at once; smaller batches reduce risk and speed feedback.

Reversibility
The ability to undo a decision or change quickly and cheaply.

Deployment frequency
How often an organisation successfully deploys changes to production.

Time to restore service
How long it takes to recover from a failure in production.

Assured conformance
Meeting governance, risk and compliance requirements, ideally through automated controls.

REFLECT

Which of the four key delivery metrics would your organisation be most embarrassed to publish? Why?

KNOWLEDGE CHECK

1. What do large studies of software delivery consistently find about speed and stability?
   - (A) Speed and stability are unrelated to each other in practice
   - (B) Faster teams are always less stable
   - (C) The best teams achieve both together
   - (D) Stability requires slowing down
2. Which metric measures time from code commit to running in production?
   - (A) Deployment frequency
   - (B) Change failure rate across all deployments
   - (C) Time to restore service
   - (D) Lead time for changes
3. Which high-velocity outcome concerns meeting regulatory and security requirements?
   - (A) Assured conformance
   - (B) Fast development
   - (C) Co-created value
   - (D) Valuable investments in innovation

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4. Why do teams that deploy rarely tend to fail more often?

(A) Their engineers are less experienced than those in fast-moving teams

(B) They have more time to test, which introduces errors

(C) Rare deployments are always automated

(D) They deploy large batches that are harder to test and roll back

5. In a regulated hospital system, how should high velocity be achieved safely?

(A) By avoiding all change

(B) By moving fast and fixing what breaks afterwards

(C) By building controls into automated pipelines

(D) By skipping approvals when deadlines are tight

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MODULE 6 · FLOW · LESSON 2 · 45 MIN

DevOps in Practice

Culture, automation, flow and feedback across build and run

- Explain DevOps as a combination of culture, practices and tooling
- Describe continuous integration, delivery and infrastructure as code
- Integrate security and service management controls into delivery pipelines

DevOps is a set of cultural values, practices and tools that bring development and operations together to deliver valuable change quickly and reliably. It is not a job title or a toolset. Organisations that buy pipeline tools but keep separate teams with conflicting goals get automation of their dysfunction, not DevOps.

CALMS: a lens on DevOps

C

Culture

Shared ownership of outcomes, trust, blamelessness and collaboration across build and run.

A

Automation

Automating builds, tests, deployments, infrastructure and compliance checks.

L

Lean

Small batches, limiting work in progress, removing waste, optimising flow.

M

Measurement

Measuring flow, quality and outcomes to guide decisions.

S

Sharing

Sharing knowledge, tools and lessons across teams.

Continuous delivery pipeline

01 **Small commit**
Developer merges a small change to the main branch

→

02 **Continuous integration**
Automated build and tests run on every change

→

03 **Security and quality gates**
Code scanning, dependency checks, policy checks

→

04 **Deploy to staging**
Production-like environment, automated acceptance tests

→

05 **Progressive production deploy**
Canary or blue-green with health checks

→

06 **Observe and learn**
Monitor outcomes; feed back into the backlog

Continuous integration, delivery and deployment

Continuous integration

- Developers merge to main at least daily
- Every change built and tested automatically
- Broken builds fixed immediately

Continuous delivery

- Every change that passes the pipeline is releasable
- Deployment to production is a business decision, one click away

Continuous deployment

- Every change that passes all checks goes to production automatically
- No manual step between commit and production

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Trunk-based workflow with infrastructure as code

```
Linux host

# small, short-lived branch for one change
git switch -c fix/timeout-retry
git commit -am "Add retry with backoff to SMS gateway client"
git push origin fix/timeout-retry   # opens a review; pipeline runs tests
# infrastructure is code too: preview the change before applying
iac plan -var-file=prod.vars -out=plan.out
iac apply plan.out
```

`iac` *stands for whichever infrastructure-as-code tool your organisation uses. The principle is that infrastructure changes are versioned, reviewed and applied through the same pipeline as application code.*

PRO TIP · PUT SERVICE MANAGEMENT INSIDE THE PIPELINE

Instead of asking engineers to fill in change forms after the fact, let the pipeline create the change record automatically with test results, scan results, approvals and deployment logs attached. Auditors get better evidence, engineers lose a manual chore and change data becomes accurate.

Shifting security and controls left

CONTROL	TRADITIONAL TIMING	DEVOPS TIMING
Security testing	Penetration test before go-live	Automated scanning on every commit, plus periodic expert testing
Dependency risks	Annual audit	Dependency and licence checks in every build
Change approval	Weekly board	Peer review plus automated policy checks; board for high risk only
Configuration compliance	Quarterly manual audit	Policy-as-code checks before deployment and continuously after

SCENARIO · DECIDE · A DEVOPS TRANSFORMATION STALLING

An insurer bought a full pipeline toolchain a year ago. Deployment frequency has not improved. Developers still hand releases to a separate operations team, which schedules them for monthly windows, and incidents are blamed on whichever team is not in the room. What should the transformation lead focus on?

- (A) Buy additional tools for release orchestration so that the monthly windows can be automated end to end
- (B) Mandate daily deployments immediately
- (C) Disband the operations team
- (D) Change team structure and incentives so teams share ownership of delivery and operation

Circle your choice. Answer and feedback: module answer key.

TEMPLATE **Pipeline control map**

Show which controls are automated in the delivery pipeline and which evidence they create.

PIPELINE STAGE	CONTROL	AUTOMATED?	EVIDENCE PRODUCED	OWNER
Build	*Dependency vulnerability scan*	*Yes*	*Scan report attached to change record*	*Platform security team*

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PIPELINE STAGE	CONTROL	AUTOMATED?	EVIDENCE PRODUCED	OWNER

Key terms

DevOps
A combination of culture, practices and tools that brings development and operations together to deliver change quickly and reliably.

CALMS
A lens for DevOps: culture, automation, lean, measurement and sharing.

Continuous integration (CI)
Frequently merging code changes into a shared main branch, with automated builds and tests on each change.

Continuous delivery (CD)
Keeping software always in a releasable state so deployment can happen at any time with low risk.

Continuous deployment
Automatically deploying every change that passes all pipeline checks to production.

Infrastructure as code (IaC)
Defining and managing infrastructure through version-controlled code rather than manual configuration.

Policy as code
Expressing compliance and security rules as code that is automatically checked.

Trunk-based development
A practice in which developers integrate small changes into a single main branch frequently.

REFLECT

Which manual control in your delivery process could be turned into an automated pipeline check, and what evidence would it produce?

KNOWLEDGE CHECK

6. What is the difference between continuous delivery and continuous deployment?

(A) Continuous deployment releases passing changes automatically

(B) Continuous delivery skips automated testing

(C) Continuous deployment requires a weekly change board to review releases

(D) They are identical terms used by different vendors

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7. An organisation buys DevOps tools but keeps separate teams with conflicting goals. What is the likely result?
   - (A) Immediate cultural change across the organisation
   - (B) Automated handovers with little real improvement
   - (C) Elimination of all incidents
   - (D) Rapid improvement in all delivery metrics
8. Which practice treats server and network configuration as version-controlled code?
   - (A) Manual configuration management with a change log
   - (B) Infrastructure as code
   - (C) Continuous integration
   - (D) Value stream mapping
9. How can change records become more accurate in a DevOps pipeline?
   - (A) The pipeline creates them with attached evidence
   - (B) A change board writes them after deployment
   - (C) They are replaced by verbal approvals
   - (D) Engineers complete them weekly from memory
10. In CALMS, what does the 'L' stand for?
    - (A) Logging
    - (B) Licensing
    - (C) Lean
    - (D) Leadership

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MODULE 6 · FLOW · LESSON 3 · 45 MIN

Site Reliability Engineering

Managing reliability as an engineering problem with SLOs and error budgets

- Define service level indicators and objectives that reflect user experience
- Use error budgets to balance reliability and feature delivery
- Reduce toil and run blameless post-incident reviews

Site reliability engineering (SRE) treats operations as a software engineering problem. Instead of aiming for perfect reliability, SRE sets explicit reliability targets that reflect what users need, measures them precisely, and uses the gap between target and perfection, the **error budget**, to decide when to move fast and when to slow down. It turns arguments between developers and operators into data-driven decisions.

From indicators to agreements

SLA	`External promise to customers` `Has consequences (credits, penalties)` `Set looser than the SLO`
SLO	`Internal target for an SLI over a window` `Example: 99.9% of requests succeed over 28 days`
SLI	`A measured indicator of user experience` `Example: proportion of successful payment requests`

Reliability calculations

MEASURE	FORMULA	WHAT IT TELLS YOU
Service level indicator	`SLI = good events ÷ valid events × 100%`	For example, payment requests completed under 2 seconds without error ÷ all valid payment requests.
Error budget	`Error budget = 1 − SLO`	A 99.9% SLO gives a 0.1% budget: about 43 minutes of full outage or 1 in 1,000 failed requests per 30 days.
Burn rate	`Burn rate = observed error rate ÷ error rate allowed by the SLO`	A burn rate of 1 uses the budget exactly over the window; 10 exhausts it ten times faster.

> **INSIGHT · 100% IS THE WRONG TARGET**
> Users cannot tell the difference between 99.99% and 100% because their phone, Wi-Fi and mobile network fail more often than that. Chasing 100% costs enormous effort and blocks all change. The right target is the one where users are happy and the business can still innovate.

Running an error budget policy

1. **Agree SLOs with product owners** Reliability targets are business decisions, informed by user research and cost.
2. **Measure continuously** Compute SLIs from real traffic; alert on fast burn rates.
3. **Within budget** Teams release features freely and can take calculated risks.
4. **Budget nearly spent** Slow releases, add extra testing, prioritise reliability fixes.

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5. **Budget exhausted** Freeze non-critical feature releases until reliability recovers; agreed in advance, not negotiated in the moment.

Core SRE practices

SLO
Service level objectives
Measured, user-centred reliability targets.

EB
Error budgets
Permission to fail within limits, used to balance speed and reliability.

T
Toil reduction
Cap manual, repetitive, automatable work, for example at 50% of an engineer's time, and automate the rest.

PIR
Blameless post-incident reviews
Learn from failures without blame and track actions to completion.

OC
Sustainable on-call
Limited pages per shift, fair rotations, compensation and recovery time.

CP
Capacity and release engineering
Plan demand, test at load, release progressively.

What counts as toil?

CHARACTERISTIC	EXAMPLE
Manual	Logging into servers to restart a service by hand
Repetitive	Rotating the same certificates every month
Automatable	Creating user accounts from a spreadsheet
Tactical and reactive	Responding to the same disk-full alert weekly
No enduring value	The system is no better after the task is done
Grows with scale	Work increases linearly as users or servers increase

SCENARIO · DECIDE · SPENDING THE ERROR BUDGET

A payments API has a 99.95% monthly availability SLO. By day 12 of the month, two incidents have consumed 90% of the error budget. The product team wants to release a major new feature tomorrow. What does a sound error budget policy suggest?

- (A) Release as planned, because the feature has already been promised to customers and marketing
- (B) Raise the SLO to 99.99% so the team focuses harder on reliability
- (C) Pause the feature, fix reliability and release when the budget recovers
- (D) Lower the SLO to 99.5% for this month so the release can proceed

Circle your choice. Answer and feedback: module answer key.

CASE STUDY · INTRODUCING SLOS AT AN E-COMMERCE PLATFORM

A Dhaka-based e-commerce platform has frequent arguments between product managers, who want weekly feature releases before sale events, and the operations team, who want release freezes after every outage. There are no agreed reliability targets. During the last 11.11 sale, checkout failed for 9% of attempts for two hours, and both sides blamed the other.

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1. How would SLOs and error budgets change these conversations?

2. Which SLIs would you define for checkout?

3. What error budget policy would you propose for sale periods?

Model answer: module answer key.

TEMPLATE

SLO specification

Define a user-centred SLO for one service.

SERVICE AND USER JOURNEY	SLI DEFINITION	SLO TARGET AND WINDOW	ERROR BUDGET	ALERTING RULE	POLICY WHEN EXHAUSTED
Wallet – send money	*Successful transfers <3 s ÷ valid transfer requests*	*99.9% over 28 days*	*0.1% (~40 min equivalent)*	*Page if 2-hour burn rate >14*	*Freeze non-critical releases; reliability sprint*

Key terms

Site reliability engineering (SRE)
An approach that applies software engineering to operations, managing reliability with measurable objectives.

Service level objective (SLO)
A target value or range for an SLI over a defined time window.

Burn rate
How fast the error budget is being consumed relative to the rate allowed by the SLO.

Post-incident review
A structured, blameless review of an incident to understand contributing factors and agree improvements.

Service level indicator (SLI)
A carefully defined quantitative measure of some aspect of the service level users experience.

Error budget
The amount of unreliability allowed by an SLO: one minus the SLO target.

Toil
Manual, repetitive, automatable operational work that has no enduring value and grows with scale.

On-call
An arrangement in which engineers are available to respond to alerts outside normal working hours.

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REFLECT

If you set an SLO for your most important service tomorrow, what would the SLI be, and who would have to agree to the target?

KNOWLEDGE CHECK

11. A service has a 99.9% SLO over 30 days. Roughly how much full downtime does the error budget allow?

(A) About 7.2 hours
(B) About 4.3 minutes
(C) About 43 hours
(D) About 43 minutes

12. Why is 100% rarely the right reliability target?

(A) Regulators forbid 100% targets in published service agreements
(B) Users can't perceive it and it blocks all change
(C) Error budgets only work above 100%
(D) It is impossible to measure reliability

13. Which of these is toil?

(A) Running a blameless post-incident review
(B) Writing an automation script
(C) Restarting the same service by hand every week
(D) Designing a new architecture for the next generation of the service

14. What should normally happen when a service exhausts its error budget?

(A) Releases pause while reliability work comes first
(B) The SLO is lowered so releases can continue
(C) Monitoring is turned off to stop alerts
(D) The on-call engineer is disciplined for the incidents that used it up

15. How should an SLA relate to the internal SLO?

(A) The SLA is looser, giving a safety margin
(B) They must always be identical so customers see the real target
(C) The SLA replaces the SLO entirely
(D) The SLA must be stricter than the SLO

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MODULE 6 · FLOW · LESSON 4 · 40 MIN

Value Streams and Flow at Scale

Managing the flow of value across teams, platforms and portfolios

- Apply value stream management to digital products
- Use flow metrics to find constraints across teams
- Use platform engineering and the theory of constraints to improve flow

Once individual teams are fast, the constraint moves: to dependencies between teams, shared environments, security reviews, data access or funding decisions. **Value stream management** looks at the full flow of work from idea to customer outcome across all teams involved, and improves the system rather than the parts. In large organisations, this is where the biggest gains in speed and value now lie.

Flow metrics for digital value streams

FT
Flow time
Elapsed time from work starting to value delivered.

FV
Flow velocity
Number of work items completed per period.

FL
Flow load
Number of items in progress across the stream (work in progress).

FE
Flow efficiency
Share of flow time in which items are actively worked on.

FD
Flow distribution
Mix of work types completed: features, defects, risk work and technical debt.

EXAMPLE · BALANCING FLOW DISTRIBUTION

A fintech in Dhaka found that 85% of completed work was new features, 5% technical debt and 2% risk and security items. Incidents and audit findings were rising. Leadership set a target distribution of 60% features, 20% debt, 10% risk and 10% defects for six months; incidents fell by 40% and feature velocity recovered within two quarters.

Theory of constraints: five focusing steps

01 **Identify** — Find the constraint that limits the whole stream →
02 **Exploit** — Get the most out of it: no idle time, no low-value work →
03 **Subordinate** — Align everything else to the constraint's pace →
04 **Elevate** — Add capacity or redesign if still limiting →
05 **Repeat** — A new constraint appears; start again

Local optimisation versus flow optimisation

Local optimisation

- Each team maximises its own throughput
- Work piles up before the slowest step
- Utilisation looks high everywhere
- Customer lead time does not improve

Flow optimisation

- Teams align to the constraint
- Work in progress is limited across the stream
- Some idle time is accepted outside the constraint
- Customer lead time falls

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Platform engineering is one of the most effective ways to remove recurring constraints. A platform team builds an internal developer platform with self-service environments, pipelines, observability and security controls. Product teams use these 'paved roads' instead of raising tickets and waiting. The platform is managed as a product, with internal customers, a roadmap and service levels.

Signs your value stream needs attention

- Teams are busy but customer lead time is not falling.
- Work waits weeks for shared environments, data or security sign-off.
- Most items cross three or more teams before release.
- Technical debt and risk work never appear in the flow distribution.
- Nobody can say how long an idea takes to reach customers.

SCENARIO · DECIDE · FINDING THE CONSTRAINT

In a telecom's digital channel, product teams deliver features in two weeks, but releases reach customers on average 9 weeks later. Mapping shows features waiting 5 weeks for a shared test environment managed by a central team with a ticket queue. What is the best first action?

(A) Ask product teams to deliver features in one week instead of two so that they reach the queue sooner
(B) Treat the test environment as the constraint and move toward self-service environments
(C) Add more product teams to increase output
(D) Introduce a weekly cross-team status meeting

Circle your choice. Answer and feedback: module answer key.

TEMPLATE **Flow metrics dashboard**

Track the flow of one value stream monthly.

MONTH	FLOW VELOCITY	FLOW TIME (MEDIAN DAYS)	FLOW LOAD (WIP)	FLOW EFFICIENCY %	FLOW DISTRIBUTION (F/D/R/DEBT %)
March	*42 items*	*37*	*118*	*11%*	*78 / 12 / 4 / 6*

Key terms

Value stream management
Managing and improving the end-to-end flow of value from idea to customer outcome across all teams involved.

Flow time
Elapsed time from when work starts on an item until value is delivered.

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Flow load
The number of work items in progress in a value stream.

Flow distribution
The proportion of completed work across types such as features, defects, risk and debt.

Constraint
The step that limits the throughput of the whole system.

Theory of constraints
A management approach that improves a system by identifying and managing its most limiting factor.

Platform engineering
Building and running internal self-service platforms that reduce friction for product teams.

Paved road
A supported, standard path that makes the right way of working the easiest way.

REFLECT

Where is the constraint in the value stream you know best? What evidence tells you it is the constraint rather than just a busy team?

KNOWLEDGE CHECK

16. According to the theory of constraints, what should happen after identifying the constraint?
    - (A) Exploit it: get the most from its capacity
    - (B) Ignore it until the next annual review
    - (C) Speed up all other steps
    - (D) Immediately buy more capacity everywhere

17. Which flow metric shows the mix of features, defects, risk and debt completed?
    - (A) Flow velocity
    - (B) Flow load
    - (C) Flow efficiency of the stream
    - (D) Flow distribution

18. Product teams are fast, but releases wait weeks for a shared environment. Why will faster product teams not help?
    - (A) The environment team will slow down further
    - (B) They are not the constraint, so work just queues
    - (C) Faster teams always produce more defects
    - (D) Product teams do not affect lead time in any measurable way

19. How should an internal developer platform be managed?
    - (A) As an optional tool nobody owns
    - (B) As a product with internal customers and a roadmap
    - (C) As a set of tickets for each team
    - (D) As a one-off infrastructure project with a fixed end date

20. What is a sign of local optimisation?
    - (A) Measuring end-to-end flow time for each item in the stream
    - (B) Busy teams everywhere, but no faster lead time
    - (C) Limiting work in progress across the stream
    - (D) Teams aligning their pace to the constraint

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MODULE 6 · FLOW

Answer key

Knowledge checks

1. C Small, frequent, automated changes improve speed and stability at the same time.

2. D Lead time for changes covers commit to production.

3. A Assured conformance covers governance, risk and compliance.

4. D Large batches carry more risk and are harder to diagnose.

5. C Automated, built-in controls let teams move quickly without bypassing safety.

6. A Continuous delivery keeps changes releasable; continuous deployment releases them automatically.

7. B Tools alone automate existing dysfunction; culture and structure must change too.

8. B IaC manages infrastructure through versioned code.

9. A Automatic creation from pipeline data is accurate and removes manual effort.

10. C CALMS: culture, automation, lean, measurement, sharing.

11. D 0.1% of 43,200 minutes ≈ 43 minutes.

12. B Beyond a point, extra reliability is invisible to users and very expensive.

13. C Manual, repetitive, automatable work with no enduring value is toil.

14. A A pre-agreed policy shifts effort to reliability until the budget recovers.

15. A Internal SLOs are tighter so the team acts before the external promise is breached.

16. A Exploit first, subordinate everything else, then elevate if needed.

17. D Flow distribution shows the balance of work types.

18. B Improving non-constraints does not change system throughput.

19. B Platforms succeed when treated as products serving internal customers.

20. B Busy teams with no improvement in customer outcomes indicate local optimisation.

Scenarios & case studies

Scenario · Assessing delivery performance
D is best. Correct. Smaller, automated changes improve all four metrics together; moving to weekly releases of the same large batches would increase failures.

Scenario · A DevOps transformation stalling
D is best. Correct. Tools cannot overcome conflicting goals and handovers. Shared ownership and blameless learning are the foundation.

Scenario · Spending the error budget
C is best. Correct. With 90% of the budget gone early, the policy favours reliability work. The decision follows a rule agreed in advance, not a debate.

Case study · Introducing SLOs at an e-commerce platform
SLOs replace opinion with an agreed, measurable definition of 'reliable enough', and the error budget turns the speed-versus-stability argument into a shared rule: release freely while budget remains, focus on reliability when it is spent. For checkout, useful SLIs include the proportion of checkout attempts that complete successfully, and the proportion completing within, for example, three seconds, both measured from real traffic. A sound policy for sale periods is to set a stricter internal SLO for the sale window, require that the budget be healthy in the two weeks before a sale for risky releases to go ahead, freeze non-critical changes during the sale itself, and agree all this with product and operations leaders in advance so that nobody negotiates under pressure.

Scenario · Finding the constraint
B is best. Correct. Improving any other step will not reduce lead time. Exploit the constraint first, then elevate it with self-service or on-demand environments.

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07

MODULE 7 · DIRECT

Direction, Planning, Governance and Digital Strategy

Set direction, govern risk, measure what matters and shape the digital future

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Cascade strategy into outcome-based objectives and a balanced portfolio
- Govern risk and compliance with proportionate, embedded controls
- Design measurement systems that guide decisions and resist gaming
- Select improvement methods and assess capability with evidence
- Build and fund a digital and IT strategy that serves the business

LESSONS

1. Direction, Strategy and Portfolio · 45 min
2. Governance, Risk and Compliance · 45 min
3. Measurement, Reporting and Decisions · 40 min
4. Continual Improvement Methods and Assessment · 40 min
5. Digital and IT Strategy · 50 min

MODULE 7 · DIRECT · LESSON 1 · 45 MIN

Direction, Strategy and Portfolio

Turning a vision into objectives, priorities and investment decisions

- Cascade vision and strategy into measurable objectives for service teams
- Use portfolio management to choose and balance investments
- Plan across strategic, tactical and operational horizons

Direction answers the questions 'where are we going, and why?'. Without it, service organisations drift: teams optimise their own priorities, budgets follow last year's pattern, and improvement efforts pull in different directions. Direction is set through vision, strategy and objectives, translated into a portfolio of investments and then into plans that teams can act on.

From vision to daily work

Vision and mission	`Where the organisation wants to be and why it exists`
Strategy	`The chosen approach to get there: where to play and how to win`
Objectives and key results	`Measurable outcomes for the next period`
Portfolio	`The products, services and initiatives funded to achieve them`
Plans and backlogs	`Tactical and operational work in teams`

Objectives and key results (OKRs)

O

Objective

A qualitative, inspiring statement of what you want to achieve. Example: 'Make digital admission effortless for every applicant'.

KR

Key results

Two to five measurable outcomes showing progress. Example: 'Online applications completed without help rise from 55% to 85%'.

I

Initiatives

The work expected to move the key results. Kept separate, because initiatives can change while the outcome stays the same.

PRO TIP · OUTCOMES, NOT OUTPUTS, IN KEY RESULTS

'Launch the new admission portal' is an output; it can be completed while nothing improves. 'Reduce application drop-out from 40% to 15%' is an outcome. Key results should describe changes in behaviour or results that matter to stakeholders.

Planning horizons

HORIZON	TYPICAL SPAN	QUESTIONS	EXAMPLE ARTEFACTS
Strategic	2–5 years	Which services and capabilities will matter? What do we stop?	Digital strategy, target operating model, capability roadmap
Tactical	Quarters to a year	Which initiatives this year? How is capacity allocated?	Portfolio plan, OKRs, budget, sourcing plan

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HORIZON	TYPICAL SPAN	QUESTIONS	EXAMPLE ARTEFACTS
Operational	Days to weeks	What do teams do next? How do we handle demand?	Backlogs, schedules, rotas, change calendar

Portfolio management cycle

01 Collect demand
Ideas, requests, regulatory needs, risks

→

02 Evaluate
Value, cost, risk, strategic fit

→

03 Prioritise and fund
Choose and balance the portfolio

→

04 Execute and monitor
Track outcomes and spending

→

05 Rebalance
Stop, continue or scale based on evidence

Balancing the portfolio

Run

- Keep current services operating
- Licences, operations, maintenance
- Often 60–70% of IT spend

Grow

- Improve and extend existing services
- New features, more users, better performance

Transform

- New business models or capabilities
- Higher risk and uncertainty; funded in stages

SCENARIO · DECIDE · TOO MANY PRIORITIES

A university IT department has 47 'top priority' projects approved by different deans. Most are half-finished and staff are exhausted. The Vice-Chancellor asks you to restore control. What do you do first?

(A) Ask each dean to reduce their projects by 20% so that the total load across the university falls fairly

(B) Hire contractors to deliver all 47 projects

(C) Continue all projects and report progress more often

(D) Set up one portfolio process that ranks projects against strategy and limits parallel work

Circle your choice. Answer and feedback: module answer key.

TEMPLATE **OKR worksheet**

Translate a strategic objective into measurable key results for a service team.

OBJECTIVE	KEY RESULT	BASELINE	TARGET	INITIATIVES	OWNER
Make digital admission effortless	*Applications completed without help*	*55%*	*85% by next intake*	*Bangla-first forms; document upload redesign*	*Head of admissions systems*

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OBJECTIVE	KEY RESULT	BASELINE	TARGET	INITIATIVES	OWNER

Key terms

Direction
Setting where an organisation is going and why, through vision, strategy and objectives.

Objectives and key results (OKRs)
A goal-setting method combining a qualitative objective with measurable key results.

Run–grow–transform
A way of classifying IT investment by whether it maintains, extends or fundamentally changes the business.

Output
A deliverable produced by work, as opposed to the outcome it is meant to achieve.

Strategy
A chosen approach for achieving long-term objectives: where to compete and how to win.

Portfolio management
The practice of selecting, balancing and governing investments in products, services and initiatives.

Planning horizon
The time span a plan covers: strategic, tactical or operational.

REFLECT

How many 'top priorities' does your organisation have right now? Who decides which one wins when they conflict?

KNOWLEDGE CHECK

1. Which key result is written as an outcome rather than an output?
   - (A) Hold five training workshops
   - (B) Reduce application drop-out from 40% to 15%
   - (C) Launch the new admission portal
   - (D) Buy a new content management system for the portal

2. Which planning horizon covers decisions about which services to stop over the next three years?
   - (A) Strategic
   - (B) Tactical
   - (C) Operational
   - (D) Daily stand-up planning within each team

3. What is the most effective response when a portfolio has far more priorities than capacity?
   - (A) Cut every project by the same percentage
   - (B) Ask all teams to work longer hours
   - (C) Rank work against strategy and limit WIP
   - (D) Start more projects so that something finishes

4. In a run–grow–transform portfolio, which category usually consumes the largest share of IT spend?
   - (A) Innovation experiments and pilots
   - (B) Grow
   - (C) Transform
   - (D) Run

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MODULE 7 · DIRECT · LESSON 2 · 45 MIN

Governance, Risk and Compliance

Directing and controlling services within risk appetite and the law

- Apply a risk management cycle to services and practices
- Design controls that are effective and proportionate
- Use the three lines model and automated compliance in service management

Every service carries risk: outages, data breaches, supplier failure, regulatory penalties, safety incidents. **Governance, risk and compliance (GRC)** ensures that the organisation knows its risks, takes them deliberately within its appetite, and can prove that it meets its legal and contractual obligations. For a service manager, GRC is not paperwork; it is the discipline that lets the organisation move fast without stumbling into avoidable harm.

Risk management cycle

01 Identify — What could go wrong, or go right?

→

02 Assess — Likelihood and impact, before and after controls

→

03 Treat — Avoid, reduce, transfer, accept, or exploit opportunities

→

04 Monitor — Track risk indicators and control effectiveness

→

05 Report — Inform governance and adjust appetite or treatment

Risk calculations

MEASURE	FORMULA	WHAT IT TELLS YOU
Risk exposure	`Exposure = likelihood × impact`	A simple ranking device; use consistent scales (e.g. 1–5 each).
Annualised loss expectancy	`ALE = single loss expectancy × annual rate of occurrence`	Example: a 2-hour outage costing Tk 50 lakh, expected twice a year, gives an ALE of Tk 1 crore.
Residual risk	`Residual risk = inherent risk − effect of controls`	The risk that remains after treatment; must be within appetite.

Risk treatment options

OPTION	MEANING	SERVICE EXAMPLE
Avoid	Stop the activity that creates the risk	Do not store card data at all; use a tokenisation provider
Reduce	Lower likelihood or impact with controls	Add a second data centre and test failover
Transfer / share	Move financial impact to another party	Cyber insurance; supplier indemnity clauses
Accept	Knowingly tolerate within appetite	Accept occasional slowness on a low-priority internal tool

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The three lines model

1

First line: management

Service owners and teams own and manage risks in daily operations, with controls built into their work.

2

Second line: risk and compliance

Specialist functions set frameworks, advise, challenge and monitor the first line.

3

Third line: internal audit

Independent assurance to the governing body that risk management and controls work.

INSIGHT · CONTROLS SHOULD BE DESIGNED INTO THE WORK

A control that depends on someone remembering to fill in a form will fail on the busiest day. Strong controls are automated, preventive and embedded: access granted by role, segregation of duties enforced by the tool, configuration checked by policy-as-code, evidence captured automatically.

Common compliance obligations for services in South Asia and beyond

- Data protection and privacy laws, including data localisation requirements.
- Central bank IT and cyber-security guidelines for banks and financial services.
- Telecom regulator rules on lawful interception, uptime and customer data.
- Payment card industry security standards for card data.
- Health-data confidentiality rules for hospitals and clinics.
- Contractual obligations in customer and supplier agreements.

SCENARIO · DECIDE · A CONTROL THAT NOBODY FOLLOWS

A bank's policy requires two people to approve every production database change. An audit finds that 60% of changes have the second approval added days after implementation. What is the best response?

(A) Enforce the second approval in the deployment tooling itself

(B) Send a reminder to all engineers about the policy and ask managers to check compliance monthly

(C) Remove the policy because nobody follows it

(D) Discipline the engineers involved

Circle your choice. Answer and feedback: module answer key.

TEMPLATE

Service risk register

Record and track risks for a service.

RISK	LIKELIHOOD (1–5)	IMPACT (1–5)	EXPOSURE	TREATMENT AND CONTROLS	OWNER	RESIDUAL WITHIN APPETITE?
Single SMS gateway fails during peak	*3*	*5*	*15*	*Reduce: second gateway with automatic failover; quarterly test*	*Head of digital channels*	*Yes after failover tested*

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RISK	LIKELIHOOD (1–5)	IMPACT (1–5)	EXPOSURE	TREATMENT AND CONTROLS	OWNER	RESIDUAL WITHIN APPETITE?

Key terms

Governance, risk and compliance (GRC)
An integrated approach to directing an organisation, managing its risks and meeting its obligations.

Inherent risk
The level of risk before any controls are applied.

Control
A means of managing risk by preventing, detecting or correcting undesired events.

Segregation of duties
Separating responsibilities so no single person can complete a sensitive action alone.

Risk
A possible event that could cause harm or loss, or make it harder to achieve objectives; also used for opportunities.

Residual risk
The level of risk remaining after controls and treatment.

Three lines model
A model distinguishing management (first line), risk and compliance functions (second line) and independent audit (third line).

Annualised loss expectancy (ALE)
Expected yearly loss from a risk: single loss expectancy multiplied by annual rate of occurrence.

REFLECT

Which control in your area depends most on human memory? How could it be built into the tools people already use?

KNOWLEDGE CHECK

5. Buying cyber insurance for a data breach is which treatment option?

(A) Transfer
(B) Avoid
(C) Accept
(D) Reduce

6. Which line provides independent assurance to the governing body?

(A) External suppliers delivering the service
(B) First line: service owners
(C) Second line: risk function
(D) Third line: internal audit

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7. An outage costs Tk 20 lakh each time and is expected three times a year. What is the annualised loss expectancy?

   (A) Tk 6 lakh
   (B) Tk 60 lakh
   (C) Tk 20 lakh
   (D) Tk 23 lakh

8. What makes a control most reliable?

   (A) Being written in a long policy
   (B) Being automated and preventive
   (C) Depending on staff memory
   (D) Being checked once a year

9. Who owns and manages risk day to day in the three lines model?

   (A) The second-line risk function alone
   (B) First-line managers and teams
   (C) The external regulator
   (D) Internal audit

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MODULE 7 · DIRECT · LESSON 3 · 40 MIN

Measurement, Reporting and Decisions

Measuring what matters and using it to steer, not to punish

- Derive meaningful metrics from goals using a structured method
- Design balanced scorecards and dashboards for different audiences
- Avoid common measurement traps such as gaming and vanity metrics

Measurement exists to support decisions. A metric that does not change any decision is noise. A metric used to punish teams will be gamed. Mature service organisations derive measures from goals, balance them so that no single number can be optimised at the expense of the others, and present them so each audience sees what it needs to act on.

Goal–question–metric

01 Goal
Reduce time patients lose due to IT problems

→

02 Questions
How often do clinicians lose time? How long? Where?

→

03 Metrics
Incidents affecting clinical workflows; minutes lost per shift; top 5 causes

→

04 Decisions
Which improvements to fund next quarter

CSFs and KPIs

CRITICAL SUCCESS FACTOR	KEY PERFORMANCE INDICATOR	TARGET
Services restored quickly when they fail	Median time to restore P1/P2 incidents	< 2 hours
Changes delivered safely	Change failure rate	< 10%
Users can help themselves	Requests resolved through self-service	> 40%
Customers value the service	Customer effort score for key journeys	≤ 2 on a 1–5 scale (lower is better)

A balanced service scorecard

Cu — Customer and user
Experience, satisfaction, effort, adoption.

Fi — Financial
Cost per service, cost per transaction, value delivered, budget adherence.

Op — Operational
Availability, restore time, change success, flow time.

Pe — People and learning
Engagement, skills coverage, attrition, on-call load.

Ri — Risk and compliance
Audit findings, open vulnerabilities, control effectiveness.

WATCH OUT · GOODHART'S LAW IN SERVICE MANAGEMENT

When a measure becomes a target, it stops being a good measure. Desks targeted on closure time close tickets early. Teams targeted on deployment count split changes artificially. Pair measures (speed with quality), review them together and use them to learn, not to rank people.

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Leading versus lagging indicators

Lagging

- Show results after the fact
- Availability last month, customer churn, audit findings
- Confirm whether goals were met

Leading

- Predict future results
- Rising error budget burn, growing backlog age, falling knowledge link rate
- Allow action before goals are missed

Designing dashboards people actually use

- One audience and one purpose per dashboard: executives, service owners, teams.
- Lead with outcomes; let users drill into operational detail.
- Show trends and targets, not just current values.
- Use consistent definitions across all reports.
- Fewer than ten measures on the main view.
- State data freshness and known data-quality issues.

SCENARIO · DECIDE · A GAMED METRIC

A telecom's service desk has improved 'tickets resolved within 4 hours' from 70% to 96% in three months. Complaints to the CEO about IT support have doubled. What is the most likely explanation and best action?

(A) Complaints are unrelated to the desk, since its metric has improved dramatically over the quarter

(B) Raise the target to 99% within 4 hours

(C) Stop measuring the desk entirely

(D) The target is being gamed; pair it with reopen rate and user effort and review them together

Circle your choice. Answer and feedback: module answer key.

TEMPLATE **Metric definition card**

Define each measure so it is consistent and trusted.

METRIC	PURPOSE / DECISION SUPPORTED	FORMULA	DATA SOURCE	TARGET	PAIRED MEASURE	OWNER
Median time to restore P1/P2	*Fund reliability improvements*	*Median(restore time − start time)*	*Incident tool + monitoring*	*< 2 h*	*Repeat incident rate*	*Head of operations*

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METRIC	PURPOSE / DECISION SUPPORTED	FORMULA	DATA SOURCE	TARGET	PAIRED MEASURE	OWNER

Key terms

Key performance indicator (KPI)
A metric used to evaluate success in meeting an objective.

Goal–question–metric
A method of deriving metrics by starting from goals and the questions that test them.

Goodhart's law
The observation that a measure used as a target tends to stop being a good measure.

Lagging indicator
A measure that reports results after they have occurred.

Critical success factor (CSF)
A condition that must be met for an objective to be achieved.

Balanced scorecard
A set of measures across several perspectives so that no single dimension dominates.

Leading indicator
A measure that predicts future performance.

Vanity metric
A number that looks impressive but does not inform decisions.

REFLECT

Name one metric your organisation reports that changes no decisions. What would you replace it with?

KNOWLEDGE CHECK

10. Which of these is a leading indicator?

(A) Annual customer churn
(B) Last quarter's availability
(C) Number of audit findings in the most recent annual audit
(D) A rising error budget burn rate

11. A desk's closure-time metric improves sharply while complaints rise. What principle explains this?

(A) The theory of constraints
(B) Little's law
(C) The law of diminishing returns
(D) Goodhart's law

12. What is the first step in the goal–question–metric method?

(A) Choose a dashboard tool
(B) List available data
(C) Set targets for each existing metric
(D) Define the goal

13. Why should measures be paired?

(A) So one cannot improve at another's expense
(B) Because single measures cannot be calculated automatically
(C) To double the number of reports
(D) To satisfy auditors

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14. Which dashboard practice is most helpful for executives?

(A) Show only technical component data
(B) Show every metric collected
(C) Lead with outcomes and trends
(D) Update once a year

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MODULE 7 · DIRECT · LESSON 4 · 40 MIN

Continual Improvement Methods and Assessment

Choosing the right improvement method and knowing where you stand

- Select improvement methods suited to different kinds of problems
- Assess practice capability and organisational maturity with evidence
- Build an improvement roadmap that balances quick wins with structural change

Continual improvement is a mindset supported by methods. No single method fits every problem: a recurring printer fault needs a different approach from a failing outsourcing contract or a slow digital product pipeline. Masters choose deliberately, combine methods where useful, and ground improvement in an honest assessment of where the organisation stands.

Improvement methods and when to use them

METHOD	CORE IDEA	BEST FOR
PDCA (plan–do–check–act)	Small, controlled cycles of change and learning	Any incremental improvement
Lean	Remove waste and improve flow from the customer's view	Slow, handover-heavy value streams
Six Sigma / DMAIC	Reduce variation and defects with statistical analysis	High-volume processes with measurable defects
A3 problem solving	One-page structured reasoning from problem to countermeasure	Coaching teams through complex problems
Kaizen events	Short, focused workshops to redesign a process	Contained processes with willing teams
Experimentation / A-B testing	Test hypotheses with real users	Digital services and uncertain user behaviour

PDCA cycle

01 Plan — Define the problem, hypothesis and measure

→

02 Do — Run a small, controlled change

→

03 Check — Compare results with the expected outcome

→

04 Act — Standardise, adapt or abandon; plan the next cycle

Lean wastes in IT services

W — Waiting
Requests queued for approval, environments or specialists.

H — Handovers
Work passed between teams, losing context each time.

R — Rework
Reopened tickets, failed changes, defects found late.

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O

Over-processing

Approvals and documentation that add no value.

I

Inventory

Large backlogs, unfinished work, unused licences.

M

Motion and switching

Engineers juggling many tasks and tools.

T

Unused talent

Skilled people restricted to narrow, repetitive work.

Capability versus maturity

Practice capability

- How well a specific practice performs
- Example: incident management at level 3 of 5
- Assessed with practice-specific criteria and evidence

Organisational maturity

- How well the whole system works together
- Governance, principles, value chain, practices and improvement
- Assessed holistically; slower to change

Running an evidence-based assessment

1. **Define scope and purpose** Which practices or services, and which decisions will the assessment inform?
2. **Choose a model** Use a consistent capability scale with clear criteria.
3. **Collect evidence** Interviews, document review, tool data, observation of real work.
4. **Rate and validate** Rate against criteria; validate with the people who do the work.
5. **Prioritise gaps** Focus on gaps that most limit value, not on raising every score.
6. **Repeat** Reassess after 6–12 months to confirm progress.

WATCH OUT · MATURITY SCORES AS GOALS

Organisations that set 'reach level 4 in all practices' as a goal often spend years on documentation that auditors like and users never feel. Raise capability where it limits value; a level 2 practice that rarely matters may be perfectly adequate.

SCENARIO · DECIDE · CHOOSING A METHOD

A garment exporter's IT team handles purchase-order integration with 40 buyers. 8% of orders fail due to data mapping errors that vary by buyer and season; failures are measurable and frequent. Which improvement method fits best?

- (A) A one-day kaizen workshop with the integration team to redesign the whole process from scratch
- (B) A/B testing of different user interfaces
- (C) A maturity assessment of all IT practices
- (D) A data-driven defect reduction method such as DMAIC

Circle your choice. Answer and feedback: module answer key.

TEMPLATE **Improvement roadmap**

Plan improvements across quick wins and structural changes.

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IMPROVEMENT	TYPE (QUICK WIN / STRUCTURAL)	METHOD	EXPECTED BENEFIT	START	OWNER
Self-service password reset	*Quick win*	*PDCA pilot with one department*	*–20% desk contacts*	*Month 1*	*Desk lead*
Product-aligned support teams	*Structural*	*Value stream redesign*	*–40% handovers*	*Month 3*	*Head of operations*

Key terms

PDCA
Plan–do–check–act: an iterative cycle for controlled improvement and learning.

DMAIC
Define, measure, analyse, improve, control: a data-driven method for reducing defects and variation.

Kaizen
Continuous, incremental improvement, often through focused team workshops.

Maturity assessment
An evidence-based evaluation of how well an organisation's management system works as a whole.

Lean
An approach that maximises customer value by removing waste and improving flow.

A3 problem solving
A structured problem-solving approach captured on a single sheet of paper.

Capability level
A rating of how well a practice performs against defined criteria.

REFLECT

Which improvement method does your organisation use by default? Name one problem it was the wrong choice for.

KNOWLEDGE CHECK

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15. Which method is best for reducing variation in a high-volume, measurable process?

(A) DMAIC
(B) A/B testing
(C) Kaizen workshop
(D) Maturity assessment

16. Reopened tickets and failed changes are which lean waste?

(A) Waiting
(B) Unused talent
(C) Inventory
(D) Rework

17. Why is 'reach level 4 in every practice' usually a poor goal?

(A) Level 4 is impossible to reach
(B) Higher levels reduce service quality
(C) It favours documentation over value
(D) Assessments are always inaccurate

18. What is the 'check' step in PDCA?

(A) Standardising the new process across the organisation
(B) Approving the plan
(C) Running the change
(D) Comparing results with the expected outcome

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MODULE 7 · DIRECT · LESSON 5 · 50 MIN

Digital and IT Strategy

Positioning the organisation for a digital future and funding the journey

- Analyse how digital technology changes markets and business models
- Choose a strategic position and align IT strategy to it
- Build a financially justified digital roadmap

A **digital strategy** describes how an organisation will use digital technology to achieve its goals and respond to how technology is changing its markets. An **IT strategy** describes how the technology function will enable that: which capabilities, platforms, sourcing and skills it needs. In digital organisations the two converge. A master service leader can explain how a service roadmap advances the organisation's strategy in board language.

Levels of digital ambition

1
Digitise
Convert existing processes to digital form: online forms, e-invoices, digital records.

2
Optimise
Use data and automation to improve existing operations: predictive maintenance, smart routing.

3
Transform
Change customer journeys and products significantly: app-only banking, telemedicine.

4
Disrupt
Create new business models or markets: platform ecosystems, embedded finance, data-driven services.

Three classic strategic positions

Operational excellence

- Lowest cost, highest reliability
- Standardised, automated services
- Example: a utility billing platform serving millions

Customer intimacy

- Tailored solutions and deep relationships
- Rich customer data and flexible services
- Example: a corporate bank serving large exporters

Product leadership

- Most innovative products
- Rapid experimentation and high-velocity delivery
- Example: a fintech launching new features weekly

EXAMPLE · DIGITAL STRATEGY IN A POWER DISTRIBUTION COMPANY

A power distribution company in Bangladesh set a strategy of 'reliable power, effortless billing'. Its digital roadmap combined smart prepaid meters (optimise), a mobile app for top-up and outage reports (transform), and analytics for loss detection (optimise). IT strategy followed: a shared data platform, partnerships with mobile financial services for payment, and a 24/7 digital operations centre.

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Strategy analysis tools

TOOL	QUESTION IT ANSWERS	EXAMPLE USE
PESTLE	Which external forces shape our future?	Data localisation law affects cloud choices
SWOT	What are our strengths, weaknesses, opportunities and threats?	Strong branch network vs weak digital skills
Five forces	How intense is competition in our market?	New digital banks lowering switching costs
Capability map	Which capabilities do we need and how mature are they?	Data analytics capability rated low
Business model canvas	How do we create, deliver and capture value?	Shift from product sales to subscriptions

Investment appraisal

MEASURE	FORMULA	WHAT IT TELLS YOU
Return on investment	`ROI = (total benefits − total costs) ÷ total costs × 100%`	Simple profitability measure; state the period and include all costs.
Payback period	`Payback = initial investment ÷ annual net benefit`	How quickly money is recovered; ignores benefits after payback.
Net present value	`NPV = Σ (net cash flow in year t ÷ (1 + r)^t) − initial investment`	Value of future benefits in today's money at discount rate r; positive NPV adds value.

Building a digital roadmap

1. **Clarify ambition** Agree the strategic position and level of digital ambition with leadership.
2. **Assess capabilities** Map current capabilities across the four dimensions.
3. **Identify initiatives** Generate options that close capability gaps and serve the strategy.
4. **Appraise and sequence** Evaluate value, cost and risk; sequence by dependencies and quick wins.
5. **Fund in stages** Fund transformation initiatives in tranches tied to evidence of value.
6. **Govern and adapt** Review the roadmap quarterly; stop, pivot or scale initiatives.

Common digital strategy mistakes

- Copying a competitor's technology without a strategic reason.
- Treating digital as an IT project rather than a business change.
- Funding everything for three years upfront with no checkpoints.
- Ignoring legacy constraints and technical debt in the plan.
- Underinvesting in data, skills and change management.

CASE STUDY · A HOSPITAL GROUP'S DIGITAL STRATEGY

A private hospital group with five sites in Dhaka and Chattogram wants to grow outpatient revenue by 30% in three years. It faces new competition from telemedicine start-ups. Its systems are fragmented: each site has its

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own patient-record system, there is no group-wide patient ID, and online booking exists only at one site. The board has approved a budget of Tk 40 crore over three years.

1. Which strategic position should the group pursue, and why?

2. Which capabilities and initiatives would you prioritise in year one?

3. How would you structure funding and governance for the roadmap?

Model answer: module answer key.

SCENARIO · DECIDE · FUNDING A TRANSFORMATION

A bank's board is asked to approve Tk 150 crore for a five-year core platform transformation, with all funds committed upfront. Benefits are uncertain and depend on customer adoption of new digital products. What should the CIO propose?

- (A) Commit the full amount upfront to signal confidence to staff, customers, suppliers and the regulator
- (B) Reduce scope to only what can be funded from this year's operating budget
- (C) Stage the funding into tranches, each released on evidence of delivered value and learning
- (D) Outsource the transformation to remove the financial risk

Circle your choice. Answer and feedback: module answer key.

TEMPLATE **Digital initiative business case summary**

Summarise an initiative for portfolio decisions.

INITIATIVE	STRATEGIC OBJECTIVE SERVED	BENEFITS (ANNUAL)	COSTS (TOTAL)	ROI / NPV	KEY RISKS	FUNDING STAGE
Group-wide online booking	*Grow outpatient revenue 30%*	*Tk 6 crore extra revenue*	*Tk 4 crore over 2 years*	*ROI ≈ 200% over 3 yrs*	*Clinician adoption; integration with 5 systems*	*Tranche 1: Tk 1.5 crore*

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Key terms

Digital strategy
How an organisation will use digital technology to achieve its goals and respond to digital change in its markets.

Digital transformation
Significant change to customer journeys, products, operations or business models enabled by digital technology.

Customer intimacy
A strategic position based on tailored solutions and deep customer relationships.

Net present value (NPV)
The value today of future net cash flows, discounted at a chosen rate, minus the initial investment.

IT strategy
How the technology function will provide the capabilities, platforms, sourcing and skills the organisation needs.

Operational excellence
A strategic position based on low cost and high reliability through standardisation and efficiency.

Product leadership
A strategic position based on offering the most innovative products.

Staged funding
Releasing investment in tranches tied to evidence of progress and value.

REFLECT

Which strategic position does your organisation actually pursue, judged by where it invests, not by what it says?

KNOWLEDGE CHECK

19. An investment of Tk 10 crore produces total benefits of Tk 15 crore over its life. What is the ROI?
    - (A) 5%
    - (B) 50%
    - (C) 150%
    - (D) 15%

20. A fintech that releases innovative features weekly is pursuing which position?
    - (A) Cost leadership through standardised services
    - (B) Product leadership
    - (C) Operational excellence
    - (D) Customer intimacy

21. Which level of digital ambition creates new business models or markets?
    - (A) Optimise
    - (B) Disrupt
    - (C) Transform
    - (D) Digitise

22. Why is staged funding preferred for uncertain transformations?
    - (A) Boards prefer smaller numbers
    - (B) It always lowers total cost
    - (C) It removes the need for a business case
    - (D) It keeps evidence-based checkpoints

23. What is a common digital strategy mistake?
    - (A) Treating digital as just an IT project
    - (B) Linking initiatives to strategic objectives
    - (C) Funding in stages
    - (D) Assessing current capabilities

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MODULE 7 · DIRECT

Answer key

Knowledge checks

1. B Drop-out rate is a change in results; the others are deliverables.

2. A Multi-year choices about the service portfolio are strategic.

3. C Ranking plus limits creates focus and lets valuable work finish.

4. D Keeping current services running typically consumes most of the budget.

5. A Insurance transfers the financial impact to another party.

6. D Internal audit is independent of management and reports to the governing body.

7. B ALE = 20 lakh × 3 = 60 lakh.

8. B Automated, preventive controls work even under pressure.

9. B The first line owns and manages risk in operations.

10. D Burn rate predicts future SLO breaches; the others report past results.

11. D A measure used as a target has been gamed.

12. D Metrics should be derived from goals, not from whatever data is available.

13. A Pairing speed with quality, for example, exposes trade-offs and gaming.

14. C Executives need outcomes and trends with the option to see detail.

15. A DMAIC uses statistical analysis to reduce defects and variation.

16. D Doing work again because it was not right the first time is rework.

17. C Capability should rise where it limits value, not uniformly.

18. D Check compares actual results with the hypothesis.

19. B ROI = (15 − 10) ÷ 10 = 50%.

20. B Rapid innovation is the hallmark of product leadership.

21. B Disruption creates new models or markets; transform changes existing journeys significantly.

22. D Tranches tied to evidence manage uncertainty and preserve options.

23. A Digital change succeeds only when the business owns it.

Scenarios & case studies

Scenario · Too many priorities

D is best. Correct. One transparent portfolio with limits on parallel work restores focus and lets the most valuable projects finish.

Scenario · A control that nobody follows

A is best. Correct. Embedding the control in the tool makes it preventive and reliable. Adjust the approval level to risk so the control remains proportionate.

Scenario · A gamed metric

D is best. Correct. A sharp improvement alongside rising complaints suggests early closure or reclassification. Paired measures make gaming visible and restore the link to outcomes.

Scenario · Choosing a method

D is best. Correct. High volume, measurable defects and variation by buyer are what statistical defect reduction is designed for.

Case study · A hospital group's digital strategy

A **customer intimacy** position fits best: the group cannot out-innovate start-ups on pure technology, but it can offer continuity of care across sites, specialists and follow-up. Year one should build the foundations: a group-wide patient ID and integrated record view, a single online booking and payment journey for all sites, and teleconsultation for follow-ups to defend against start-ups. Data quality and clinician adoption must be in scope. Funding should be staged, for example Tk 12 crore in year one with release of later tranches on evidence such as online booking share and repeat-visit rate. A digital steering group with clinical and finance leaders should review the roadmap quarterly against OKRs.

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Scenario · Funding a transformation

C is best. Correct. Staged funding manages uncertainty, keeps the board in control and allows the programme to pivot or stop.

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08

MODULE 8 · LEAD

AI, Automation and Leading Service Transformation

Apply AI responsibly, analyse complex situations and lead lasting change

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Select and justify AI and automation use cases across service management
- Govern AI with proportionate oversight, fairness checks and evaluation
- Analyse complex situations and justify recommendations with transparent criteria
- Lead adoption so that new ways of working stick and benefits are realised
- Integrate the whole ZITIL Service Value Loop in real, messy cases

LESSONS

1. AI and Automation in Service Management · 45 min
2. Responsible AI in Service Operations · 45 min
3. Analysing Complex Situations · 45 min
4. Leading Adoption and Organisational Change · 45 min
5. Master Practice: Integrated Case and Exam Approach · 60 min

MODULE 8 · LEAD · LESSON 1 · 45 MIN

AI and Automation in Service Management

Where intelligent automation creates value across the service lifecycle

- Identify high-value AI and automation use cases across practices
- Distinguish types of automation and choose the right one
- Build a business case for automation using realistic benefits

AI has moved from experiment to everyday service operations. Language models summarise incidents and draft knowledge articles; machine-learning models classify tickets and predict failures; virtual agents resolve routine requests in Bangla and English at midnight. Yet many organisations automate the wrong things, or automate badly. A master service leader treats AI as one powerful option within a broader automation strategy, applied where it creates measurable value and governed so it stays safe.

High-value use cases by practice

PRACTICE	AI OR AUTOMATION USE CASE	TYPICAL BENEFIT
Service desk	Virtual agent for password resets, status queries and how-to questions	20–40% of contacts resolved without an agent
Incident management	Automatic classification, routing and summary of incident timelines	Faster triage; less reassignment
Monitoring and event	Event correlation and anomaly detection (AIOps)	Fewer alerts; earlier detection
Problem management	Clustering similar incidents to reveal hidden problems	Problems found before they become major
Knowledge management	Drafting articles from resolved tickets; answer generation from approved sources	Higher link rate; fresher knowledge
Change enablement	Risk scoring of changes from history and test data	Right-sized approval; lower failure rate
Service request	End-to-end fulfilment orchestration	Same-day fulfilment; lower cost

Types of automation

Autonomous / agentic	`AI agents plan and act across tools within guardrails` `Highest potential, highest governance need`
Intelligent automation	`Machine learning or language models make judgements: classify, predict, summarise`
Orchestration	`Workflows coordinate many systems and teams end to end`
Task automation	`Scripts, robotic process automation, runbook automation for single tasks`

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INSIGHT · AUTOMATE THE DECISION, NOT JUST THE KEYSTROKE

Scripting a password reset saves minutes. Designing role-based access so most access requests need no human decision saves days. The biggest automation gains usually come from redesigning the flow first, then applying automation and AI to what remains.

Choosing what to automate

1. **Map the value stream** Find where time, effort and errors concentrate.
2. **Simplify first** Remove steps and approvals that add no value.
3. **Score candidates** Volume, repeatability, rule clarity, error cost, data availability, risk.
4. **Choose the automation type** Simple rules → task automation; many systems → orchestration; judgement on unstructured data → AI.
5. **Design human oversight** Decide where humans review, approve or can override.
6. **Pilot and measure** Start with a limited scope; compare outcomes with the baseline.

Automation economics

MEASURE	FORMULA	WHAT IT TELLS YOU
Hours saved	`Hours saved per year = volume × minutes saved per item ÷ 60`	Basis for capacity released; verify after go-live.
Automation ROI	`ROI = (annual benefit − annual run cost) × years − build cost, ÷ total cost`	Include maintenance, model monitoring and licence costs.
Containment rate	`Containment = conversations resolved by virtual agent without human ÷ total conversations × 100%`	Pair with satisfaction and repeat contacts to avoid trapping users.

WATCH OUT · THE VIRTUAL AGENT THAT TRAPPED USERS

A telecom operator launched a chatbot that resolved 55% of conversations on paper. Customer complaints doubled: users could not reach a human, and the bot confidently gave wrong tariff information. Containment must be paired with resolution quality, satisfaction and an easy route to a person.

SCENARIO · DECIDE · SELECTING THE FIRST AI USE CASE

A hospital IT department has budget for one AI pilot this year. Options: an AI model to predict server failures (few failures in history), a virtual agent for staff password and access queries (9,000 contacts a month, clear rules), an AI tool to write clinical discharge summaries, and AI-based change risk scoring (300 changes a year).

(A) AI-generated clinical discharge summaries, because they would save doctors the most time across the whole hospital

(B) Predictive server failure, because it is the most advanced

(C) Change risk scoring for 300 changes a year

(D) A virtual agent for staff password and access queries

Circle your choice. Answer and feedback: module answer key.

TEMPLATE **Automation candidate scorecard**

Compare automation candidates objectively.

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CANDIDATE	VOLUME / YEAR	RULE CLARITY (1–5)	DATA AVAILABLE (1–5)	RISK IF WRONG (1–5)	EST. HOURS SAVED	RECOMMENDED TYPE
Password and access queries	*108,000*	*5*	*4*	*2*	*9,000*	*Virtual agent + orchestration*

Key terms

AIOps
The use of machine learning and analytics on operational data to detect, correlate and resolve issues.

Orchestration
Coordinating automated tasks across multiple systems into an end-to-end workflow.

Intelligent automation
Automation that uses AI to handle judgement, such as classification, prediction or summarisation.

Containment rate
The share of conversations handled entirely by a virtual agent without human help.

Virtual agent
An AI-driven conversational interface that answers questions and fulfils requests.

Robotic process automation (RPA)
Software that mimics human actions in user interfaces to automate repetitive tasks.

Agentic AI
AI systems that can plan and take actions across tools to achieve goals, within defined limits.

REFLECT

Which task in your service area is done thousands of times a year with clear rules? What is stopping it from being automated?

KNOWLEDGE CHECK

1. Which automation type best suits judgement on unstructured text, such as classifying free-text tickets?
   - (A) Orchestration
   - (B) Task automation with fixed scripts
   - (C) Manual triage by a senior engineer
   - (D) Intelligent automation

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2. What should normally happen before automating a process?
   - (A) Wait until all staff agree
   - (B) Buy the most advanced AI platform
   - (C) Automate every step exactly as it is
   - (D) Simplify it and remove non-value steps

3. A chatbot shows 60% containment, but complaints are rising. What measure should be paired with containment?
   - (A) Resolution quality and satisfaction
   - (B) The size of the bot's training data set
   - (C) Bot response speed
   - (D) Number of chatbot conversations

4. Why is a high-volume, rule-based request a good first AI pilot?
   - (A) Clear benefit, low risk, plenty of data
   - (B) It is the most technically advanced option
   - (C) It requires no human oversight
   - (D) It will replace the service desk entirely

5. A desk handles 50,000 requests a year and automation saves 6 minutes each. How many hours are saved?
   - (A) 300,000 hours
   - (B) 5,000 hours
   - (C) 833 hours
   - (D) 50,000 hours

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MODULE 8 · LEAD · LESSON 2 · 45 MIN

Responsible AI in Service Operations

Governing AI so it is accurate, fair, secure and accountable

- Identify the specific risks of AI in service management
- Design human oversight and guardrails proportionate to risk
- Evaluate AI performance with suitable measures before and after deployment

AI in service operations can be wrong in ways that are confident, fast and hard to notice. A language model may invent a plausible but false fix; a classifier may route Bangla-language tickets less accurately than English ones; an agent with broad permissions may take an action no human intended. **Responsible AI** is the set of principles, controls and practices that keep AI useful, safe and accountable. In service management, it is part of governance, not an optional extra.

AI risks in service management

RISK	EXAMPLE	CONTROL
Hallucination	Virtual agent invents a refund policy	Ground answers in approved knowledge; cite sources; refuse when unsure
Bias and unfairness	Lower routing accuracy for Bangla tickets	Test performance by language and user group; retrain with balanced data
Privacy leakage	Personal data in prompts sent to external models	Data minimisation, masking, approved providers, data residency checks
Prompt injection and misuse	Malicious text in a ticket instructs an agent to grant access	Treat input as untrusted, restrict tool permissions, human approval for sensitive actions
Over-automation	Agent restarts production database without approval	Action limits, approval gates, kill switch, audit logs
Model drift	Classifier accuracy falls after new services launch	Continuous monitoring and periodic re-evaluation

Principles for responsible AI in services

Ac
Accountability
Every AI system has a named owner accountable for its outcomes.

Tr
Transparency
Users know when they are dealing with AI and can reach a human.

Fa
Fairness
Performance is checked across languages, user groups and regions.

Pr
Privacy and security
Data use is minimised, protected and lawful.

Sa
Safety and reliability
Systems are tested, monitored and fail safely.

Hu
Human oversight
Humans can review, override and stop AI actions in proportion to risk.

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Levels of human oversight

Human in the loop

- AI suggests; a human decides every time
- For high-risk actions: access grants, production changes, customer compensation

Human on the loop

- AI acts; humans monitor and can intervene
- For medium-risk actions with good monitoring: ticket routing, standard fulfilment

Human out of the loop

- AI acts autonomously within narrow limits
- For low-risk, reversible actions: tagging, summarising, clearing temp files

Evaluating AI classifiers and assistants

MEASURE	FORMULA	WHAT IT TELLS YOU
Precision	`Precision = true positives ÷ (true positives + false positives)`	Of the items the model flagged, how many were right.
Recall	`Recall = true positives ÷ (true positives + false negatives)`	Of the items that should have been flagged, how many the model found.
Answer accuracy	`Accuracy = answers judged correct and grounded ÷ answers evaluated × 100%`	Measured on a representative test set reviewed by experts, before and after release.

Deploying an AI capability safely

1. **Define purpose and owner** What problem, what scope, who is accountable?
2. **Assess risk** Data sensitivity, impact of errors, regulatory obligations.
3. **Prepare data and knowledge** Clean, current, approved sources; privacy controls.
4. **Evaluate offline** Test set covering languages, edge cases and adversarial inputs.
5. **Pilot with oversight** Limited users, human review, clear feedback route.
6. **Monitor in production** Quality, fairness, drift, cost and incidents; review regularly.

PRO TIP · TREAT AI SYSTEMS AS CONFIGURATION ITEMS AND SERVICES

Register AI models and agents in your service and configuration records with owners, versions, data sources and dependencies. Changes to prompts, models or tool permissions go through change enablement. AI incidents are logged and reviewed like any other. This brings AI under the same disciplined management as the rest of your estate.

SCENARIO · DECIDE · AN AGENT WITH TOO MUCH POWER

An IT team wants to give an AI agent permission to reset passwords, grant access to any system and restart servers, so that it can 'resolve tickets end to end'. A pilot showed it resolved 70% of tickets correctly. What should the service manager recommend?

(A) Limit permissions by risk: allow low-risk actions, require human approval for access grants and restarts

(B) Approve full permissions, since 70% accuracy is better than many junior engineers achieve in their first year

(C) Reject AI agents entirely

(D) Allow full permissions but review logs monthly

Circle your choice. Answer and feedback: module answer key.

TEMPLATE **AI system register entry**

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Record each AI capability used in service management.

AI SYSTEM	PURPOSE	OWNER	DATA SOURCES	OVERSIGHT LEVEL	KEY RISKS AND CONTROLS	LAST EVALUATION
Desk virtual agent v3	*Answer staff IT questions; reset passwords*	*Service desk manager*	*Approved knowledge base; identity system*	*On the loop; human for access grants*	*Hallucination → grounded answers; injection → restricted tools*	*92% grounded accuracy, May*

Key terms

Responsible AI
Principles and practices that keep AI systems accurate, fair, secure, transparent and accountable.

Prompt injection
An attack in which malicious input instructs an AI system to ignore its rules or take unintended actions.

Model drift
Degradation of a model's performance as real-world data changes over time.

Recall
The proportion of actual positives that a model correctly identifies.

Hallucination
A confident but false or unsupported output from a generative AI model.

Human in the loop
An oversight model in which a human approves each AI decision or action.

Precision
The proportion of positive predictions that are correct.

Grounding
Constraining AI answers to approved, cited sources of information.

REFLECT

If an AI system in your service made a harmful mistake tomorrow, who would be accountable, and how would you find out it had happened?

KNOWLEDGE CHECK

6. A virtual agent states a refund policy that does not exist. What is this risk called?

(A) Model drift
(B) Prompt injection
(C) Hallucination
(D) Bias against a language group

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7. Which oversight level suits AI granting access to sensitive systems?
   - A) Human out of the loop
   - B) Human in the loop
   - C) No oversight if accuracy exceeds 70%
   - D) Monthly log review only
8. A ticket classifier flags 100 tickets as security incidents; 80 are correct. There were 120 real security incidents. What is the recall?
   - A) 80%
   - B) About 83%
   - C) 100%
   - D) About 67%
9. How should changes to an AI agent's prompts or tool permissions be handled?
   - A) Through change enablement
   - B) By users in the chat window
   - C) Informally, since prompts are just text
   - D) Only by the AI vendor
10. Why should AI routing accuracy be tested separately for Bangla and English tickets?
    - A) To double the size of the test set
    - B) To detect unfair gaps between user groups
    - C) Because regulators require exactly two languages
    - D) Because Bangla tickets are always less important

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MODULE 8 · LEAD · LESSON 3 · 45 MIN

Analysing Complex Situations

Thinking like a master: framing problems, weighing options and justifying choices

- Frame a complex service problem using systems thinking
- Classify situations by complexity and choose a fitting approach
- Compare options with transparent, weighted criteria and justify a recommendation

At master level the question is rarely 'what is the definition of X?'. It is 'given this messy situation, with these stakeholders, constraints and history, what should we do, and why?'. Good answers combine a clear problem frame, an understanding of the type of situation, a set of realistic options and a transparent justification. This lesson gives you a toolkit for doing that under pressure.

A structured analysis method

01 Frame — What is really the problem, for whom, and how do we know?

→

02 Map the system — Stakeholders, dimensions, value streams, constraints

→

03 Classify — Is the situation clear, complicated, complex or chaotic?

→

04 Generate options — At least three, including 'do less' and 'do differently'

→

05 Evaluate — Weighted criteria, risks and principles

→

06 Recommend and plan — Choose, justify, plan feedback and adoption

Types of situation and how to act

Cl — Clear

Cause and effect are obvious. Apply known good practice. Example: a password reset.

Co — Complicated

Cause and effect need expert analysis. Analyse, then respond. Example: diagnosing a performance bottleneck.

Cx — Complex

Cause and effect are visible only in hindsight. Probe with safe experiments, sense, then respond. Example: changing a team's culture.

Ch — Chaotic

No clear cause and effect; urgent harm. Act to stabilise first, then sense. Example: a ransomware attack in progress.

INSIGHT · MOST TRANSFORMATION PROBLEMS ARE COMPLEX

Leaders often treat organisational change as complicated: hire experts, design the perfect process, roll it out. But how people respond is not predictable. Complex problems need small experiments, fast feedback and the willingness to amplify what works and dampen what does not.

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Symptoms versus system causes

Symptom-level thinking

- 'The desk is slow, so hire more agents'
- 'Changes fail, so add approvals'
- 'Users complain, so run a survey'

System-level thinking

- Why is demand so high? What could shift left?
- Why do changes fail? Batch size, testing, environments?
- Where in the journey does value break, and what causes it back stage?

Weighted decision matrix

MEASURE	FORMULA	WHAT IT TELLS YOU
Weighted score	`Score(option) = Σ (weight of criterion × rating of option on criterion)`	Make weights explicit and agreed before scoring; test how sensitive the result is to weight changes.

Example: choosing a service desk model for a garment group

CRITERION (WEIGHT)	INSOURCE CENTRAL (RATING 1–5)	OUTSOURCE FULLY (1–5)	HYBRID: INTERNAL L1 + PARTNER L2 (1–5)
User experience in Bangla on factory floor (30%)	5	2	4
Cost over 3 years (25%)	2	5	4
24/7 coverage for three shifts (20%)	3	5	4
Knowledge retention (15%)	5	1	4
Speed to implement (10%)	2	4	3
Weighted total	3.65	3.25	3.90

Habits of strong justifications

- Link the recommendation to stakeholder value and strategy.
- Name the trade-offs openly; do not pretend there are none.
- Use evidence and data, and state assumptions.
- Refer to principles: start where you are, think holistically, keep it simple.
- Explain how you will know early if you are wrong.

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SCENARIO · DECIDE · CLASSIFYING THE SITUATION

A ransomware attack has encrypted file servers at a university during exam week. Exam results and timetables are unavailable, and the attack may still be spreading. As the senior service leader, what approach fits first?

(A) Convene a working group to analyse root cause in depth before taking any action that could destroy evidence

(B) Act immediately to contain and stabilise, then assess and plan recovery

(C) Run small experiments to see which systems recover

(D) Follow the standard incident process step by step

Circle your choice. Answer and feedback: module answer key.

TEMPLATE **Option evaluation worksheet**

Compare options transparently and record the justification.

OPTION	BENEFITS	COSTS AND RISKS	FIT WITH PRINCIPLES	WEIGHTED SCORE	DECISION AND RATIONALE
Hybrid desk (internal L1 + partner L2)	*Bangla support on floor; 24/7 cover*	*Two contracts to manage*	*Starts where we are; holistic*	*3.90*	*Recommended: best balance of experience and cost*

Key terms

Systems thinking
Understanding a situation as a set of interacting parts and feedback loops rather than isolated events.

Complex situation
A situation in which cause and effect can be understood only in hindsight and outcomes emerge from interactions.

Weighted decision matrix
A tool that scores options against weighted criteria to support transparent decisions.

Safe-to-fail experiment
A small, contained trial whose failure causes acceptable harm and produces learning.

Problem framing
Defining what the real problem is, for whom, and how it is evidenced, before seeking solutions.

Chaotic situation
A situation with no clear cause and effect and urgent harm, requiring immediate action to stabilise.

Sensitivity analysis
Testing how much a decision changes when assumptions or weights change.

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REFLECT

Think of a decision you made recently that went wrong. Did you treat a complex situation as if it were merely complicated?

KNOWLEDGE CHECK

11. Changing a team's culture is best described as which type of situation?
    - (A) Clear
    - (B) Chaotic
    - (C) Complex
    - (D) Complicated
12. What is the right first response in a chaotic situation?
    - (A) Act to stabilise
    - (B) Analyse root cause
    - (C) Run experiments
    - (D) Apply standard practice
13. Why should criterion weights be agreed before options are scored?
    - (A) So weights are not bent to fit a favourite
    - (B) Because scoring is impossible otherwise
    - (C) To shorten the meeting
    - (D) Because weights must always be equal
14. 'Changes fail, so add approvals' is an example of what?
    - (A) Systems thinking
    - (B) Symptom-level thinking
    - (C) Problem framing
    - (D) Sensitivity analysis
15. Which element makes a recommendation stronger at master level?
    - (A) Claiming the option has no downsides
    - (B) Naming trade-offs and early-warning signs
    - (C) Avoiding numbers to keep it simple
    - (D) Using as many frameworks as possible

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MODULE 8 · LEAD · LESSON 4 · 45 MIN

Leading Adoption and Organisational Change

Making new ways of working stick, not just launch

- Plan organisational change that addresses awareness, motivation and ability
- Engage sponsors, managers and champions to drive adoption
- Measure adoption and sustain new behaviours

Most service management transformations fail not because the design was wrong, but because people did not adopt it. New processes are bypassed, tools are used as old ones were, and within a year the organisation drifts back. **Organisational change management** is the discipline of helping people move from the current way of working to the new one, so that intended benefits are actually realised.

Individual change journey

01 Aware
I know what is changing and why

→

02 Willing
I want to take part; I see what is in it for me and for users

→

03 Knowing
I know how to work the new way

→

04 Able
I can do it in real work, with support and practice

→

05 Sustained
It is now normal; reinforcement prevents slipping back

Leading organisational change: eight moves

1 Create urgency
Show the real cost of the current state with evidence users and leaders recognise.

2 Build a guiding coalition
Sponsors and respected practitioners who will lead visibly.

3 Shape a clear vision
Describe the future simply: what will be different for users and staff.

4 Communicate relentlessly
Many channels, many times, two-way, in plain language and local languages.

5 Remove obstacles
Fix tools, policies and incentives that block the new way of working.

6 Win early
Deliver visible improvements in weeks, not years, and celebrate them.

7 Build on gains
Use credibility from early wins to tackle harder structural changes.

8 Anchor in culture
Embed new behaviours in hiring, measures, rituals and leadership habits.

Stakeholder roles in adoption

ROLE	WHAT THEY DO	COMMON FAILURE
Executive sponsor	Sets direction, funds, removes barriers, speaks publicly	Announces then disappears

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ROLE	WHAT THEY DO	COMMON FAILURE
Middle managers	Translate change into daily priorities and coach teams	Not consulted, so quietly resist
Change champions	Peers who model, support and gather feedback	Chosen for availability, not influence
Practitioners	Use the new practices every day	Trained once, never supported afterwards
Users and customers	Experience the result	Informed only after go-live

WATCH OUT · RESISTANCE IS INFORMATION

Resistance usually signals a real concern: fear of losing skills or status, workload during transition, or a design that does not fit real work. Treat it as feedback. Listen, adjust the design where the concern is valid, and support people where it is about uncertainty.

Adoption measures

MEASURE	FORMULA	WHAT IT TELLS YOU
Adoption rate	`Adoption = target users working the new way ÷ total target users × 100%`	Measure by behaviour, not by training attendance.
Proficiency	`Proficiency = transactions done correctly in the new way ÷ total transactions × 100%`	Shows whether people can do it well, not just whether they try.
Benefit realisation	`Benefits achieved ÷ benefits planned × 100%`	The final test that the change created value.

Sustaining change

- Update measures and incentives to reward new behaviours.
- Retire old tools and paths so people cannot slip back easily.
- Keep champions active after go-live with a regular community.
- Include the new ways of working in induction for new hires.
- Review adoption data monthly for at least six months.

SCENARIO · DECIDE · A ROLLOUT THAT STALLED

Six months after launching a new incident and problem process across a telecom operator, adoption is 35%. Many engineers still fix issues through personal contacts and log tickets afterwards. Training attendance was 95%. Managers were not involved in the design. What should the transformation lead do?

- (A) Run the same training again for all engineers, since attendance is no proof that people learned the process
- (B) Mandate disciplinary action for engineers who bypass the process
- (C) Declare the project complete because the process is live
- (D) Engage managers to understand obstacles, fix practical blockers and set expectations through them

Circle your choice. Answer and feedback: module answer key.

CASE STUDY · ADOPTING A NEW SERVICE PLATFORM ACROSS TEN FACTORIES

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A garment group replaces email-based IT support with a service platform covering ten factories in Gazipur, Savar and Chattogram. Head office staff adopt it quickly. On factory floors, adoption is under 20% after three months: supervisors still phone the IT officer they know, the portal is English-only, and many line supervisors share one computer per floor. Factory managers see the platform as 'head office's project'.

1. Which stages of the individual change journey are blocked, and for whom?

2. What adoption actions would you take in the next 60 days?

3. How would you measure whether the change is working?

Model answer: module answer key.

TEMPLATE **Change impact and adoption plan**

Plan support for each affected group.

GROUP	WHAT CHANGES FOR THEM	CONCERNS	ENGAGEMENT ACTIONS	ADOPTION MEASURE	OWNER
Network engineers	*Must log incidents before acting; swarm on P1s*	*Slower fixes; loss of autonomy*	*Co-design swarm rules; quick logging from chat*	*% incidents logged before work starts*	*Head of network operations*

Key terms

Organisational change management
The discipline of helping people transition to new ways of working so that intended benefits are realised.

Change champion
A respected peer who models, supports and gathers feedback on a change.

Executive sponsor
A senior leader accountable for a change, who sets direction, funds it and removes obstacles.

Guiding coalition
A group of influential people with enough authority and credibility to lead a change.

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Adoption rate
The proportion of target users who have actually adopted a new way of working.

Reinforcement
Actions such as recognition, measures and incentives that sustain new behaviours.

Resistance
Behaviours that slow or oppose a change, often signalling legitimate concerns.

REFLECT

Recall a change you were asked to adopt. What would have made you move faster from 'aware' to 'able'?

KNOWLEDGE CHECK

16. Training attendance is 95%, but adoption is 35%. Which stage of the individual journey is most likely blocked?
    - (A) Sustainment after full adoption
    - (B) Willingness or ability
    - (C) Knowledge of the vision statement
    - (D) Awareness

17. Who most directly shapes daily priorities and behaviour in teams during change?
    - (A) External consultants
    - (B) The vendor
    - (C) The board
    - (D) Middle managers

18. How should a change leader treat resistance?
    - (A) As something to ignore
    - (B) As disloyalty to be punished
    - (C) As proof the change is wrong
    - (D) As information

19. Which is the best measure of adoption?
    - (A) Target users actually working the new way
    - (B) Training attendance
    - (C) Go-live date achieved
    - (D) Number of emails sent

20. Why retire old tools after a new practice is live?
    - (A) To force immediate perfection
    - (B) To save licence costs only
    - (C) So people cannot slip back
    - (D) Because auditors require it

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MODULE 8 · LEAD · LESSON 5 · 60 MIN

Master Practice: Integrated Case and Exam Approach

Bringing every stage of the loop together in one real-world challenge

- Apply the whole ZITIL Service Value Loop to an integrated case
- Prioritise interventions across practices, people and technology
- Use a disciplined method to answer master-level scenario questions

This final lesson brings the eight stages of the loop together. Real organisations do not present problems labelled 'incident' or 'strategy'. They present a tangle: outages, angry customers, a stretched team, a demanding regulator, a new AI tool someone already bought. Your job as a master practitioner is to see the system, choose where to intervene first, and lead the organisation through the change.

The ZITIL Service Value Loop as a diagnostic

Value	`Who are the stakeholders and what outcomes, costs and risks matter?`
Model	`What do the four dimensions and value streams reveal?`
Restore and Control	`How well are incidents, problems, changes and service levels managed?`
Journey and Flow	`What do customers experience? Where does work wait?`
Direct and Lead	`Is there direction, governance, measurement and a credible adoption plan, including for AI?`

CASE STUDY · INTEGRATED CASE: A MOBILE FINANCIAL SERVICES PROVIDER UNDER PRESSURE

A mobile financial services provider in Dhaka serves 18 million customers and 250,000 agents. In the last quarter it suffered four major outages, two during festival peaks. The central bank has asked for a resilience improvement plan within 90 days. Customer NPS fell from +22 to −5. Engineering deploys monthly in large batches with a 30% change failure rate. The service desk handles 1.2 million contacts a month, 40% about failed transactions and PIN issues. Operations receives 20,000 alerts a day. A team has already bought an AI chatbot licence but has no plan. Staff morale is low and two senior engineers resigned last month.

1. Using the ZITIL Service Value Loop, what are the three most important system-level problems?

2. Which interventions would you prioritise in the first 90 days, and why?

3. How would you use AI responsibly here?

4. How would you measure progress and report to the board and the regulator?

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5. How would you lead adoption and protect staff during the change?

Model answer: module answer key.

Answering master-level scenario questions

1. **Read the last sentence first** Is it asking what to do FIRST, NEXT, BEST or MOST effectively? The qualifier changes the answer.
2. **Identify the stage and the stakeholders** Which part of the loop is at stake, and whose value matters most?
3. **Find the real problem** Separate symptoms from causes. Which constraint or failure is driving the situation?
4. **Eliminate extremes** Options that ignore risk, freeze everything, blame individuals or buy tools without a plan are rarely best.
5. **Prefer system fixes** Choose options that address causes, apply principles and keep value in view.
6. **Check timing** A good action at the wrong time is a wrong answer: stabilise before optimising, simplify before automating.

Common distractor patterns in master exams

PATTERN	WHY IT IS USUALLY WRONG
Add more approvals or meetings	Adds delay; rarely addresses the cause of failure
Buy a tool	Tools automate the current system, including its flaws
Blame or discipline individuals	Ignores system causes and suppresses reporting
Optimise a non-constraint	Local improvement that does not change the outcome
Freeze or stop everything	Removes risk temporarily but builds a bigger problem
Report the same measure more often	More data on the wrong measure does not help

PRO TIP · THINK IN SEQUENCES

Many master-level answers are about order: restore before diagnose, stabilise before transform, measure before claiming improvement, simplify before automating, involve managers before mandating. When two options both sound good, ask which must come first.

SCENARIO · DECIDE · WHAT TO DO FIRST

You join a government hospital network as head of IT services. In your first week you find: no service catalogue, a major incident every week, an outsourced desk with an expiring contract in eight months, and a ministry request for a digital strategy in six months. What do you prioritise first?

(A) Stabilise major incidents with clear roles and problem management on the top causes

(B) Write the digital strategy first, since the ministry deadline is the most visible commitment and will shape all other work

(C) Start re-tendering the desk contract immediately

(D) Build a complete service catalogue

Circle your choice. Answer and feedback: module answer key.

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Master practitioner checklist for any situation

☐ I can state who the stakeholders are and what value means to each

☐ I have looked at all four dimensions, not just technology

☐ I have traced the relevant value stream end to end

☐ I have separated symptoms from system causes

☐ I know whether the situation is clear, complicated, complex or chaotic

☐ I have at least three options, including a simpler one

☐ My recommendation names its trade-offs and early-warning measures

☐ I have a plan for adoption, not just implementation

☐ AI and automation are used where they add value, with proportionate oversight

Key terms

Integrated case
A realistic scenario combining several practices, stakeholders and constraints, requiring a prioritised, justified response.

Distractor
An answer option that sounds plausible but is weaker than the best option.

Resilience plan
A plan describing how an organisation will prevent, withstand and recover from disruption.

Sequencing
Choosing the order of actions so that each one builds on the last.

Intervention
A deliberate action taken to change a system's behaviour or outcomes.

Stabilise before transform
The principle of restoring control of service failures before attempting large-scale change.

Board pack
A concise set of reports and decisions prepared for the governing body.

REFLECT

Apply the ZITIL Service Value Loop to your own organisation. Which stage is weakest today, and what is the first intervention you would lead?

KNOWLEDGE CHECK

21. A question asks what a manager should do FIRST after a major outage caused by a recent change. Which kind of answer is usually best?

(A) Freeze all changes for three months
(B) Buy a new change management tool
(C) Restore service first, then investigate
(D) Discipline the engineer who made the change

22. An organisation suffers weekly major incidents and has been asked for a digital strategy. What should a new leader prioritise first?

(A) Launching an AI chatbot
(B) Writing the strategy
(C) Re-tendering every contract
(D) Stabilising major incidents

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23. Which option pattern is usually a distractor in master-level questions?

(A) Defining SLOs with the business
(B) Reducing batch size with automated testing
(C) Running a blameless review
(D) Adding more approvals to reduce change failure

24. In the integrated case, why should the chatbot launch be paused?

(A) The regulator bans chatbots
(B) The licence was too expensive
(C) Chatbots never work in financial services
(D) It lacks a use case, grounding and oversight

25. Why should alert noise be reduced before adding AIOps correlation?

(A) Alert noise is a security risk only
(B) Correlation is not useful in finance
(C) AIOps cannot process more than 1,000 alerts
(D) AI fed noisy alerts adds cost, not clarity

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MODULE 8 · LEAD

Answer key

Knowledge checks

1. D Classifying unstructured text needs machine-learning or language-model judgement.

2. D Optimise before automating to avoid speeding up waste.

3. A Containment alone can hide users trapped by a bot that does not solve their problem.

4. A Volume, clarity and low risk make value measurable and safe.

5. B 50,000 × 6 ÷ 60 = 5,000 hours.

6. C A confident, unsupported output is a hallucination.

7. B High-risk actions need a human decision each time.

8. D Recall = 80 ÷ 120 ≈ 67%. Precision is 80 ÷ 100 = 80%.

9. A Prompts and permissions change behaviour and must be controlled.

10. B Fairness checks reveal whether some users get worse service.

11. C Human responses are unpredictable; cause and effect are seen only in hindsight.

12. A Chaos requires immediate action to contain harm.

13. A Agreeing weights first prevents biased, back-filled justifications.

14. B It treats the visible symptom without asking why changes fail.

15. B Honest trade-offs and feedback plans build credibility.

16. B People know about the change; motivation, obstacles or support in real work are the gaps.

17. D Managers translate change into everyday expectations.

18. D Resistance often reveals valid design or support issues.

19. A Adoption is behaviour in real work, not activity around the change.

20. C Removing the old path reinforces the new behaviour.

21. C Restoration first; blame, tools and long freezes are classic distractors.

22. D Stabilise before transform; credibility and capacity come first.

23. D Extra approvals add delay without addressing the cause of failure.

24. D Responsible AI starts with a clear use case, grounding and oversight.

25. D Optimise before automating: clean signals make AI useful.

Scenarios & case studies

Scenario · Selecting the first AI use case
D is best. Correct. High volume, clear rules, low risk and measurable benefit make it an ideal first pilot. Clinical summaries carry patient-safety risk beyond IT's remit, and the other two lack data volume.

Scenario · An agent with too much power
A is best. Correct. Proportionate oversight captures the benefit for low-risk work while protecting against costly errors and misuse on high-risk actions.

Scenario · Classifying the situation
B is best. Correct. This is a chaotic situation: urgent harm with unclear cause and effect. Stabilising comes before analysis.

Scenario · A rollout that stalled
D is best. Correct. Managers shape daily behaviour. Their absence from design explains much of the gap; involving them and removing real obstacles addresses the cause.

Case study · Adopting a new service platform across ten factories
Factory supervisors are aware of the platform but not willing (it is seen as head office's project) and not able (English-only portal, limited computer access). Factory managers have not been engaged as sponsors. In 60 days: make each factory manager a visible sponsor with local targets; recruit respected supervisors as champions; add Bangla interfaces and a messaging-app or phone-to-ticket channel so supervisors can log issues without a shared computer; let the local IT officer log calls on supervisors' behalf at first while showing them the new way; and share quick wins such as faster machine-PC fixes on the floor. Measure adoption as the share of factory incidents logged through any platform channel, resolution time on factory floors, repeat contacts and supervisor satisfaction, reviewed monthly by factory.

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Case study · Integrated case: a mobile financial services provider under pressure

System problems: (1) Unsafe change flow: large monthly batches with 30% failure drive outages, so the Control and Flow stages are the main source of harm. (2) Poor detection and restoration: 20,000 alerts a day hide real signals and major incidents run without clear roles (Restore). (3) No outcome-based direction: the organisation lacks SLOs and resilience targets agreed with the business and regulator (Direct). Underneath, people are overloaded and leaving (Model: organisations and people).

First 90 days: declare a festival-period change policy and an emergency change authority; introduce a major incident model with commanders and communications leads; define SLOs for send-money and cash-out and start error-budget reporting; begin reducing batch size on the most critical service with automated tests and canary releases; run an alert clean-up to cut noise by at least 70%; launch problem management on the top five outage causes. These target the biggest risks fast and give the regulator credible evidence.

AI: pause the chatbot launch until a use case is defined. Start with grounded answers for transaction-status and PIN queries, linked to real transaction data, with an easy route to a human and Bangla and English evaluation. Use AIOps correlation only after alert noise is reduced.

Measurement: board and regulator pack with SLO attainment, major incidents and restore time, change failure rate, lead time, NPS and customer effort for failed-transaction journeys, plus progress on the resilience plan. Leading indicators: error-budget burn and alert actionability.

Adoption: visible CEO sponsorship; involve engineers in designing the change and incident models; cap on-call load and protect improvement time; recognise early wins publicly; retain key engineers with clear career paths and reduced toil.

Scenario · What to do first

A is best. Correct. Weekly major incidents harm patients and destroy credibility; stabilising creates the trust and capacity needed for the catalogue, sourcing decision and strategy.

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REFERENCE

Glossary

A

A3 problem solving A structured problem-solving approach captured on a single sheet of paper. (Direct)

Abandonment rate The share of contacts where the user gives up before being answered. (Restore)

Adoption rate The proportion of target users who have actually adopted a new way of working. (Lead)

Agentic AI AI systems that can plan and take actions across tools to achieve goals, within defined limits. (Lead)

AIOps The use of machine learning and analytics on operational data to detect, correlate and resolve issues. (Lead)

Alert fatigue Desensitisation caused by too many non-actionable alerts, leading to missed real issues. (Restore)

Annualised loss expectancy (ALE) Expected yearly loss from a risk: single loss expectancy multiplied by annual rate of occurrence. (Direct)

Assured conformance Meeting governance, risk and compliance requirements, ideally through automated controls. (Flow)

Automation rate The share of requests fulfilled with no manual intervention. (Control)

Availability The ability of a service to perform its agreed function when required, usually expressed as a percentage of agreed service time. (Control)

B

Back stage The internal activities, systems and partners that support front-stage interactions. (Journey)

Balanced scorecard A set of measures across several perspectives so that no single dimension dominates. (Direct)

Baseline A measured starting point used to judge whether an improvement actually changed anything. (Value)

Batch size The amount of change delivered at once; smaller batches reduce risk and speed feedback. (Flow)

Benefit owner A business person accountable for realising a defined benefit. (Journey)

Blameless review A review of a failure that seeks system causes and improvements rather than individuals to punish. (Model)

Blue-green deployment Running two identical environments and switching traffic from the old to the new, with the ability to switch back instantly. (Control)

Board pack A concise set of reports and decisions prepared for the governing body. (Lead)

Burn rate How fast the error budget is being consumed relative to the rate allowed by the SLO. (Flow)

C

CALMS A lens for DevOps: culture, automation, lean, measurement and sharing. (Flow)

Canary release Routing a small share of users or traffic to a new version and expanding only if health measures stay within limits. (Control)

Capability level A rating of how well a practice performs against defined criteria. (Direct)

Capacity allocation Reserving fixed shares of a team's capacity for different types of work. (Journey)

Capture as you solve An approach in which knowledge is searched, reused, improved and created as part of everyday work rather than as a separate project. (Restore)

Change The addition, modification or removal of anything that could have a direct or indirect effect on services. (Control)

Change authority The person or group that authorises a change, chosen according to the change's risk. (Control)

Change champion A respected peer who models, supports and gathers feedback on a change. (Lead)

Change enablement The practice of maximising successful changes by assessing risk, authorising changes and managing the change schedule. (Control)

Change failure rate The percentage of changes that lead to degraded service or require remediation. (Control)

Change schedule A calendar of planned and past changes used to coordinate work and avoid conflicts. (Control)

Chaotic situation A situation with no clear cause and effect and urgent harm, requiring immediate action to stabilise. (Lead)

Co-creation of value The idea that value appears only when provider and consumer both contribute: the provider supplies resources, the consumer adopts and uses them. (Value)

Cognitive load The amount of mental effort a team must spend to understand and operate what it owns. (Journey)

Complex situation A situation in which cause and effect can be understood only in hindsight and outcomes emerge from interactions. (Lead)

Configuration item (CI) Any component that needs to be managed in order to deliver an IT service. (Control)

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Configuration management database (CMDB) A database used to store configuration records and their relationships throughout their lifecycle. (Control)

Constraint The step that limits the throughput of the whole system. (Flow)

Containment rate The share of conversations handled entirely by a virtual agent without human help. (Lead)

Continual improvement Recurring activity at all levels to keep services and practices aligned with changing needs. (Value)

Continual improvement register (CIR) A single, prioritised record of improvement ideas with owners, status and measured benefits. (Value)

Continuous delivery (CD) Keeping software always in a releasable state so deployment can happen at any time with low risk. (Flow)

Continuous deployment Automatically deploying every change that passes all pipeline checks to production. (Flow)

Continuous integration (CI) Frequently merging code changes into a shared main branch, with automated builds and tests on each change. (Flow)

Contributing factor A condition that, together with others, allowed an incident to happen or made it worse. (Restore)

Control A means of managing risk by preventing, detecting or correcting undesired events. (Direct)

Cost of delay The economic impact of delaying a piece of work by a unit of time. (Journey)

Critical success factor (CSF) A condition that must be met for an objective to be achieved. (Direct)

Customer effort score (CES) A measure of how easy customers found it to get something done. (Journey)

Customer intimacy A strategic position based on tailored solutions and deep customer relationships. (Direct)

Customer journey The complete end-to-end experience a customer has with a service provider across all stages and touchpoints. (Journey)

D

Delegated authority Decision rights passed by governance to management within defined boundaries. (Value)

Demand Need or desire for products and services from internal and external consumers. (Value)

Deployment frequency How often an organisation successfully deploys changes to production. (Flow)

Deployment management The practice of moving new or changed components into live environments. (Control)

Deployment pipeline An automated sequence of build, test and deployment steps from code commit to production. (Control)

DevOps A combination of culture, practices and tools that brings development and operations together to deliver change quickly and reliably. (Flow)

Digital strategy How an organisation will use digital technology to achieve its goals and respond to digital change in its markets. (Direct)

Digital transformation Significant change to customer journeys, products, operations or business models enabled by digital technology. (Direct)

Direction Setting where an organisation is going and why, through vision, strategy and objectives. (Direct)

Discovery Automated scanning of networks and platforms to identify devices, software and relationships. (Control)

Distractor An answer option that sounds plausible but is weaker than the best option. (Lead)

DMAIC Define, measure, analyse, improve, control: a data-driven method for reducing defects and variation. (Direct)

E

Early-life support A period after go-live in which extra resources and monitoring support a new or changed service until it stabilises. (Journey)

Emergency change A change that must be implemented as soon as possible, using an expedited authorisation path. (Control)

Employee experience (EX) How employees perceive their tools, workplace and support. (Journey)

Engage The value chain activity of understanding stakeholder needs and maintaining relationships with them. (Model)

Error budget The amount of unreliability allowed by an SLO: one minus the SLO target. (Flow)

Error control Managing known errors, including periodic reassessment and initiating permanent fixes through change. (Restore)

Event Any change of state that has significance for the management of a service or component. (Restore)

Event correlation Grouping related events to identify a single underlying cause. (Restore)

Exception event An event indicating that a service or component is failing or not operating normally. (Restore)

Executive sponsor A senior leader accountable for a change, who sets direction, funds it and removes obstacles. (Lead)

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Exit plan A pre-agreed approach for moving a service away from a supplier, including data return and knowledge transfer. (Model)

Experience level agreement (XLA) An agreement focused on user experience and outcomes, complementing operational SLA targets. (Control)

Explicit knowledge Knowledge that has been written down or recorded and can be easily shared. (Restore)

F

Feature flag A configuration switch that turns functionality on or off without redeploying code. (Control)

Feedback loop A mechanism by which the results of an action are observed and used to adjust the next action. (Value)

First-contact resolution (FCR) The percentage of contacts resolved during the first interaction without a callback or handover. (Restore)

Five whys A root-cause technique that repeatedly asks why a failure happened until an actionable cause is found. (Restore)

Flow distribution The proportion of completed work across types such as features, defects, risk and debt. (Flow)

Flow efficiency Touch time as a percentage of lead time. (Model)

Flow load The number of work items in progress in a value stream. (Flow)

Flow time Elapsed time from when work starts on an item until value is delivered. (Flow)

Focus on value The principle that all activities should link back, directly or indirectly, to value for stakeholders. (Value)

Follow-the-sun A desk model in which teams in different time zones hand over work to provide 24-hour support. (Restore)

Four dimensions Organisations and people; information and technology; partners and suppliers; value streams and processes: the perspectives that must all be considered for any service. (Model)

Front stage The parts of a service that customers see and interact with directly. (Journey)

Fulfilment The delivery of what was requested. (Control)

G

Go/no-go criteria Agreed, objective conditions that must be met before a release proceeds. (Control)

Goal–question–metric A method of deriving metrics by starting from goals and the questions that test them. (Direct)

Golden signals Latency, traffic, errors and saturation: four measures that describe the health of most services. (Restore)

Goodhart's law The observation that a measure used as a target tends to stop being a good measure. (Direct)

Governance The means by which an organisation is directed and controlled, through evaluating, directing and monitoring. (Value)

Governance, risk and compliance (GRC) An integrated approach to directing an organisation, managing its risks and meeting its obligations. (Direct)

Governing body The person or group accountable at the highest level for the performance and conformance of the organisation. (Value)

Graceful degradation Designing a service to keep providing reduced but useful functionality when a component fails. (Journey)

Grounding Constraining AI answers to approved, cited sources of information. (Lead)

Guiding coalition A group of influential people with enough authority and credibility to lead a change. (Lead)

Guiding principle A recommendation that guides an organisation in all circumstances, regardless of changes in goals, strategies, type of work or structure. (Value)

H

Hallucination A confident but false or unsupported output from a generative AI model. (Lead)

High-velocity IT The application of service management in digital organisations where fast delivery, resilience and responsible technology use matter together. (Flow)

Holistic thinking Considering the whole system, including all dimensions and their interactions, rather than one part in isolation. (Value)

Human in the loop An oversight model in which a human approves each AI decision or action. (Lead)

I

Impact The effect of an incident on the business: users affected, revenue, safety, compliance or reputation. (Restore)

Incident An unplanned interruption to a service or reduction in its quality. (Restore)

Incident commander The person who leads a major incident response and makes decisions without personally fixing components. (Restore)

Information and technology The dimension covering the information a service uses and creates and the technology that supports it. (Model)

Information quality The fitness of information for its purpose, judged by accuracy, completeness, timeliness, accessibility, security and relevance. (Model)

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Infrastructure as code (IaC) Defining and managing infrastructure through version-controlled code rather than manual configuration. (Flow)

Inherent risk The level of risk before any controls are applied. (Direct)

Integrated case A realistic scenario combining several practices, stakeholders and constraints, requiring a prioritised, justified response. (Lead)

Intelligent automation Automation that uses AI to handle judgement, such as classification, prediction or summarisation. (Lead)

Intervention A deliberate action taken to change a system's behaviour or outcomes. (Lead)

IT asset Any financially valuable component that can contribute to the delivery of an IT product or service. (Control)

IT asset management The practice of planning and managing the full lifecycle of IT assets to maximise value, control costs and manage risks. (Control)

IT strategy How the technology function will provide the capabilities, platforms, sourcing and skills the organisation needs. (Direct)

Iteration A small, time-boxed piece of work that produces a usable result and invites feedback. (Value)

J

Journey map A visual representation of a customer's stages, touchpoints, feelings and pain points. (Journey)

K

Kaizen Continuous, incremental improvement, often through focused team workshops. (Direct)

Key performance indicator (KPI) A metric used to evaluate success in meeting an objective. (Direct)

Knowledge base A searchable repository of articles, runbooks and other explicit knowledge. (Restore)

Knowledge management The practice of maintaining and improving the effective, efficient and convenient use of information and knowledge. (Restore)

Known error A problem that has been analysed and has a documented cause or workaround but is not yet resolved. (Restore)

L

Lagging indicator A measure that reports results after they have occurred. (Direct)

Lead time Total elapsed time from the arrival of demand to the realisation of value. (Model)

Lead time for changes Time from code being committed to it running successfully in production. (Flow)

Leading indicator A measure that predicts future performance. (Direct)

Lean An approach that maximises customer value by removing waste and improving flow. (Direct)

Licence compliance The match between software entitlements owned and software actually deployed or used. (Control)

Link rate The share of resolved tickets that reference a knowledge article. (Restore)

Little's law A queueing relationship: average lead time equals average work in progress divided by average throughput. (Journey)

M

Major incident An incident with significant business impact that needs a separate, coordinated response. (Restore)

Maturity assessment An evidence-based evaluation of how well an organisation's management system works as a whole. (Direct)

Mean time to restore service (MTRS) The average elapsed time from loss of service to its restoration. (Restore)

Model drift Degradation of a model's performance as real-world data changes over time. (Lead)

Moment of truth A touchpoint at which the customer forms or changes their opinion of the service strongly. (Journey)

Multi-sourcing Using several suppliers for different parts of a service or portfolio. (Model)

N

Net present value (NPV) The value today of future net cash flows, discounted at a chosen rate, minus the initial investment. (Direct)

Net Promoter Score (NPS) A measure of loyalty: the percentage of promoters minus the percentage of detractors. (Journey)

Normal change A change that must be scheduled, assessed and authorised according to its risk. (Control)

O

Objectives and key results (OKRs) A goal-setting method combining a qualitative objective with measurable key results. (Direct)

Observability The ability to understand a system's internal state from its external outputs, typically metrics, logs and traces. (Restore)

Omnichannel Providing consistent support across phone, chat, portal, email and other channels with a shared record. (Restore)

On-call An arrangement in which engineers are available to respond to alerts outside normal working hours. (Flow)

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Onboarding Bringing a customer and their users into a service so they can start realising value. (Journey)

Operability The ease with which a service can be deployed, monitored, scaled and recovered. (Journey)

Operational excellence A strategic position based on low cost and high reliability through standardisation and efficiency. (Direct)

Operational level agreement (OLA) An agreement between internal teams that supports the delivery of an SLA. (Control)

Opportunity A possibility to add value for stakeholders or improve the organisation, not yet requested by a consumer. (Value)

Orchestration Coordinating automated tasks across multiple systems into an end-to-end workflow. (Lead)

Organisational agility The ability of an organisation to move and adapt quickly, flexibly and decisively. (Value)

Organisational change management The discipline of helping people transition to new ways of working so that intended benefits are realised. (Lead)

Organisational culture The shared values, beliefs and behaviours that shape how people work, decide and react to failure. (Model)

Organisational resilience The ability of an organisation to anticipate, absorb and recover from disruption. (Value)

Organisations and people The dimension covering structures, roles, culture, skills and capacity. (Model)

Outcome A result the consumer wants to achieve, such as faster admissions or fewer failed payments. (Value)

Output A deliverable produced by work, as opposed to the outcome it is meant to achieve. (Direct)

P

Partners and suppliers The dimension covering other organisations involved in the service and how relationships with them are managed. (Model)

Paved road A supported, standard path that makes the right way of working the easiest way. (Flow)

PDCA Plan–do–check–act: an iterative cycle for controlled improvement and learning. (Direct)

Performance-oriented culture A culture in which information flows freely, failures lead to enquiry and new ideas are welcomed. (Model)

Persona A realistic description of a type of customer or user, used to guide design. (Journey)

PESTLE A checklist of external factors: political, economic, social, technological, legal and environmental. (Model)

Planning horizon The time span a plan covers: strategic, tactical or operational. (Direct)

Platform engineering Building and running internal self-service platforms that reduce friction for product teams. (Flow)

Platform team A team that provides internal self-service capabilities, such as pipelines and monitoring, to other teams. (Journey)

Policy as code Expressing compliance and security rules as code that is automatically checked. (Flow)

Portfolio management The practice of selecting, balancing and governing investments in products, services and initiatives. (Direct)

Post-incident review A structured, blameless review of an incident to understand contributing factors and agree improvements. (Flow)

Practice A set of organisational resources designed for performing work or accomplishing an objective. (Value)

Precision The proportion of positive predictions that are correct. (Lead)

Priority The relative importance of an incident, usually derived from impact and urgency. (Restore)

Proactive problem management Identifying and resolving problems before they cause incidents. (Restore)

Problem A cause, or potential cause, of one or more incidents. (Restore)

Problem control Analysing problems, documenting workarounds and known errors, and prioritising them. (Restore)

Problem framing Defining what the real problem is, for whom, and how it is evidenced, before seeking solutions. (Lead)

Product leadership A strategic position based on offering the most innovative products. (Direct)

Product team A long-lived team that owns a product or service end to end, including building and running it. (Journey)

Prompt injection An attack in which malicious input instructs an AI system to ignore its rules or take unintended actions. (Lead)

R

Recall The proportion of actual positives that a model correctly identifies. (Lead)

Reinforcement Actions such as recognition, measures and incentives that sustain new behaviours. (Lead)

Relationship management The practice of establishing and nurturing links between an organisation and its stakeholders at strategic and tactical levels. (Journey)

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Release management The practice of making new and changed services and features available for use. (Control)

Request model A predefined approach for fulfilling a type of request, including scope, approvals, steps and service levels. (Control)

Residual risk The level of risk remaining after controls and treatment. (Direct)

Resilience plan A plan describing how an organisation will prevent, withstand and recover from disruption. (Lead)

Resistance Behaviours that slow or oppose a change, often signalling legitimate concerns. (Lead)

Responsible AI Principles and practices that keep AI systems accurate, fair, secure, transparent and accountable. (Lead)

Reversibility The ability to undo a decision or change quickly and cheaply. (Flow)

Risk A possible event that could cause harm or loss, or make it harder to achieve objectives; also used for opportunities. (Direct)

Risk appetite The amount and type of risk an organisation is willing to take in pursuit of its objectives. (Value)

Robotic process automation (RPA) Software that mimics human actions in user interfaces to automate repetitive tasks. (Lead)

Role-based access Granting access automatically based on a person's job role rather than individual approvals. (Control)

Run–grow–transform A way of classifying IT investment by whether it maintains, extends or fundamentally changes the business. (Direct)

Runbook A step-by-step guide for performing an operational task or responding to a known situation. (Restore)

S

Safe-to-fail experiment A small, contained trial whose failure causes acceptable harm and produces learning. (Lead)

Segregation of duties Separating responsibilities so no single person can complete a sensitive action alone. (Direct)

Sensitivity analysis Testing how much a decision changes when assumptions or weights change. (Lead)

Sequencing Choosing the order of actions so that each one builds on the last. (Lead)

Service A means of enabling outcomes for a consumer while the provider carries specified costs and risks on the consumer's behalf. (Value)

Service catalogue Structured information about the services and offerings available to consumers. (Control)

Service configuration management The practice of ensuring accurate, reliable information about the configuration of services and the CIs that support them is available when needed. (Control)

Service design package The set of information handed to those who will run and support a service, describing how it works and how to operate it. (Journey)

Service desk The single point of contact between the service provider and users for incidents, requests and communication. (Restore)

Service integration Coordinating multiple suppliers so they deliver one coherent end-to-end service with clear accountability. (Model)

Service level agreement (SLA) A documented agreement between a provider and a customer that identifies services and expected service levels. (Control)

Service level indicator (SLI) A carefully defined quantitative measure of some aspect of the service level users experience. (Flow)

Service level management The practice of setting business-based targets for service performance and assessing, monitoring and managing delivery against them. (Control)

Service level objective (SLO) A target value or range for an SLI over a defined time window. (Flow)

Service offering A formal description of one or more services as presented to a consumer group, combining goods, access to resources and service actions. (Value)

Service owner The person accountable for a service's performance and value across its life. (Model)

Service relationship The cooperation between provider and consumer, including provision, consumption and relationship management activities. (Value)

Service request A request from a user for something that is a normal, pre-defined part of service delivery. (Control)

Service request management The practice of supporting agreed service quality by handling all pre-defined, user-initiated requests effectively and in a user-friendly way. (Control)

Service review A regular meeting between provider and customer to review performance, experience and improvements. (Control)

Service value chain An operating model of six interconnected activities that respond to demand and create value. (Model)

Service value system (SVS) A model of how all components and activities of an organisation work together to turn opportunity and demand into value. (Value)

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Shadow IT Technology acquired or used without the knowledge or control of the IT function. (Control)

Shift-left Moving resolution of work to lower-cost, faster levels, ideally to users themselves through self-service. (Restore)

Silo A team or department that optimises its own work and measures at the expense of the end-to-end flow of value. (Value)

Site reliability engineering (SRE) An approach that applies software engineering to operations, managing reliability with measurable objectives. (Flow)

Sourcing model The way an organisation obtains a capability: insourcing, outsourcing, single sourcing, multi-sourcing or partnership. (Model)

Sponsor The person or body that authorises budget for service consumption. (Value)

Stabilise before transform The principle of restoring control of service failures before attempting large-scale change. (Lead)

Staged funding Releasing investment in tranches tied to evidence of progress and value. (Direct)

Standard change A low-risk, pre-authorised change that is well understood and follows a defined procedure. (Control)

Standardisation Using common platforms, tools and methods across an organisation to reduce cost, risk and complexity. (Model)

Start where you are The principle of observing and reusing what already exists before building something new. (Value)

Strategic supplier A supplier whose services are critical and hard to replace, managed through joint governance and shared goals. (Model)

Strategy A chosen approach for achieving long-term objectives: where to compete and how to win. (Direct)

Swarming An approach in which the people needed to resolve an incident collaborate on it early, instead of escalating in tiers. (Restore)

Synthetic monitoring Running scripted transactions against a service to test its behaviour as a user would experience it. (Restore)

Systems thinking Understanding a situation as a set of interacting parts and feedback loops rather than isolated events. (Lead)

T

T-shaped professional A person with deep expertise in one area and broad knowledge across several others. (Model)

Tacit knowledge Knowledge held in people's experience and intuition, hard to write down and shared mainly through practice. (Restore)

Technical debt The future cost created by choosing a quick or limited technical solution now instead of a better one. (Model)

Theory of constraints A management approach that improves a system by identifying and managing its most limiting factor. (Flow)

Three lines model A model distinguishing management (first line), risk and compliance functions (second line) and independent audit (third line). (Direct)

Throughput The number of work items completed per unit of time. (Journey)

Time to restore service How long it takes to recover from a failure in production. (Flow)

Toil Manual, repetitive, automatable operational work that has no enduring value and grows with scale. (Flow)

Total cost of ownership (TCO) The full cost of a service over its life, including acquisition, operation, transition and exit. (Model)

Touch time The time during which work is actually being done on an item. (Model)

Touchpoint Any interaction between a customer and the provider's people, systems or communications. (Journey)

Trunk-based development A practice in which developers integrate small changes into a single main branch frequently. (Flow)

Trusted adviser A relationship level in which the provider understands the customer's strategy and proposes solutions proactively. (Journey)

U

Underpinning contract (UC) A contract with an external supplier that supports the delivery of agreed service levels. (Control)

Unplanned work Work that arrives without being scheduled, such as incidents and urgent escalations. (Journey)

Urgency How quickly an incident must be resolved to avoid greater impact. (Restore)

Utility The functionality of a service: what it does to support an outcome; fitness for purpose. (Value)

V

Value chain activity One of plan, improve, engage, design and transition, obtain/build, and deliver and support. (Model)

Value realisation Confirming that expected outcomes and benefits have actually been achieved. (Journey)

Value stream A series of steps, combining value chain activities and practices, that turns a specific type of demand into value. (Model)

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Value stream management Managing and improving the end-to-end flow of value from idea to customer outcome across all teams involved. (Flow)

Value stream mapping A technique for visualising the steps, times and waste in a value stream to design a better future state. (Model)

Value streams and processes The dimension covering how work is organised and coordinated to create value. (Model)

Vanity metric A number that looks impressive but does not inform decisions. (Direct)

Vendor lock-in Dependence on a supplier that makes switching costly or impractical. (Model)

Virtual agent An automated conversational assistant that answers questions or fulfils requests through chat or voice. (Restore)

W

Warranty Assurance that a service meets agreed requirements for availability, capacity, security and continuity; fitness for use. (Value)

Waste Any activity or step that consumes resources without adding value for the consumer. (Value)

Watermelon reporting Reports that look green because component targets are met while the consumer's experience is red. (Model)

Weighted decision matrix A tool that scores options against weighted criteria to support transparent decisions. (Lead)

Weighted shortest job first (WSJF) A prioritisation method that divides cost of delay by job size. (Journey)

Work in progress (WIP) Work that has started but is not yet finished. (Journey)

Workaround A solution that reduces or eliminates the impact of an incident or problem for which a full resolution is not yet available. (Restore)

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CAPSTONE

Capstone: Lead a service management transformation around the ZITIL Service Value Loop

Choose a real organisation you know, or a realistic one (for example: a mobile financial services provider in Dhaka suffering repeated weekend outages, a 900-bed hospital in Chattogram moving to a new clinical system, a garment group running ERP across ten factories, or a public university launching digital admissions). Diagnose its current service management capability, design the target state across all eight stages of the loop, and write the adoption plan that its executive team could approve and its practitioners could follow.

Deliverables

☐ Service value map: consumers, sponsors, outcomes, costs, risks and the services that matter most

☐ Four-dimensions assessment with current and target maturity and the evidence behind each rating

☐ Two value-stream maps (one for restoring service, one for delivering a change) with flow metrics and waste

☐ Incident, major-incident and problem design: classification, roles, communication and known-error flow

☐ Change enablement model: change types, delegated authority, automated checks and a release/deployment approach

☐ Service level framework: SLAs, OLAs, experience measures (XLAs) and a reporting pack

☐ Customer-journey map with moments of truth and improvement opportunities

☐ DevOps/SRE plan: SLOs, error budgets, toil targets and on-call model for one critical service

☐ Governance, measurement and digital strategy pack: objectives, KPIs, risk appetite and investment roadmap

☐ AI and automation plan with use cases, guardrails and benefits, plus a 12-month adoption and change plan

Self-assessment rubric

CRITERION	DEVELOPING	PROFICIENT	EXCELLENT
Value & principles	○ Activity described, value unclear	○ Value and stakeholders defined	○ Outcomes, costs and risks balanced for each stakeholder
Holistic system design	○ Technology only	○ All four dimensions covered	○ Dimensions, value chain and streams linked with evidence
Restoring service	○ Ad hoc firefighting	○ Defined incident and problem flow	○ Fast, measured restoration with proactive problem control
Control & flow of change	○ Single approval board for everything	○ Change types and release plan	○ Risk-based, automated, delegated change with low failure rate
Customer experience	○ Internal view only	○ Journey mapped	○ Journey improved with experience measures and feedback loops
High-velocity practice	○ Not addressed	○ DevOps/SRE ideas listed	○ SLOs, error budgets and flow metrics drive decisions
Direction & governance	○ No measures	○ KPIs and governance roles	○ Strategy, risk, investment and measures form one system

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CRITERION	DEVELOPING	PROFICIENT	EXCELLENT
AI & leading adoption	○ Tools only	○ Use cases and change plan	○ Responsible AI, staged adoption and sustained behaviour change

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CAPSTONE

Planning pages

Sketch your design, timeline or architecture here.

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EXAM PREPARATION

Practice exam · 60 questions

The official online exam draws 45 questions at random from a separate, secret bank that tests the same skills, with a 90-minute limit and a 85% pass mark. Practise with these questions first: time yourself, mark your answers, then check the key and read every explanation, including for the questions you got right.

Value — Service Value System and Guiding Principles

1. A telecom operator's IT department reports 99.99% server availability, yet enterprise customers say the business messaging service is unreliable because messages arrive hours late. What should the service manager conclude FIRST?

   (A) Component availability does not prove the warranty that consumers experience

   (B) Customers are mistaken, because the servers were measurably available all quarter

   (C) Utility is missing because the service lacks features

   (D) The sponsor has under-funded the server estate

2. The finance director of a hospital funds a new pharmacy system, the head pharmacist defines its requirements and pharmacists use it daily. The pharmacists want faster screens, the finance director wants lower licence costs. How should the service owner proceed?

   (A) Prioritise the pharmacists, because as users they are closest to the daily work

   (B) Prioritise the sponsor because they control the budget

   (C) Let the supplier decide which requirement wins

   (D) Make the speed–cost trade-off visible and agree it with customer and sponsor

3. A university installs an excellent e-learning platform, but six months later only 12% of courses use it. Which action is MOST likely to increase value?

   (A) Mandate use of the platform and switch off all other teaching tools this term

   (B) Invest in onboarding, lecturer support and feedback to raise adoption

   (C) Upgrade the servers to improve platform availability

   (D) Add more features so lecturers find it more attractive

4. An IT director wants to apply the guiding principles to a decision about outsourcing the service desk. Which approach is BEST?

   (A) Test the option against every principle and record how tensions were resolved

   (B) Use the principles only after the contract is signed

   (C) Apply only 'keep it simple', since outsourcing is complex

   (D) Work through the principles in order, completing each one fully before starting the next

5. A bank's change process requires nine signatures for every change, including updates to a printer driver. Change volume is falling and teams bypass the process. Which guiding principle most directly points to the fix?

   (A) Collaborate and promote visibility

   (B) Start where you are

   (C) Keep it simple and practical

   (D) Focus on value

6. A governing body notices that most IT decisions are escalated to it, causing delays of weeks. What should it do?

   (A) Meet more frequently so decisions are made sooner

   (B) Appoint an external consultant to make the escalated decisions on its behalf

   (C) Ask managers to stop escalating any decisions

   (D) Define risk appetite and delegate authority within clear limits

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7. After a year of improvement work, an organisation has 230 ideas in its register, few owners and no measured benefits. What is the BEST next step?

   (A) Move the register to a better tool with more fields for scoring, status and ownership
   (B) Ask each team to keep its own separate register
   (C) Stop collecting ideas until all 230 are implemented
   (D) Triage the register, give owners to top items and measure benefits

8. A diagnostic traces a customer request through the value system and finds the longest delay is waiting for a supplier who has no service level agreement. Which response is MOST holistic?

   (A) Treat the supplier agreement as part of end-to-end service design
   (B) Increase internal team size so that the team can compensate for supplier delays
   (C) Report the supplier delay as outside IT's responsibility
   (D) Escalate each delayed request to the supplier's account manager

Model — Four Dimensions and the Service Value Chain

9. A bank deploys a new loan-origination platform. Technology tests pass, but loan officers keep using spreadsheets because the new screens do not match how they assess applications. Which dimension was MOST neglected during design?

   (A) Partners and suppliers
   (B) Value streams and processes
   (C) Information and technology
   (D) External legal factors affecting loan approvals

10. A government agency must decide whether to outsource its citizen identity service. Which consideration should weigh MOST heavily?

    (A) Outsourcing reduces the number of staff the agency must manage
    (B) Several suppliers have bid, which ensures competition
    (C) The service is core and highly sensitive, which favours keeping control in-house
    (D) A supplier offers a monthly price 20% below current internal running costs, with a five-year price freeze

11. Three suppliers meet all their service levels, but the end-to-end service fails weekly. What should the service owner do FIRST?

    (A) Ask users to log failures directly with each supplier
    (B) Run a joint, cross-supplier problem investigation
    (C) Add financial penalties to each supplier contract
    (D) Replace the supplier with the lowest satisfaction score

12. A team maps a value stream and finds 4 hours of work in a 15-working-day lead time. What does this MOST strongly indicate?

    (A) Staff are working too slowly and need individual productivity targets for each step in the stream
    (B) More people should be hired for the steps with the most work
    (C) The stream is efficient because the work itself is short
    (D) Most lead time is waiting, so reducing queues and handovers will help most

13. An IT director proposes standardising all departments on one collaboration platform, while the research division wants freedom to experiment with new data tools. What is the BEST approach?

    (A) Standardise commodity collaboration and allow governed variety for research tools
    (B) Let every department choose its own collaboration platform
    (C) Postpone the decision until all departments agree unanimously
    (D) Force every department, including research, onto the single platform so that support costs fall

14. After repeated outages, a CIO asks why engineers never raise concerns before changes. Staff say people who reported problems in the past were criticised. Which action addresses the root cause?

    (A) Introduce a stricter approval form so that every change records its risks before approval
    (B) Leaders adopt blameless reviews and visibly reward early reporting
    (C) Run technical training for all engineers
    (D) Hire an external auditor to review every change

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15. A value stream for new staff onboarding starts when HR raises a request and ends when IT closes its ticket. Users complain they still cannot work on day one. What should be changed FIRST?

(A) Redefine the stream's end as the new employee working productively
(B) Move onboarding responsibility entirely to HR
(C) Set a target for IT to close tickets faster
(D) Add a satisfaction survey that users complete when IT closes the onboarding ticket

Restore — Restoring Service: Desk, Incident, Problem, Monitoring and Knowledge

16. A service desk reports average speed of answer of 20 seconds and 92% of tickets closed within a day, yet a survey shows users find IT support exhausting. What should the desk manager investigate FIRST?

(A) Whether more tickets can be closed within the first hour
(B) Whether average speed of answer can be reduced to ten seconds by adding agents at peak times
(C) How much effort users spend: repeat contacts, handovers and reopened tickets
(D) Whether the survey sample is large enough to trust

17. A major incident has affected online banking for 25 minutes. Engineers from four teams are on the bridge, each fixing different things, and nobody is sending updates. What should happen NEXT?

(A) Appoint an incident commander and a communications lead
(B) Close the bridge and let teams work separately
(C) Ask each team to write its own update for customers about its part of the fix
(D) Escalate to the CIO to take technical control

18. During a major incident, the team identifies that a configuration change made 40 minutes earlier is the likely cause. Rolling it back takes five minutes; full diagnosis may take hours. What should the incident commander do?

(A) Ask the change author to explain and justify the change in writing before acting
(B) Continue diagnosis until the exact fault is proven
(C) Roll back the change, then continue diagnosis through problem management
(D) Wait for the change advisory board to approve a rollback

19. The same file-share outage has occurred eight times in two months. Each time the desk restarts a service and closes the incident. What is the MOST appropriate action?

(A) Ask users to save files locally to avoid the file share
(B) Raise the incident priority so it gets more attention
(C) Open a problem record and analyse the recurring cause
(D) Create a script so the desk can restart the service faster

20. A problem analysis finds that a storage fault has a reliable two-minute workaround, while a permanent fix would require a risky platform migration costing a large sum. What should the problem manager recommend?

(A) Hide the workaround so engineers are forced to find a fix
(B) Document the known error and workaround, and review the decision periodically
(C) Close the problem because a workaround exists
(D) Implement the migration immediately because known errors must always be fixed

21. An operations team receives 9,000 alerts a day and acts on fewer than 2%. Leaders want to buy an AI tool to prioritise alerts. What should be done FIRST?

(A) Buy the AI tool now so it can start learning patterns from all of the existing alerts
(B) Remove non-actionable alerts and tune thresholds to service impact
(C) Hire more engineers to review every alert
(D) Disable all alerts except those for the core database

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22. A university portal showed healthy infrastructure throughout a two-hour period when students could not submit assignments due to an authentication provider fault. Which improvement BEST prevents this blind spot?

(A) Ask the authentication provider to send weekly availability and incident reports
(B) Add synthetic transactions that test login and submission end to end
(C) Add memory and CPU monitoring to every portal server
(D) Increase dashboard refresh frequency

23. A desk's knowledge base contains 4,000 articles, but agents rarely use it and the link rate is 6%. What is the MOST effective way to improve it?

(A) Delete the knowledge base and start a wiki
(B) Build article search, reuse and improvement into every ticket workflow
(C) Make article creation a monthly performance target for each agent on the desk
(D) Commission a contractor to rewrite all 4,000 articles

Control — Controlling Change and Service Levels

24. A bank's change process requires CAB approval for every change, including 180 routine monthly patch deployments that have never failed. Lead time is three weeks. What should the change manager do FIRST?

(A) Stop patching until the CAB backlog has been cleared
(B) Ask the CAB to meet twice a week to approve changes faster
(C) Require detailed, separately reviewed rollback plans for every one of the patch deployments
(D) Convert the proven patch deployments into a pre-authorised standard change

25. An actively exploited vulnerability is announced at 23:00 for an internet-facing system. The next CAB meeting is in six days. What is the MOST appropriate action?

(A) Patch without any authorisation and record nothing
(B) Use the emergency change path with the named emergency authority
(C) Wait for the CAB, because security changes are high risk
(D) Ask the vendor's security team to approve the change formally on the bank's behalf

26. A digital bank wants to release a new loan feature to 2% of customers, observe behaviour, and expand if results are good, without redeploying code. Which technique BEST supports this?

(A) Feature flags controlling exposure by customer segment
(B) A phased rollout by branch over six months, starting in Dhaka
(C) A big-bang deployment on a weekend
(D) Blue-green deployment of the whole application

27. New staff at a hospital wait an average of seven days for system access. Most requests need three approvals that are almost never refused. What should be done FIRST?

(A) Ask managers to approve requests within 24 hours
(B) Train new staff to submit requests earlier
(C) Simplify the request model, replacing routine approvals with role-based access
(D) Automate the existing three-step approval workflow exactly as it is today, with reminders

28. A university plans a CMDB covering every device, cable and software setting across campus. The goal is faster incident routing and change impact analysis. What should the configuration manager recommend?

(A) Start with cables and patch panels, as they are the hardest items to discover automatically
(B) Replace the CMDB with a spreadsheet for each department
(C) Scope the CMDB to the services, CIs and relationships those decisions need
(D) Proceed with the full scope to avoid missing anything

29. A supplier-hosted payroll service meets its 99.9% availability SLA, but payroll staff lose a full day each month because batch jobs fail on the 25th. What is the BEST improvement to the service level framework?

(A) Add a target for payroll completing on schedule
(B) Ask the supplier to report availability weekly instead of monthly
(C) Remove payroll from the SLA reports
(D) Raise the availability target to 99.99%

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30. An internal team supporting a customer-facing SLA has no agreed targets, and its delays repeatedly cause SLA breaches. What should the service level manager put in place?

(A) An underpinning contract with that team
(B) A penalty clause in the customer SLA
(C) An OLA with that team, aligned to the SLA
(D) A new SLA directly with the end users

Journey — Creating, Delivering and Supporting Services Along the Customer Journey

31. A development team plans to hand a new claims service to operations with no runbooks and alerts only on CPU. Launch is in two weeks and cannot move. What should the service manager arrange?

(A) A formal objection to launch, recorded in the risk register only
(B) Responsibility for all claims issues passing permanently to developers
(C) A full handover on launch day as originally planned
(D) Early-life support with joint ownership and exit criteria

32. A team finishes 15 items a week and has 90 items in progress. Leaders want lead time under two weeks. What is the MOST effective lever?

(A) Start all new requests immediately on arrival
(B) Add a daily status meeting covering every item in progress
(C) Increase engineer utilisation to 100%
(D) Reduce work in progress to around 30 items or less

33. A support team spends 55% of its time on unplanned work, and its improvement projects never finish. What should be done FIRST?

(A) Move all improvement projects to a separate, dedicated improvement team
(B) Ban unplanned work during improvement weeks
(C) Increase planned project commitments to force focus
(D) Reserve improvement capacity and cut top interrupt sources

34. A wallet provider sees many new users abandon the service after their first failed cash-out at an agent. Which approach BEST identifies how to fix this?

(A) Increase cash-out fees to fund new incentives for the agents who hold float
(B) Send a satisfaction survey to all users
(C) Run an app usability test with new users
(D) Map the journey and link the touchpoint to its back-stage causes

35. Each department in a hospital meets its own waiting-time targets, but patients spend over three hours per visit. What should the service manager do?

(A) Ask patients to arrive later in the day when departments are quieter
(B) Add more staff to each department
(C) Measure and manage the end-to-end patient journey
(D) Tighten each department's targets

36. Enterprise customers give a cloud provider high satisfaction on support tickets but a negative NPS. What should the account manager do FIRST?

(A) Offer immediate discounts to every enterprise customer to rebuild goodwill
(B) Hold account reviews to explore relationship-level concerns
(C) Stop running NPS surveys
(D) Rely on CSAT, since it is measured more often

37. A board approves a new HR system expecting 30% fewer manual HR transactions. Two years later nobody knows if this happened. Which practice would have prevented this?

(A) A larger project contingency budget
(B) Weekly IT availability and incident reports sent to the board
(C) Named business benefit owners with a baseline and tracking
(D) A longer system warranty period from the supplier

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38. A head of IT wants to move from being seen as an order taker to a trusted adviser by business units. Which action is MOST effective?

(A) Publish more detailed SLA reports each month

(B) Respond to every business request faster than before, with daily updates

(C) Reduce the cost of all IT services by 10%

(D) Assign relationship managers who learn each unit's strategy

Flow — High-Velocity IT: DevOps, SRE and Value Streams

39. A digital bank deploys quarterly. Each release contains hundreds of changes and roughly one in three releases causes a major incident. The CTO proposes a stricter release approval board. What is the BEST alternative?

(A) Move to small, frequent, automated deployments with progressive rollout

(B) Add a second approval board for releases touching core systems

(C) Outsource testing to a specialist supplier and keep the cycle

(D) Reduce releases to twice a year so that each one gets far more testing time

40. A company bought a complete DevOps toolchain, yet development and operations still blame each other for incidents and deploy monthly. What should the transformation lead address FIRST?

(A) A second toolchain dedicated to the operations team's own release needs

(B) A mandate to deploy daily from next month

(C) Shared goals, ownership and blameless learning across both teams

(D) More detailed handover documents between the two teams

41. An auditor requires evidence that every production change was tested and reviewed. Engineers complain that filling in change forms takes hours. What is the MOST effective solution?

(A) Generate change records from the pipeline with evidence

(B) Ask the auditor to accept verbal confirmation from team leads

(C) Reduce the number of fields on the change form

(D) Hire administrators to complete change forms for engineers

42. A payments service with a 99.9% SLO has used its entire monthly error budget in the first ten days. The product owner wants to launch a campaign feature next week. What should happen according to a sound SRE approach?

(A) Launch the feature, since the SLO is internal and not contractual

(B) Lower the SLO for the rest of the month so the launch can proceed

(C) Increase the SLO to 99.99% to raise urgency

(D) Pause non-critical releases and prioritise reliability, as agreed

43. An operations team spends about 70% of its time on manual, repetitive tasks and has no time for engineering improvements. What should the SRE lead do?

(A) Accept the toil as an unavoidable cost of running production systems at scale

(B) Cap toil, measure it and prioritise automating the largest sources

(C) Move the manual tasks to the service desk

(D) Hire more operators to absorb the manual work

44. Product teams in a retailer complete features in two weeks, but customers see them only after ten weeks because of a shared security review queue. Which action improves customer lead time MOST?

(A) Ask product teams to complete features in one week

(B) Add three more product teams

(C) Report each product team's velocity weekly to leadership and the board

(D) Fix the security review constraint with automated checks

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45. Six months of flow data shows 90% of completed work is new features, while incidents and audit findings rise steadily. What should the value stream leader do?

(A) Rebalance flow toward debt, risk and defect work
(B) Measure only feature velocity to keep teams focused
(C) Stop all feature work until every audit finding is closed
(D) Keep the distribution, since features create the most value

Direct — Direction, Planning, Governance and Digital Strategy

46. A CIO's annual plan lists 'deploy new ERP' and 'migrate to cloud' as key results. The CEO says she cannot see how IT supports the company's growth goal. How should the CIO revise the plan?

(A) Rewrite key results as business outcomes the initiatives should move
(B) Add delivery dates and budgets to each initiative
(C) Present the plan to the board without the CEO
(D) Add more technical detail about the ERP and cloud projects and their architecture

47. Different directors in a manufacturer have each approved IT projects directly. IT has 38 active projects and finishes almost none on time. What should the head of IT propose FIRST?

(A) A rule that each director can approve at most three projects
(B) Contractors to deliver the backlog of approved projects
(C) A project management office that tracks all 38 projects
(D) One ranked portfolio with a limit on active projects

48. A risk assessment shows an internet banking outage would cost Tk 1 crore per occurrence and is expected twice a year. A resilience upgrade costing Tk 60 lakh a year would cut occurrences to once every two years. What does this analysis suggest?

(A) The upgrade should be rejected because it does not remove all risk
(B) The decision cannot be made until a full penetration test of internet banking is complete
(C) The upgrade is not justified because outages are unlikely
(D) Justified: expected annual loss falls far more than the cost

49. An audit finds that privileged access to production systems is reviewed by managers only when they remember, and many reviews are missing. What is the MOST effective improvement?

(A) Make access reviews an annual audit activity
(B) Remind managers by email each quarter to complete their access reviews on time
(C) Remove privileged access from all engineers
(D) Automate access reviews with enforced revocation

50. A service desk's first-contact resolution rose from 50% to 85% in two months, but reopened tickets tripled. What should the service manager do?

(A) Replace first-contact resolution with average handling time as the desk's main target
(B) Pair the measure with quality data and investigate closures
(C) Stop measuring reopened tickets, as they distract staff
(D) Celebrate the result and raise the target to 90%

51. An IT director wants a maturity programme with the goal of reaching level 4 in all 34 practices within two years. What is the BEST advice?

(A) Proceed, since higher maturity is always better
(B) Reach level 4 first in practices that are easiest to document
(C) Target capability improvements where gaps most limit value
(D) Skip assessment and buy a new tool for each practice

52. A logistics company's leadership wants to 'go digital' because a competitor launched an app. No strategic objective has been defined. What should the head of digital do FIRST?

(A) Commission a copy of the competitor's app
(B) Clarify the strategy digital should support
(C) Buy a digital platform suite now so the company is ready for any later strategic choice
(D) Run a hackathon to generate app ideas

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53. A board is asked to commit Tk 200 crore upfront for a five-year digital transformation with uncertain adoption. Which funding approach BEST manages the risk?

(A) Approve the full amount to provide certainty
(B) Fund the first year and decide later without criteria
(C) Release funding in stages based on evidence of value
(D) Reject the transformation as too risky

Lead — AI, Automation and Leading Service Transformation

54. A service desk handles 30,000 password and PIN-unlock requests a month with clear rules. Leadership also wants AI for complex diagnostic tickets with little historical data. Which AI initiative should come FIRST?

(A) Both at once, to show the board real ambition and spread the licence cost across more use cases
(B) A virtual agent for password and PIN requests
(C) Neither until all processes reach high maturity
(D) An AI model for complex diagnostic tickets

55. An AI agent can resolve tickets end to end, including granting system access. In testing it was right 85% of the time. How should its permissions be set?

(A) Grant all permissions, since 85% beats many junior staff
(B) Low-risk actions alone; human approval for access grants
(C) Block the agent from all actions until it reaches 100%
(D) Allow all actions, with a full review of the agent's action logs once a quarter

56. A ticket-routing model performs well for English tickets but routes 30% of Bangla tickets to the wrong team. What should the service owner do?

(A) Retrain with balanced data and monitor by language
(B) Ask Bangla-speaking users to write their tickets in English from now on
(C) Accept it, since most tickets are in English
(D) Turn off the model for everyone

57. A university's new service management process has 95% training attendance but only 30% adoption after four months. Department managers were not involved in the design. What should the change lead do NEXT?

(A) Introduce penalties for staff who bypass the process
(B) Report the rollout as complete because the process is live
(C) Involve managers, uncover practical obstacles and fix them
(D) Repeat the training for all staff

58. During exam week, ransomware is spreading across a university's servers. Several senior staff want to hold a meeting to analyse how the attacker got in before acting. What should the IT service leader do?

(A) Contain and stabilise now, preserving evidence
(B) Hold the analysis meeting first so that no hasty decisions destroy the forensic trail
(C) Follow the normal incident queue in order of logging
(D) Run small experiments to see which systems are affected

59. A head of IT must choose between three desk-sourcing options. Each stakeholder prefers a different one. Which approach gives the MOST defensible recommendation?

(A) Ask the incumbent supplier which option it recommends, since it knows the desk best
(B) Choose the option the most senior stakeholder prefers
(C) Agree weighted criteria, then score and test sensitivity
(D) Choose the cheapest option to protect the budget

60. An organisation has weekly major incidents, a falling NPS, a 30% change failure rate, 20,000 daily alerts and a newly bought AI chatbot with no plan. Which FIRST set of actions is MOST appropriate?

(A) Stabilise incidents, make change safer and cut alert noise first
(B) Replace the operations team with a managed service provider under a new outcome-based contract
(C) Launch the chatbot immediately to improve NPS
(D) Commission a two-year maturity assessment

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EXAM PREPARATION

Practice exam answers & explanations

1. A Warranty is about the service meeting agreed performance as consumers experience it. Components can be up while the end-to-end service fails. Nothing suggests missing features or funding.

2. D Different roles value different things. The service owner's job is to surface the trade-off and get an agreed decision, not to pick a side alone or hand the decision to a supplier.

3. B Value is co-created; low adoption is the constraint. More features or capacity do not address it, and a sudden mandate without support usually produces resistance and poor use.

4. A The principles are lenses to apply together before deciding. They are not sequential steps, not to be used selectively, and are most useful before commitment.

5. C The process applies the same heavy path to every case; simplifying it and designing for exceptions is the core of keeping it simple and practical. Other principles are relevant but less direct.

6. D Unclear boundaries pull decisions upward. Delegation within a defined appetite speeds decisions safely. Meeting more often treats the symptom; banning escalation removes control.

7. D The register's value comes from prioritisation, ownership and measurement. A new tool or fragmented registers do not fix those gaps; refusing new ideas discourages engagement.

8. A Partners are part of the system that delivers value. Fixing the relationship and agreement addresses the cause; escalation per case and extra staff treat symptoms, and disowning the problem ignores the consumer's experience.

9. B The platform does not fit how work actually flows. That is the value streams and processes dimension. Technology passed its tests; nothing points to suppliers or legal factors.

10. C Strategic focus and risk dominate for core, sensitive services. Price and headcount matter but should not override control of a critical capability.

11. B The cause lies in the interactions; a joint investigation finds it. Penalties and replacement do not address the integration gap, and routing users to suppliers fragments the service further.

12. D Flow efficiency is very low (about 3%), which shows waiting waste dominates. Speeding up work or hiring staff barely changes lead time.

13. A Commodities benefit from standardisation; differentiating, experimental work benefits from controlled variety. Extremes either stifle innovation or multiply cost and risk.

14. B The issue is cultural: information is suppressed. Leadership behaviour must change. Forms, training and auditors do not make people feel safe to speak up.

15. A Ending at an internal milestone hides the consumer's experience. Redefining the end exposes the remaining gaps. Faster closure or surveys measure the wrong endpoint.

16. C Speed and closure measures can look good while users still repeat themselves and get passed around. User effort reveals the real experience. Faster answers or closures optimise the wrong things.

17. A A major incident needs one decision-maker and structured communication. Separate updates confuse customers; executives should not run technical work; separating teams loses coordination.

18. C Restoration comes first. A quick rollback of the probable cause restores service, and evidence is preserved for later analysis. Waiting for proof or formal approval prolongs impact.

19. C Recurring incidents point to an underlying problem. Faster restarts or higher priority treat symptoms, and changing user behaviour avoids the cause.

20. B Error control weighs fix cost and risk against workaround cost. Keeping the known error documented and reviewed is sound. Closing it loses visibility; forcing a risky fix ignores the trade-off.

21. B Optimise before automating: clean up noise first. An AI tool trained on mostly useless alerts adds cost and complexity. More staff or blanket disabling both miss the cause.

22. B Synthetic user journeys reveal failures in dependencies that component metrics miss. Weekly reports are too slow; more component metrics show the same green.

23. B Knowledge becomes useful when it is part of the work. A rewrite project goes stale; creation targets produce low-quality articles; changing tools does not change behaviour.

24. D Proven, low-risk, repeatable changes are ideal standard changes. More meetings or paperwork keep the bottleneck, and stopping patches raises security risk.

25. B Emergency changes use an expedited authority so they can be implemented quickly yet remain controlled and recorded. Waiting leaves exposure; skipping control creates other risks.

26. A Feature flags separate release from deployment and allow precise targeting. Blue-green switches all traffic; big bang and branch phasing do not target customer segments.

27. C Optimise before automating. Removing low-value approvals through role-based access addresses the cause; automating or chasing approvals keeps the waste.

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28. C Scoping by decisions keeps data accurate and useful. Over-scoping produces untrusted data; spreadsheets fragment information.

29. A The agreement must reflect the outcome that matters: payroll completed on time. Higher availability or more frequent reporting of the same measure misses the issue.

30. C OLAs set targets between internal teams to support SLAs. Underpinning contracts are for external suppliers; penalties do not fix internal alignment.

31. D Early-life support manages the gaps while keeping the date. Handing over as planned shifts risk to users; objecting without action changes nothing; permanent developer ownership may overload the team.

32. D Little's law: lead time = WIP ÷ throughput = 6 weeks now. Cutting WIP to 30 or fewer gives about 2 weeks. Higher utilisation and starting more work make queues worse.

33. D Protected capacity plus problem management on the biggest sources of interruption reduces unplanned load sustainably. Bans are impossible for incidents; separating improvement removes ownership.

34. D The pain appears at a front-stage moment of truth but is caused back stage. Journey mapping connects them. App tests, surveys and fees miss the operational cause.

35. C The waiting occurs between departments; only an end-to-end view addresses it. Local targets and staffing improve parts, not the journey.

36. B The two measures look at different levels. Relationship issues such as reliability or value need account conversations. Ignoring or dropping NPS hides the issue; blanket discounts may not fix causes.

37. C Value realisation needs owners, baselines and tracking. Warranty, availability and contingency do not show whether benefits arrived.

38. D Trusted advisers understand the customer's strategy and propose solutions. Faster responses, cheaper services and more reports improve the order-taker role but do not change it.

39. A Large batches drive failure. Smaller, automated, progressive changes improve speed and stability together. More approvals or longer cycles increase batch size and risk.

40. C The constraint is cultural and structural. Tools, mandates and better handover documents leave the conflicting goals and handovers in place.

41. A Pipeline-generated records give stronger evidence with less effort. Administrators and fewer fields keep a manual process; verbal confirmation weakens control.

42. D Error budget policies are agreed in advance so these decisions are not renegotiated under pressure. Ignoring or moving the target undermines the system.

43. B SRE caps toil and invests the freed time in automation. Hiring or moving work keeps the toil growing with scale.

44. D Only improving the constraint reduces end-to-end lead time. Faster or more teams add to the queue; reporting velocity measures non-constraints.

45. A Neglected debt and risk work produce incidents and findings. A deliberate rebalance protects long-term flow. Stopping all features or ignoring the trend are extremes.

46. A Outputs like deployments do not show contribution to growth. Outcome-based key results connect IT work to business goals. More detail or dates still describe outputs.

47. D One ranked portfolio with a WIP limit creates focus. Tracking or contractors leave the overload in place; per-director quotas ignore relative value.

48. D ALE falls from Tk 2 crore to Tk 50 lakh a year, saving Tk 1.5 crore against Tk 60 lakh cost. Residual risk is acceptable; no control removes all risk.

49. D Automated, enforced controls do not depend on memory. Reminders and annual audits remain weak; removing all privileged access stops operations.

50. B A sudden rise in one measure with falling quality signals gaming. Paired measures and investigation reveal the cause. Raising the target or hiding quality data makes it worse.

51. C Capability should rise where it limits value. Uniform targets drive documentation over value, and tools do not create capability.

52. B Strategy must come before technology choices. Copying, buying platforms or generating ideas without direction produces unfocused investment.

53. C Evidence-based tranches keep options open and control risk. Full commitment removes checkpoints; rejection ignores strategic need; funding without criteria lacks governance.

54. B High volume, clear rules and measurable benefit make the password and PIN use case the right first step. Complex diagnostics lack data, doing both at once spreads effort, and waiting for maturity forgoes clear value.

55. B Oversight should match the impact of errors. A 15% error rate is acceptable for low-risk, reversible actions but not for access grants. Quarterly review is too late; demanding 100% forgoes safe benefits.

56. A Unequal performance across user groups is a responsible-AI failure. Retraining and monitoring by language fixes it. Asking users to change language shifts the burden; switching off for all loses benefits.

57. C Managers shape daily behaviour and were excluded. Their involvement and removal of real obstacles addresses the gap. Training was not the issue; penalties produce hidden workarounds.

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58. A A spreading attack is a chaotic situation: stabilise first. Analysis can follow once harm is contained. Experimentation and normal queues are too slow.

59. C Transparent, agreed criteria and sensitivity checks make the decision defensible to all stakeholders. Seniority, price alone or supplier advice introduce bias.

60. A Stabilising service and making change safer address the causes of harm; cleaner signals and a defined AI use case follow. Launching the bot, a long assessment or wholesale outsourcing do not address the urgent causes.

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